INTERNATIONAL ROADMAP FOR DEVICES AND SYSTEMS™

2024 EDITION

EXECUTIVE SUMMARY

THE IRDS IS DEVISED AND INTENDED FOR TECHNOLOGY ASSESSMENT ONLY AND IS WITHOUT REGARD TO ANY COMMERCIAL CONSIDERATIONS PERTAINING TO INDIVIDUAL PRODUCTS OR EQUIPMENT.

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FOREWORD

The IEEE International Roadmap for Devices and Systems (IRDS) is the continuation and extension of the National Technology Roadmap for Semiconductors/International Technology Roadmap for Semiconductors (NTRS/ITRS) with a record of uninterrupted publications spanning from 1992 to 2015. The IRDS has followed several of the successful methodologies demonstrated by the NTRS/ITRS with additional expansion to system integration, data centers and Internet communications requirements as well as quantum computing and quantum communications. The IRDS has been previously published in 2016, 2017, 2018, 2020, 2021, 2022, and 2023.

Historically the draft of all the chapters of the ITRS used to be collected in October. Afterwards, the Executive Summary was assembled and delivered to the Semiconductor Industry Association (SIA) Board at their mid-November annual meeting.

In the subsequent months major results and forecasts for the following year were delivered in key conferences like the International Electron Devices Meeting (IEDM) (mid-December) and International Solid-State Circuits Conference (ISSCC) (late-February). In addition, multiple company announcements were made in January outlining their vision for the year that had just started. Historically, the publication of ITRS was timed consistently with the SIA board meeting.

With the transition to IEEE in May 2016 this constraint no longer existed, and the publication schedule evolved towards release of the IRDS at the beginning of 2Q of the following year. As a result, all the information that previously was included as part of the roadmap update and scheduled to be published the following year has now become available for the IRDS publication in real time. As a result, the 2019 IRDS was renamed the 2020 IRDS.

Additionally, historically the NTRS/ITRS and IRDS have predicted specific problems to occur early on, allowing enough time for research and supplier organizations to develop possible solutions. For instance, in 1998 the transformation of the planar silicon gate CMOS to strained silicon, high-κ/metal-gate and FinFET was clearly articulated, and these key milestones were accomplished in 2003, 2007 and 2011, respectively. Starting with the 2020 IRDS a section dedicated to Industry Highlights and IRDS messages was introduced to better show the relations between the two organizations.

ACKNOWLEDGMENTS

International Roadmap Committee

Europe—Francis Balestra, Enrico Sangiorgi

Japan—Yoshihiro Hayashi, Norikatsu Takaura, Hidemi Ishiuchi

U.S.A.— Paolo Gargini-chair, Tom Conte-vice-chair

The outstanding work by the members of the International Focus Teams is acknowledged in each of their roadmap chapters.

The chairs and co-chairs of these teams are as follows:

Application Benchmarking—Tom Conte
Systems and Architectures—Kirk Bresniker and Stephen Dukes
Outside Systems Connectivity—Michael Garner
More Moore—Mustafa Badaroglu
Lithography—Mark Neisser and Harry Levinson
Factory Integration—Supika Mashiro and James Moyne
Yield—Slava Libman and Dan Wilcox
Beyond CMOS—An Chen, Shamik Das and Matt Marinella
Cryogenic Electronics and Quantum Information Processing—Scott Holmes
Packaging Integration—Dev Gupta
Metrology—Yaw Obeng, Umberto Celano
Environment, Safety, Health, and Sustainability—Leo Kenny and Steve Moffat
More than Moore—Alan O'Riodan
Autonomous Machine Computing -Shaoshan Liu and Jean-Luc Gaudiot
Mass Memory Storage - Tom Coughlin and Roger Hoyt

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IRDS IEEE Societies and Councils Support Organizations

IEEE COMPUTER SOCIETY

IEEE rebooting COMPUTING

IEEE COMPUTER SOCIETY TFRC
Task Force on Rebooting Computing

Supporting Councils and Societies

IEEE CSC
Council on Superconductivity

CEDA
IEEE Council on Electronic Design Automation

ELECTRON DEVICES SOCIETY

IEEE Signal Processing Society

IEEE MAGNETICS

pels
IEEE POWER ELECTRONICS SOCIETY
Powering a Sustainable Future

CAS
IEEE CIRCUITS AND SYSTEMS SOCIETY

SSCS
IEEE SOLID-STATE CIRCUITS SOCIETY
IC Innovation

2024 EXECUTIVE SUMMARY

1. INTRODUCTION

IRDS MISSION

Identify the roadmap of electronic industry from devices to systems and from systems to devices.

IRDS STRUCTURE

This initiative focuses on an International Roadmap for Devices and Systems (IRDS) through the work of International Focus Teams (IFT) closely aligned with the advancement of the devices and systems industries. Led by the International Roadmap Committee (IRC), IFTs collaborated in the development of the 2022 IRDS roadmap, and engaged with other segments of the IEEE, such as the Rebooting Computing Initiative (RCI), Electron Devices Society (EDS), Computer Society (CS), Communication Society (ComSoc), and other societies as well as with related industry and academic communities like the System and Device Roadmap Committee of Japan (SDRJ) and the European SINANO Institute (ESI) in complementary activities to help ensure alignment and consensus across a range of stakeholders, such as:

- Academia
- Consortia
- Industry
- National laboratories

IEEE, the world's largest technical professional organization dedicated to advancing technology for humanity, through the IEEE Standards Association (IEEE-SA) Industry Connections (IC) program, supports the IRDS to ensure alignment and consensus across a range of stakeholders to identify trends and develop the roadmap for all the related technologies in the computer industry.

The scope of the IRDS™ of the fundamental building blocks in the electronics industry spanning from devices to systems and from systems to devices, in which the AI-centric IoT-social-infrastructures with high-speed network communication will be built on the semiconductor device and process technologies, as illustrated in Figure ES1.

The 2024 IRDS Focus Teams are as follows:

1. Systems and Architectures (SA)
2. More Moore (MM)
3. Outside System Connectivity (OSC)
4. Beyond CMOS (BC)
5. Cryogenic Electronics and Quantum Information Processing (CEQIP)
6. Packaging Integration (PI)
7. Factory Integration (FI)
8. Lithography (L)
9. Yield Enhancement (YE)
10. Metrology (M)
11. More than Moore (MtM) white paper
12. Environment, Safety, Health, and Sustainability (ESH/S) as Environmental Sustainability of Semiconductor Facilities (ESSF)

Two additional chapters have been added. These are:

13. Mass Data Storage and Non-volatile Memory
14. Autonomous Machine Computing

Each of the IFTs are focusing not only on technical roadmaps in their specific areas but also on cross-boundary areas among the systems and the devices, or the devices and the fabrication process technologies. The major revision of the

IRDS roadmaps is updated every two years, while minor revisions are done every year if necessary. Here, the IRDS 2024 Edition is the material resulting from revisions to the 2023 IRDS. Under management of the international roadmap committee (IRC) of the international representatives, the technical points in each IFT have been selected and discussed internally, and the results are being released to public through the IEEE IRDS homepage (irds.ieee.org).

The scope of the IRDS™ of the fundamental building blocks in the electronics industry spanning from devices to systems and from systems to devices under the international collaboration to discuss the focus points. The new ecosystem of the electronics' industry based on semiconductor technologies.

IEEE SPONSORS

The IRDS is sponsored by the IEEE Rebooting Computing Initiative in consultation and support from many IEEE Operating Units and Partner organizations, including the following:

CASS—Circuits and Systems Society
CEDA—Council on Electronic Design Automation
CS—Computer Society
CSC—Council on Superconductivity
EDS—Electron Devices Society
EPS— Electronics Packaging Society
MAG—Magnetics Society
NTC—Nanotechnology Council
PELS—Power Electronics Society
RS—Reliability Society
SSCS—Solid State Circuits Society
SRC—Semiconductor Research Corporation
IEEE Standards Association

INTERNATIONAL SPONSORS AND COOPERATION

There are several international roadmap efforts directly aligned with the IRDS:

- The SINANO Institute http://www.sinano.eu/
- The System Device Roadmap Committee of Japan (SDRJ) https://www.sdrj.jp/
- The International Electronics Manufacturing Initiative (iNEMI) http://www.inemi.org/

2. THE BIG PICTURE

2.1. GDP DEFINITION AND CONSIDERATIONS

Gross domestic product (GDP) is a key economic indicator that represents the total monetary value of all finished goods and services produced within a country's borders in a specific period. It is typically broken down into four main components:

- Consumption—Expenditures by households on goods and services.
- Investment—Spending on capital goods that will be used for future production
- Government Spending—Total government expenditures on final goods and services.
- Net Exports—The value of a country's exports minus its imports.

Understanding the composition of GDP across different countries provides insight into their economic structures and growth drivers. Below is a summary of the GDP components for the leading GDP leaders of United States, China, Germany and Japan over the past decade.

Country	Consumption (% GDP)	Investment (% GDP)	Government Spending (% GDP)	Net Exports (% GDP)
United States	~68% (Household) / ~70% (Final)	~21%	~17%	~−3%
China	~39% (Household) / ~56% (Final)	~42%	~17%	~+1%
Germany	~52%	~21%	~19%	~+8%
Japan	~55%	~24%	~20%	~+1%

Figure ES-1. Composition of GDP compiled from multiple sources

United States (2015-2024):

- The U.S. economy is predominantly driven by consumption, consistently accounting for approximately 68–70% of GDP. Investment contributes around 21%, government spending about 17%, and net exports have historically been negative, at around -3% reflecting a trade deficit.

China (2015-2024):

- China has been transitioning from an investment and export-driven economy to one more reliant on consumption. However, consumption remained at 39% (real consumer spending) and 56% (considering government support for multiple consumers' needs paid for consumers in other regions) respectively of GDP, with investment at 42%, indicating a continued emphasis on investment-led growth. Net exports remain positive.

Germany (2015-2024)

- Germany has been averaging at 52% consumption with investments from industry and government have remain constant around 20%, Net exports have remained at 8% value.

Japan (2015-2024):

- Japan's economy features a high consumption rate, contributing around 55% of GDP. Investment accounts for approximately 24%, government spending about 20%, and net exports have a modest 1% negative contribution.

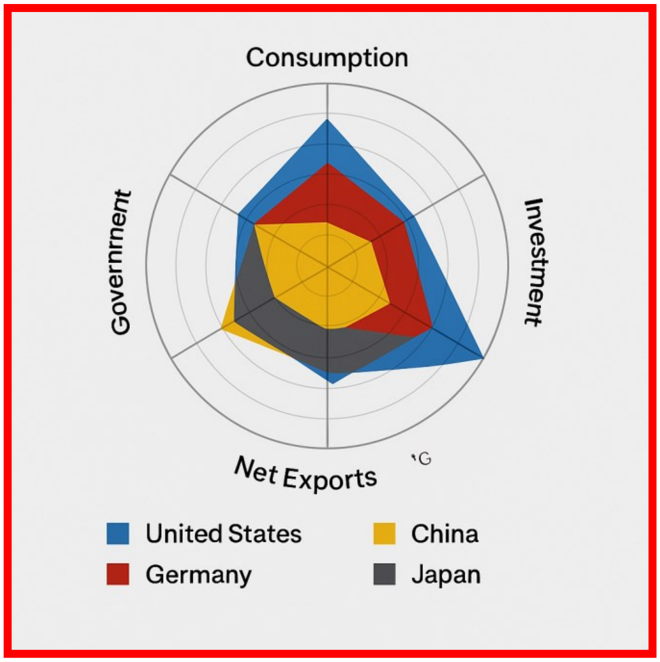


Figure ES-2. Comparative trends in World GDP

This following chart summarizes how the world economy collapsed during the pandemic to almost -3% but it rebounded to over +6% in 2021. This proves that the global economy developed in the last half of the previous century is extremely flexible and capable of rebounding even from dramatic worldwide negative events like the pandemic.

Year	GDP Real (Inflation adj.)	GDP growth
2013	$70,960,600	2.87%
2014	$73,182,700	3.13%
2015	$75,472,500	3.13%
2016	$77,596,100	2.81%
2017	$80,274,300	3.45%
2018	$82,909,000	3.28%
2019	$85,127,600	2.68%
2020	$82,677,400	-2.88%
2021	$87,927,500	6.35%
2022	$90,774,600	3.24%
2023	$93,346,700	2.83%

Source: https//www.worldometers.info/gdp/#gdpyear

Figure ES-3. World GDP trends for the past ten years

However, international tension over trade issues is compelling most economists to be overly cautious about the second half of 2025 and this is reflected in the latest estimates of the International Monetary Fund (IMF).

World Economic Outlook Growth Projections

(Real GDP, annual percent change)	2024	2025 (Projections)	2026 (Projections)
World Output	3.3	2.8	3.0
Advanced Economies	1.8	1.4	1.5
United States	2.8	1.8	1.7
Euro Area	0.9	0.8	1.2
Germany	-0.2	0.0	0.9
France	1.1	0.6	1.0
Italy	0.7	0.4	0.8
Spain	3.2	2.5	1.8
Japan	0.1	0.6	0.6
United Kingdom	1.1	1.1	1.4
Canada	1.5	1.4	1.6
Other Advanced Economies	2.2	1.8	2.0

Source: International Monetary Fund (IMF)

Top 10 Economies by Nominal GDP

IMF World Economic Outlook (October 2025)

Rank	Country	GDP (USD)	Growth (%)
1	United States	$30.62	2.00%
2	China	$19.40	4.80%
3	Germany	$5.01	0.20%
4	Japan	$4.28	1.10%
5	India	$4.13	6.60%
6	United Kingdom	$3.96	1.30%
7	France	$3.36	0.70%
8	Italy	$2.54	0.50%
9	Russia	$2.54	0.60%
10	Canada	$2.28	1.20%

Revised on October 2025 by IMF

Figure ES-4. World GDP may decline if political tensions continue to affect the leading economies

Below are some GDP highlights of the top four economies in Figure ES-5. All the numbers are adjusted for inflation. Nominal reported GDPs are generally higher.

	US GDP	Growth	China GDP	Growth	Germany GDP	Growth	Japan GDP	
Year	Million USD	%	Million USD	%	Million USD	%	Million USD	%
2015	18,295,000	3.13%	11,061,600	7.04%	3,423,570	1.65%	4,444,930	1.56%
2016	18,627,000	1.81%	11,819,200	6.85%	3,502,130	2.29%	4,478,440	0.75%
2017	19,085,700	2.46%	12,640,300	6.95%	3,597,250	2.72%	4,553,470	1.68%
2018	19,651,900	2.97%	13,493,400	6.75%	3,637,410	1.12%	4,582,760	0.64%
2019	20,159,600	2.58%	14,296,400	5.95%	3,673,340	0.99%	4,564,330	-0.40%
2020	19,723,600	-2.16%	14,616,400	2.24%	3,522,910	-4.10%	4,375,040	-4.15%
2021	20,917,900	6.06%	15,851,300	8.45%	3,652,210	3.67%	4,487,020	2.56%
2022	21,443,400	2.51%	16,319,000	2.95%	3,702,230	1.37%	4,529,850	0.95%
2023	22,062,600	2.89%	17,175,700	5.25%	3,692,370	-0.27%	4,605,910	1.68%

Figure ES-5. A decade of quantitative GDP trends of leading economies

2.2. ECONOMIC HIGHLIGHTS OF THE PAST FIVE YEARS

ECONOMIC SHOCKS, RECOVERY, AND THE SEMICONDUCTOR RENAISSANCE

Gross domestic product serves as a vital indicator of the nations' economic health, reflecting the total value of goods and services produced over a specific time. Since 2019, the global economy has undergone extraordinary disruptions and transformations, most notably due to the COVID-19 pandemic. This introduction explores the trajectory of GDP from 2019 to the present, analyzing key factors that influenced its changes, particularly focusing on the semiconductor industry's role during the crisis and recovery period. It also highlights policy responses like the CHIPS Acts around the world and evaluates current trends and prospects.

- Global GDP Trends in 2019: A Pre-Pandemic Baseline

In 2019, the global economy was growing modestly, albeit with growing concerns. The International Monetary Fund (IMF) reported global GDP growth of about 2.68%, a slowdown from previous years due to trade tensions (notably between the U.S. and China), Brexit uncertainties, and a decline in manufacturing output in key economies like Germany and Japan. The U.S. economy grew to 2.58%, bolstered by consumer spending and a strong labor market. China saw a growth rate of 8.45%, continuing its trend of gradual deceleration although still outpacing the US and Japan. Semiconductor markets were stable but still supported by soft demand from smartphones and sluggish automotive sectors, though emerging technologies like AI and 5G were laying groundwork for future growth.

- The COVID-19 Pandemic and Global Recession (2020)

The outbreak of COVID-19 in early 2020 sent shockwaves through the global economy. Lockdowns, supply chain disruptions, and mass layoffs caused GDP to plummet across most economies. Global GDP contracted to -2.88% in 2020, marking the worst recession since the Great Depression. The U.S. GDP shrank to -2.16%, and the Eurozone economy contracted by 6.4%. In contrast, China rebounded quickly, growing to 6% in 2021, largely due to early containment and stimulus efforts.

During this period, semiconductor supply chains were heavily disrupted. Remote work, online education, and telehealth created an unexpected surge in demand for electronics, while factory shutdowns and logistics bottlenecks constrained supply. The result was a severe global chip shortage by late 2020, exposing the fragility of the semiconductor value chain.

- Recovery and Rebound (2021–2022): Uneven but Strong

As vaccines were rolled out in 2021, economies began reopening. Global GDP grew by 6.35% in 2021, an impressive rebound led by stimulus spending and pent-up consumer demand. The U.S. economy grew 6.06%, its fastest pace since 1984. Europe also saw recovery, albeit uneven due to vaccine disparities and supply issues. Semiconductors became central to post-pandemic recovery: automotive, consumer electronics, and cloud infrastructure all drove up demand. However, supply chain fragility persisted. Industries from carmakers to smartphone producers faced production halts due to chip shortages. These bottlenecks fueled inflationary pressures as goods became scarce, further challenging global GDP growth.

In response, countries began to prioritize semiconductor self-sufficiency. The U.S. CHIPS Act (Creating Helpful Incentives to Produce Semiconductors) was introduced and signed into law in August 2022, aiming to restore domestic chip production and reduce reliance on East Asian suppliers. The legislation included $52 billion in subsidies and tax incentives for semiconductor manufacturing, and R&D. Europe and Japan quickly reacted by introducing similar programs albeit aimed at their specific markets. Several other countries followed the trend.

- Stabilization and Inflationary Challenges (2023)

2023 marked a transitional year as economies tried to balance post-pandemic recovery with inflation control. The global GDP grew by about 2.83%, down from 2021 highs of 6.35% but indicating stability. However, several challenges loomed ahead, and fears of inflation prompted government actions. Interest rate hikes by central banks (e.g., U.S. Federal Reserve and European Central Bank (ECB)) to fight inflation. Energy price volatility, especially in Europe due to the Ukraine war. Overall moderating consumer demand as borrowing costs rose.

The semiconductor industry faced a temporary slowdown in early 2023, driven by high inventory levels caused by supply reaction to the 2020–2022 shortage and reduced consumer electronics demand. However, longer-term drivers—like AI, electric vehicles (EVs), 5G expansion, and industrial automation—continued to stimulate investment in chip fabrication and innovation.

The U.S. CHIPS Act funds began to be disbursed, with major players like Intel, TSMC, and Samsung each investing billions in new U.S.-based fabs. Europe and Japan launched similar initiatives spurring new semiconductor plants investments, all reflecting a strategic realignment of the semiconductor supply chain.

- Present Status (2024–2025): A Balanced but Cautious Outlook

By early 2025, global GDP growth is projected at 2.9–3.2%, with signs of cautious consumer optimism. Inflation has eased in many regions, but geopolitical tensions and supply chain risks remain. However, should international trade tension continue to increase the successful recovery of the past three years could quickly precipitate the word economy into a widespread recession. See latest IMF forecast for 2H2025 above.

2.3. THE ELECTRONICS INDUSTRY AND THE TRANSFORMATIVE ROLE OF SEMICONDUCTORS

INTRODUCTION

The electronics industry stands at the core of modern civilization, supporting nearly every aspect of daily life and global commerce. The key business drivers of the electronics industry will be referenced multiple times in this ES report and so it is useful to clarity this classification from the very beginning. This industry is broadly divided into six major segments:

1. Computers
2. Communications
3. Industrial
4. Consumer electronics
5. Automotive
6. Government/military

The common thread uniting all these segments is the semiconductor—a microscopic marvel that has enabled rapid technological advancement over the last several decades. This introduction to the 2024IRDS Executive Summary outlines the definitions of each segment, how semiconductors have shaped their development and the resulting societal effects.

COMPUTERS

This segment includes desktops, laptops, servers, and data centers used in personal, business, and scientific computing. Semiconductors form the backbone of all computing systems. Central processing units (CPUs), graphics processing units (GPUs), memory (RAM), and storage devices (e.g., SSDs) are all powered by advanced microchips. Semiconductor scaling has allowed computing power to double approximately every two years (as per Moore's Law), enabling more powerful and compact systems.

Societal Impact

The proliferation of computers has revolutionized education, business, and research. Tasks that once took days can now be completed in seconds. Data centers powered by semiconductors support cloud computing, artificial intelligence, and big data analytics, empowering innovation in every industry.

COMMUNICATIONS

This segment covers devices and infrastructure used for transmitting data, including smartphones, routers, fiber-optic systems, satellites, and wireless networks. Communication devices rely heavily on semiconductor technologies for signal processing, frequency modulation, data encoding, and antenna control. Key components include RF (radio frequency) chips, baseband processors, and networking ASICs (application-specific integrated circuits).

Societal Impact

Semiconductors have made global communication instantaneous and ubiquitous. Mobile networks, broadband internet, and satellite communications have enabled globalization, social connectivity, remote work, and real-time access to information worldwide.

INDUSTRIAL

This segment encompasses manufacturing systems, automation equipment, robotics, power grids, and infrastructure control systems. Microcontrollers, power electronics, sensors, and programmable logic devices control everything from assembly line robots to smart power meters. These chips offer precision, energy efficiency, and real-time data processing essential for automation and monitoring.

Societal Impact

Semiconductors have driven the evolution of Industry 4.0—where smart factories use interconnected systems to optimize production and maintenance. This results in greater efficiency, lower emissions, and more responsive supply chains. It also supports safer working environments through automation.

CONSUMER ELECTRONICS

This segment includes personal devices such as televisions, smartwatches, tablets, gaming consoles, smart home devices, and wearable tech. These devices rely on semiconductors for processing, user interface management, image and sound rendering, wireless connectivity, and power management. Chips enable voice assistants, touchscreens, biometric sensors, and immersive graphics.

Societal Impact

Consumer electronics have reshaped how people live, work, and entertain themselves. They enable personalized experiences, on-demand media consumption, home automation, and health tracking. They've also fostered a culture of digital interaction and convenience.

AUTOMOTIVE

The automotive segment covers electronic systems in vehicles, including safety systems, infotainment, navigation, electric powertrains, and autonomous driving technologies. Modern vehicles use hundreds of chips, including engine control units (ECUs), power management ICs, light detection and ranging (LiDAR) sensors, radar systems, and advanced driver-assistance systems (ADAS). With electric vehicles (EVs), power semiconductors play a vital role in managing battery systems and motors.

Societal Impact

Semiconductors have enabled safer, cleaner, and smarter transportation. Features like automatic braking, lane assistance, and adaptive cruise control have significantly reduced accidents. Electrification reduces dependency on fossil fuels, supporting sustainability goals.

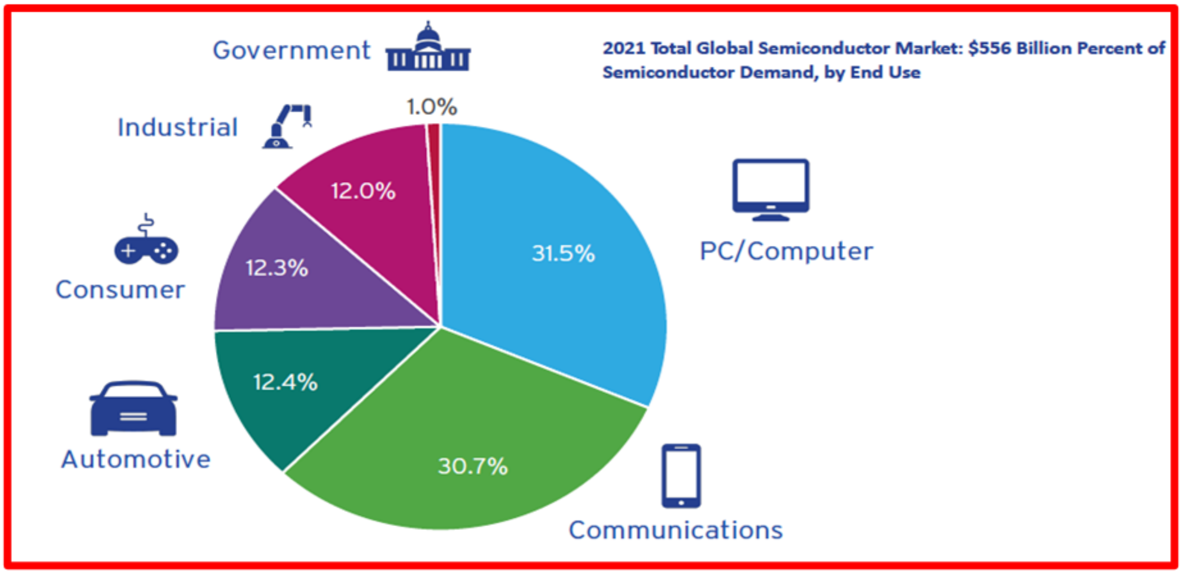
GOVERNMENT AND MILITARY

This segment involves aerospace systems, defense electronics, public infrastructure monitoring, and secure communications. Defense and government applications require robust, high-performance, and secure chips. They are used in radar, navigation systems, satellite communications, encryption modules, and aerospace control systems.

Societal Impact

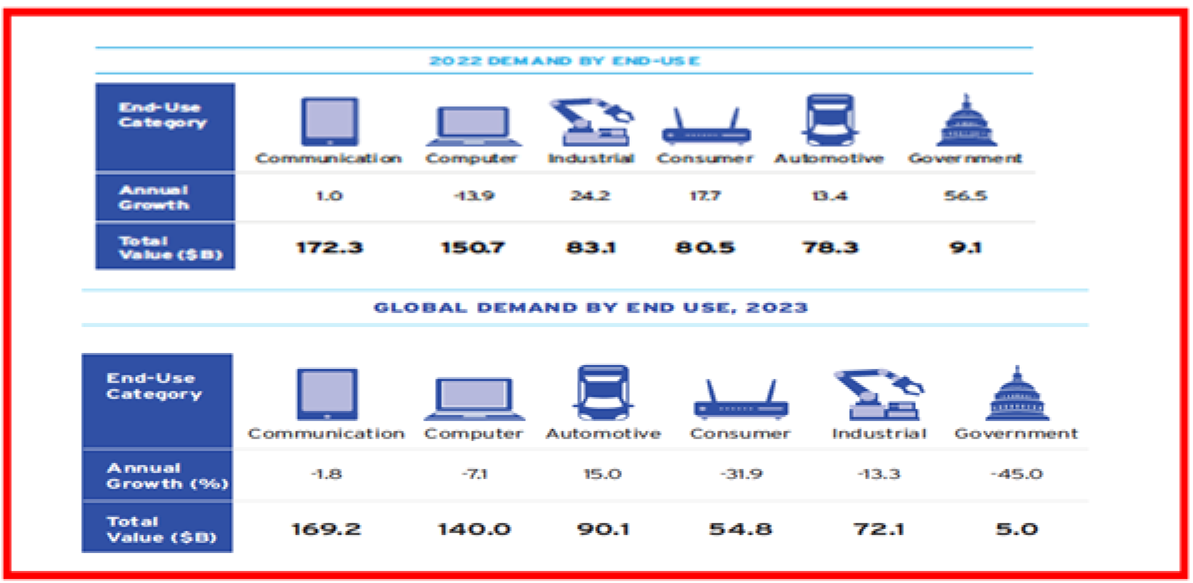
Semiconductors ensure national security and the integrity of critical infrastructure. They also enable space exploration, weather forecasting, and disaster response, with broad implications for public safety and knowledge advancement.

These charts indicate where all the semiconductors were used in each of the above categories from 2021 to 2023. It is relevant to observe that since 2022 the semiconductor content in communications has overtaken content in computing products.



Source: SIA

Figure ES-6. Distribution of semiconductor demand by end users in 2021



2022 DEMAND BY END-USE

End-Use Category	Communication	Computer	Industrial	Consumer	Automotive	Government
Annual Growth	1.0	-13.9	24.2	17.7	13.4	56.5
Total Value ($B)	172.3	150.7	83.1	80.5	78.3	9.1

GLOBAL DEMAND BY END USE, 2023

End-Use Category	Communication	Computer	Automotive	Consumer	Industrial	Government
Annual Growth (%)	-1.8	-7.1	15.0	-31.9	-13.3	-45.0
Total Value ($B)	169.2	140.0	90.1	54.8	72.1	5.0

Source: SIA

Figure ES-7. Distribution of semiconductor demand by end users in 2022 and 2023

SUMMARY

Semiconductors are the foundational technology that has enabled the modern digital era. From everyday consumer devices to highly specialized defense systems, semiconductors have powered innovation, productivity, and connectivity. Their impact extends far beyond individual industries, shaping economies, societal structures, and global interactions. As the demand for processing power, energy efficiency, and connectivity continues to grow, semiconductor technology will face both opportunities and challenges. Emerging fields such as quantum computing, neuromorphic chips, 6G communication, and edge AI will define the next frontier of innovation. Simultaneously, geopolitical and supply chain dynamics will shape how semiconductor ecosystems evolve globally. In conclusion, semiconductors will remain a cornerstone of progress—transforming not only industries but the very way humanity interacts with technology and each other.

3. THE ELECTRONICS INDUSTRY IN REVIEW FROM 2019 TO THE PRESENT—AND THE ROAD AHEAD

3.1. INTRODUCTION

Let's focus now in particular on the electronics industry. The global electronics industry has always thrived on innovation, efficiency, and adaptability. From semiconductors to system integration, its role in shaping modern society is foundational. Yet, from 2019 to the present, the industry has undergone some of the most significant disruptions and transformations in its history—sparked by a pandemic, supply chain chaos, rapid AI breakthroughs, and sectoral reinvention. This chapter charts an overview of the industry's evolution over this pivotal period and offers a forward-looking view into its future trajectory.

2019: THE CALM BEFORE THE DIGITAL STORM

In 2019, the electronics industry was in a mature but relatively stable phase: shipments of Smartphones had reached saturation in developed markets. PC shipments were flat or declining, seen largely as utilitarian. Semiconductor demand was moderate after a cyclical high in 2018. AI buzz was alive but only within the experts 'community and commercial returns remained limited—most real-world deployments were narrow and brittle.

At this point reports on NVIDIA's A100 GPU already in development were circulating but it was far from headline news. Several specialized AI chips were also being built, but AI still hadn't found a compelling "killer app" outside of tech experts.

2020–2021: THE PANDEMIC-DRIVEN DIGITAL EXPLOSION

When COVID-19 swept across the world, the electronics industry was both disrupted and catalyzed.

Unusual PC Growth

After nearly a decade of stagnation, PC demand exploded in 2020–2021. Driven by remote communications including work, virtual classrooms, and digital lifestyles, shipments surged. IDC estimated a 13% year-over-year (YoY) increase in PC shipments in 2020, followed by continued strength in 2021. Laptops with webcams, adequate CPUs, and better connectivity became essential tools, not optional gadgets. This resurgence reshaped consumer tech expectations, leading to longer-term upgrades in productivity hardware and peripherals. (Figure ES-19). It is fair to say that communications via PCs and smart phones kept societies from collapsing in complete chaos and kept innumerable human activities moving forward.

Communications and Cloud Boom

During this period broadband upgrades, fiber rollout, and 5G deployment accelerated. Data center build-outs surged to support Zoom, Netflix, gaming, and e-commerce. Equipment vendors scrambled to scale up, exposing semiconductor shortages, especially in analog, RF, and legacy nodes.

AI'S QUIET MATURATION AND SUDDEN SPOTLIGHT

AI's Long Road and Early Disappointments

The concept of AI had been formulated as far back as the 40s but no meaningful demonstrations had occurred since it required a massive parallel manipulation of data and lack of an adequate technology to do so had prevented AI from reaching any practical results. It was finally a group of students at Stanford University that demonstrated in 2009 that a GPU developed for parallel processing of data to create high quality and fast-moving images in computer games could be utilized to enable AI. By 2018 the design of a customized GPU for AI application had been realized and published but still more needed to be done. From 2015 to 2020, deep learning showed promise but failed to transform mainstream business processes. Many AI startups folded or pivoted. Hype cycles rose and fell around self-driving cars, voice assistants, and general AI. Despite this, finally NVIDIA launched the A100 GPU in May 2020, targeting data centers and AI training. It was a beast in performance—but at the time, outside of hyperscalers, few noticed AI developments and, most of all, at the time, societies were in the middle of the pandemic epidemic trying to solve fundamental existence problems. Most enterprises weren't ready for large-scale AI inference or training but then came H100—and ChatGPT. *When ChatGPT launched in late 2022 it reached the 100 million users in two months*, a record never even approached by any of the most popular applications (Figure ES-8).

This changed everything. Suddenly, the world saw the value of AI at scale. Demand for generative AI models and inference workloads soared, and everyone began asking: *What hardware powers this?* And then role of NVIDIA's H100 that was launched in 2022 was finally understood in 2023. With this, AI training clusters with thousands of H100s became coveted assets. Hyperscalers, enterprises, and even governments raced to secure GPU capacity. The A100 was the stealthy pioneer but the H100 was the rock star that made everyone look back and realize how quietly but quickly AI hardware had grown up. This is just one element highlighting the new role of electronics in transforming society.

More details on this subject will be highlighted later in this Executive Summary in Section 9, in the section at this link.

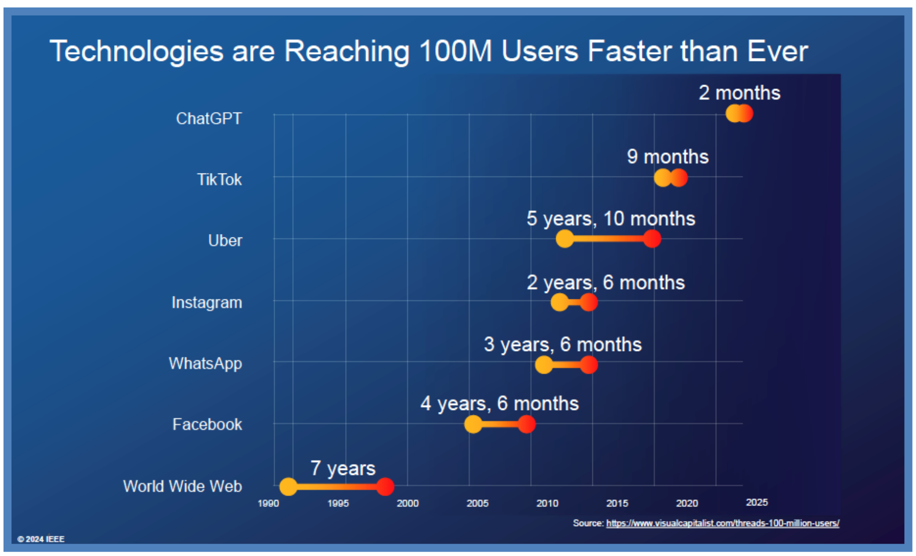


Figure ES-8. ChatGPT marks the new speed record in consumer adoption

The real trailblazer announcing the advent of AI to the consumers at large was ChatGPT, which reached the 100 million users in only two months!

3.2. SECTORAL SHIFTS AND THE EXPANDING ROLE OF ELECTRONICS

AUTOMOTIVE: FROM MECHANICS TO MECHATRONICS

However, the success of AI heralded by the hyperscalers does not represent the whole story.

The global automotive industry has experienced significant transformations over the past decade, influenced by technological advancements, shifting consumer preferences, and the growing emphasis on sustainability. This overview of automobile production trends in the United States, China, Japan, and Europe from 2019 to 2023, examines the increasing integration of electronics in vehicles, and highlights the rise of EV sales in these countries during the same period.

REGISTRATIONS OR SALES OF NEW VEHICLES - ALL TYPES						
REGION	2,019	2,020	2,021	2,022	2,023	TOTAL
AMERICA	25,389,730	20,817,484	22,003,539	20,876,860	23,247,296	112,334,909
CHINA	25,796,931	25,311,069	26,314,263	26,863,745	30,093,698	134,379,706
JAPAN	5,195,216	4,598,615	4,448,340	4,201,320	4,779,086	23,222,577
EUROPE+	20,930,134	16,714,115	16,882,486	15,079,901	17,898,967	87,505,603
TOTAL	77,312,011	67,441,283	69,648,628	67,021,826	76,019,047	357,442,795

EUROPE= EU 27+EFTA+UK

The European Free Trade Association. The European Free Trade Association (EFTA):
The intergovernmental organiZation of Iceland, Liechtenstein, Norway and Switzerland

Source: Statista

Figure ES-9. Number of vehicles sold in the past five years by region

China has consistently led global vehicle production, with significant growth observed in 2023. The United States, Japan, and Germany have maintained stable production levels, with some fluctuations due to economic conditions and global events. Overall, the automobile industry annual sales have presently stabilized around the 70M units value. However, since new features are continuously added the electronics content has been increasing at a steady state.

ELECTRONICS CONTENT IN AUTOMOBILES

Over the past decade, the automotive industry has seen a substantial increase in the integration of electronic components, enhancing vehicle safety, performance, and user experience. The following table illustrates the estimated average electronics content per vehicle in the United States, China, Japan, and Germany from 2015 to 2024, expressed as a percentage of the total vehicle cost.

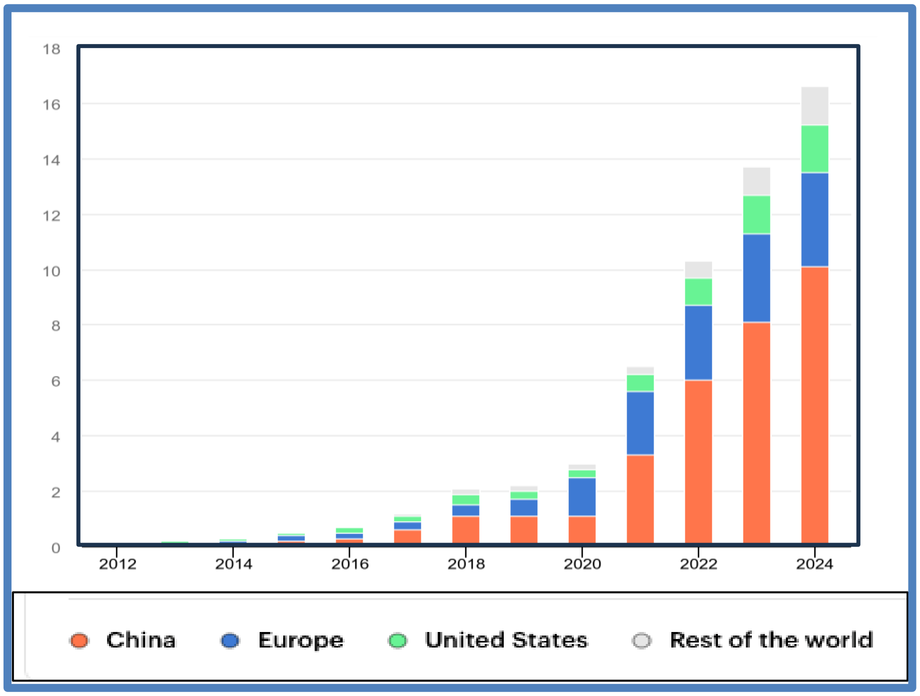
Year	United States	China	Japan	Germany
2015	18%	15%	20%	22%
2016	19%	16%	21%	23%
2017	20%	17%	22%	24%
2018	22%	18%	23%	25%
2019	23%	19%	24%	26%
2020	25%	21%	26%	28%
2021	26%	22%	27%	29%
2022	28%	24%	29%	31%
2023	30%	26%	31%	33%
2024	32%	28%	33%	35%

Source: Wall Street Journal (WSJ)

Figure ES-10. Relative value of electronics content in automobiles

The rise in electronics content is driven by advancements in safety systems, infotainment, connectivity, and the introduction of advanced driver assistance systems (ADAS). Manufacturers have increasingly focused on integrating sophisticated electronic systems to meet consumer demand for enhanced features and to comply with stringent safety regulations.

The automotive industry has indeed become one of the largest consumers of semiconductors (see charts above) ranking third after surpassing consumer electronics in semiconductor content. In 2023. For instance, EVs require 2–3x the semiconductor content of traditional vehicles. Cameras, radar, lidar, and powerful SoCs are the backbone of ADAS and autonomy. Automakers are becoming software companies—Tesla led the way, but now everyone is building centralized compute platforms. China is dominating this market by far.



Source: International Energy Agency (IEA)

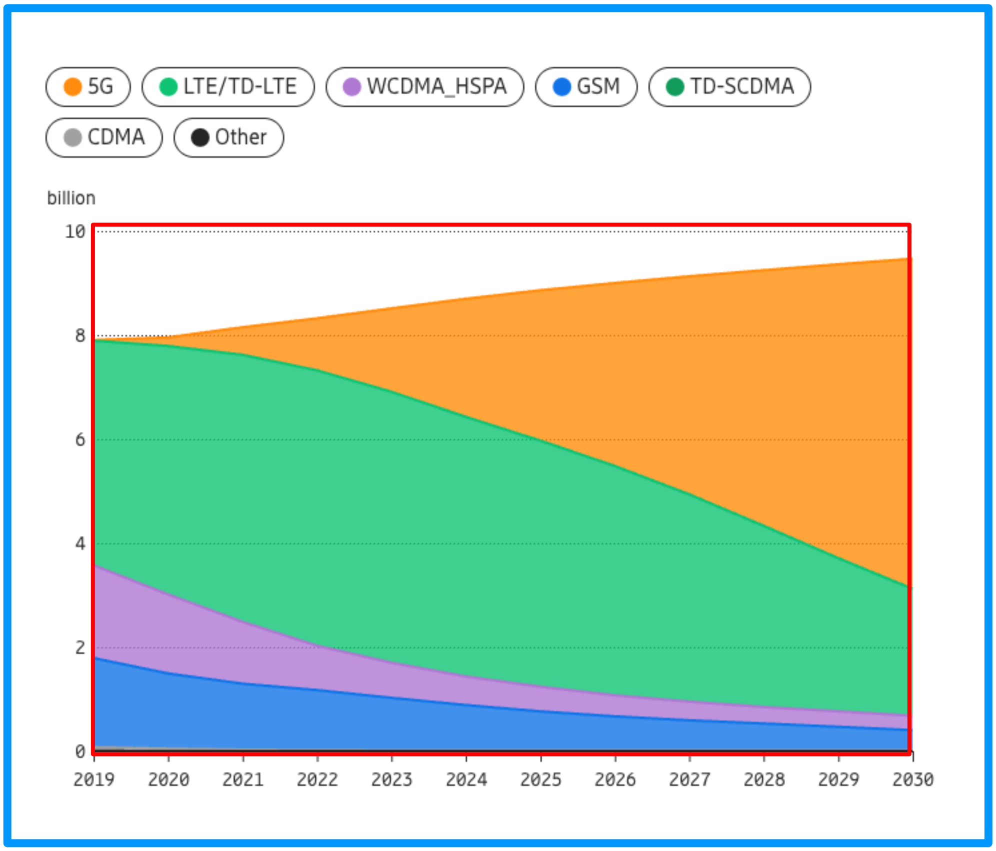
Figure ES-11. Number of units in millions of EV Automobile sales

COMMUNICATIONS

The landscape of global communications has undergone significant transformations since 2019, marked by the rapid deployment of (1) 5G networks, the emergence of (2) low earth orbit (LEO) satellite constellations, and the continued expansion of (3) fiber optic infrastructure. These advancements have collectively enhanced connectivity, reduced latency, and increased data transmission capacities worldwide.

5G Penetration

The 5G rollouts created demand for RF chips, beamforming arrays, and high-frequency filters. Fiber backhauls, small cells, and network slicing placed new demands on network silicon. Telecom operators partnered with hyperscalers to virtualize network functions (virtualized radio access network (vRAN), Open RAN), further increasing cloud-electronics convergence. The rollout of 5G technology has been a pivotal development in the telecommunications sector. By the third quarter of 2024, global 5G connections surpassed 2 billion, reflecting a 48% year-over-year growth. Projections indicate that 5G connections will reach 8.4 billion by 2029, accounting for 59% of all global wireless technologies. North America has been at the forefront of 5G adoption, with 264 million connections by the end of Q3 2024, representing 37% of all wireless connections in the region.



Source: Eriksson

Figure ES-12. Total number of connections by technology type

ROLE OF LOW EARTH ORBIT SATELLITE

LEO satellites typically orbit between 500 km to 2,000 km above Earth's surface. The orbital period is roughly 90 to 120 minutes per orbit, depending on altitude. However, this implies that a LEO satellite remains in "line-of-sight" for communications in any location only for a short time for and this limitation requires multiple satellites operating in the same orbit to assure continuity of communications. Due to proximity to Earth, LEO satellites offer much lower latency than geostationary (GEO) satellites and it can be as low as 20–50 ms, depending on routing and ground station proximity. When used, they reduce reliance on ground stations and improve latency consistency. As an example, Starlink users report 50 Mbps to 250 Mbps downlink speeds, with some bursts beyond that.

Feature	LEO Satellites	GEO Satellites	Terrestrial Fiber
Altitude	500–2,000 km	~35,786 km	Ground level
RTT Latency	20–50 ms	~600 ms	~10–30 ms
Bandwidth	50–250 Mbps (user)	Lower per-user speeds	Multi-Gbps
Coverage	Global (with enough sats)	Global (but sparse)	Regional

RTT: Round Trip Time

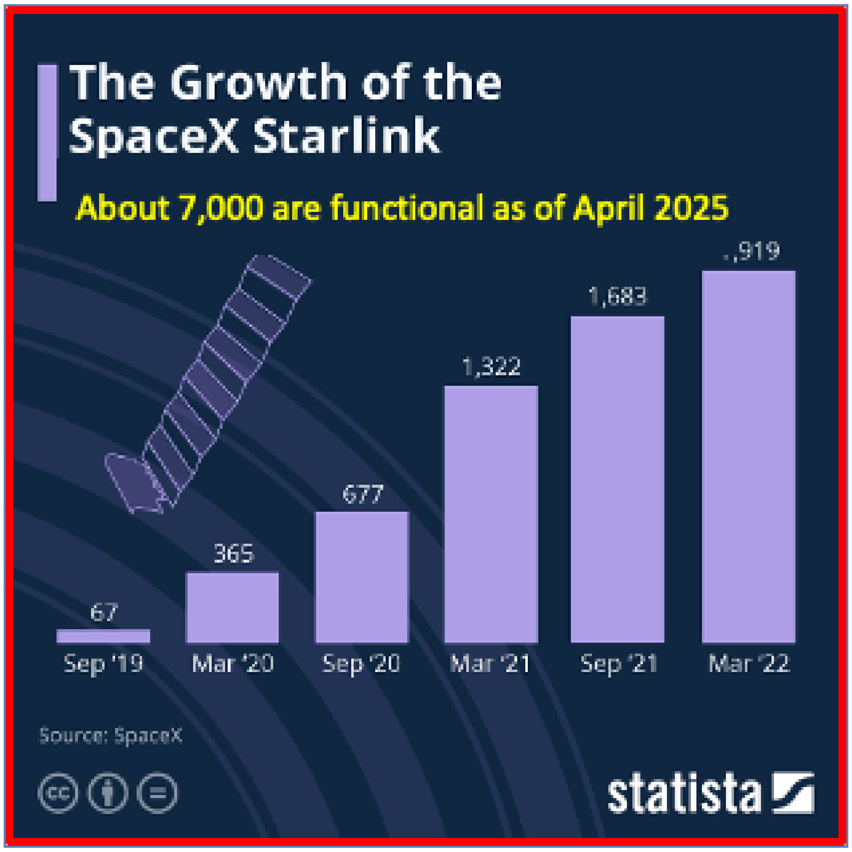
Figure ES-13. Key attributes of satellite and ground communications

However, rain, clouds, and terrain can affect throughput and stability of communications. Nevertheless, LEO satellite constellations have emerged as a transformative force in global communications, offering high-speed internet access to remote and underserved areas. SpaceX's Starlink, for instance, has deployed a cumulative number of over 7,000 satellites since 2019 and, as of September 2024, serves more than 4 million subscribers across 100 countries. As an example of global communications, in Africa, MTN Group's collaboration with Lynk Global led to the continent's first satellite voice call using a standard smartphone in March 2025. This development underscores the potential of LEO satellites to bridge connectivity gaps in rural regions.

Mission	Number of Satellites
Communications	>8000
Earth Observation	>1000
Technology Development	>300
Navigation	>100
Space Science	>100

Source: Kongsberg

Figure ES-14. Number of LEO satellites deployed to date and their function

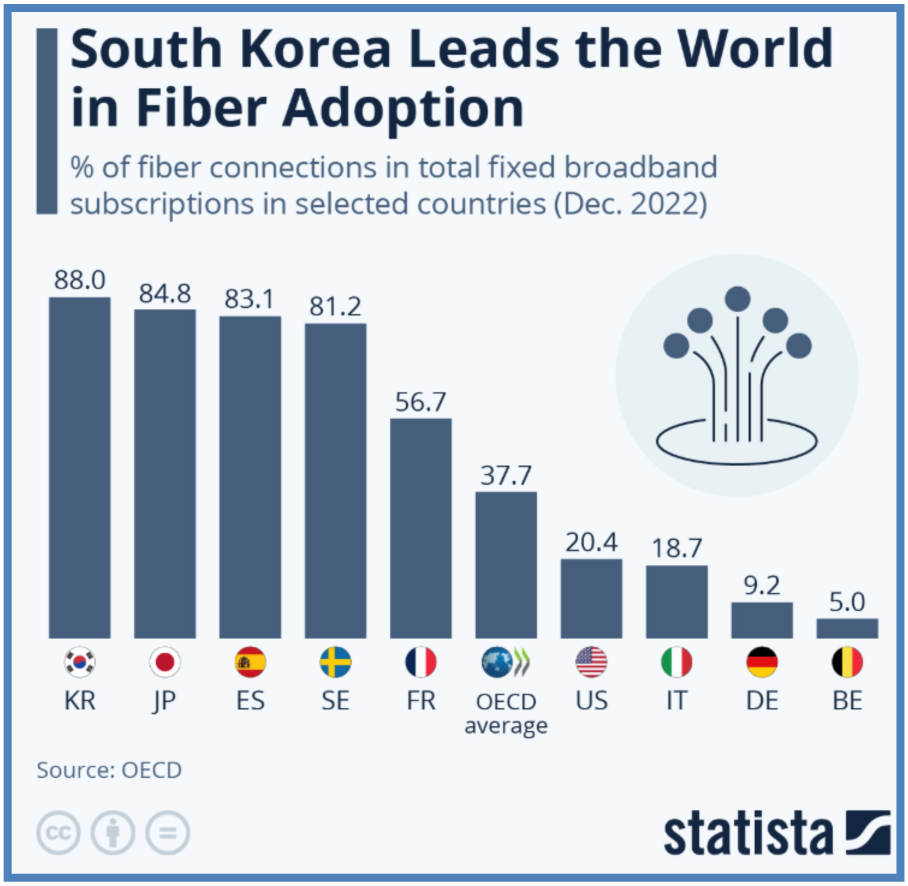


Source: Statista

Figure ES-15. Observing the growth in the number of Starlink satellites in the past

FIBER OPTICS

Fiber optic technology continues to be a cornerstone of high-speed internet infrastructure, offering unparalleled data transmission capabilities. While specific global deployment figures since 2019 are varied, numerous countries have invested heavily in expanding fiber networks to meet the growing demand for faster and more reliable internet services.



Source: Statista

Figure ES-16. Advancements in number of subscribers of fiber optic infrastructure

Conclusion

Since 2019, the communications sector has witnessed remarkable advancements through the proliferation of 5G networks, the rise of LEO satellite constellations, and the ongoing development of fiber optic infrastructure. These technologies have collectively enhanced global connectivity, offering higher speeds, reduced latency, and expanded access, particularly in previously underserved regions. As these trends continue, they are poised to further revolutionize the way individuals and industries interact and communicate worldwide.

INDUSTRIAL AND EDGE COMPUTING

Factory automation, predictive maintenance, and real-time vision inspection all require smart edge devices. Edge AI has become a frontier—low-power inference at the edge using specialized chips.

Industrial and edge computing involve processing data closer to its source in industrial environments, reducing latency and enhancing real-time decision-making. The key components can be categorized into hardware, software, connectivity, security, and data management.

Hardware Components

- Edge Nodes/Edge Servers: Compute devices at the network edge that process data locally before sending it to centralize systems.
- Industrial Gateways: Connect IoT devices and industrial machines to edge servers and the cloud.
- Sensors and Actuators: Collect environmental, machine, and operational data.
- Embedded Systems: Controllers and processors designed for real-time applications.
- AI Accelerators: Specialized hardware (e.g., GPUs, TPUs) for machine learning inference at the edge.

Software Components

- Edge AI/ML Models: Run predictive analytics and anomaly detection close to the data source.
- Containerized Applications: Lightweight applications deployed at the edge (e.g., Kubernetes, Docker).
- Real-time Operating Systems (RTOS): OS optimized for low-latency processing.
- Middleware and APIs: Facilitate communication between edge devices and cloud services.

Connectivity and Networking

- 5G and Private LTE: High-speed, low-latency communication.
- Industrial Ethernet and Fieldbus: Standardized industrial networking protocols (e.g., Modbus, PROFINET).
- LPWAN (LoRa, NB-IoT): Low-power, long-range communication for IoT devices.

Security and Management

- Zero Trust Security Frameworks: Identity-based security for edge devices.
- Secure Boot and Hardware Encryption: Protect firmware and data integrity.
- Device and Network Management Platforms: Ensure remote monitoring, updates, and compliance.

Data Processing & Storage

- Edge Databases: Lightweight databases optimized for edge storage.
- Time-Series Data Management: Capturing high-frequency sensor data.
- Streaming Analytics: Processing real-time event streams (e.g., Apache Kafka, AWS Greengrass).

Components of Industrial and Edge Computing

Category	Indicator
Hardware	Number of Edge Nodes / Servers
Software	AI/ML Inference at Edge
Connectivity	5G / LPWAN Adoption Rate
Security	Zero Trust Implementations
Data Management	Latency in Edge Processing < 5 ms response time

Figure ES-17. Summary of categories of industrial and edge computing

However, a quiet revolution is taking place in the industrial world. Robotics is emerging as a transformative force across multiple sectors, revolutionizing industries, enhancing productivity, and reshaping societal structures. As automation and artificial intelligence continue to advance, the economic impact of robotics has grown substantially, particularly in industrial production, healthcare, logistics, and service sectors. It is forecasted that the role of robotics in society and industry, with a specific focus on their contribution to the GDP of the United States, China, and Japan.

Here's a snapshot of where robotics is gaining traction:

- Industrial automation: Robots are replacing manual tasks on factory floors with speed and precision.
- Healthcare: Surgical and assistive robots are improving outcomes and reducing human error.
- Autonomous vehicles: Robotic systems are at the heart of navigation, perception, and control in self-driving tech.
- Retail and logistics: Warehouse robotics and customer service bots are streamlining operations.
- Exploration: Robots are operating in extreme environments—deep seas, disaster zones, and other planets

Total market size is expected to grow from $73 Billion in 2025 to $185 Billion by 2030. Here are the industries that are most aggressively adopting robotics.

Sector	Share of New Robot Installations
Electronics	28%
Automotive	25%
Metal & Machinery	12%
Plastics & Chemicals	4%
Food & Beverage	3%

Source: https://joingenius.com/statistics/robotics-industry-statistics

Figure ES-18. Current deployment rates of robotics in leading industries

ROBOTICS AND SOCIETY

Robots have significantly influenced society by improving efficiency, reducing human labor in hazardous environments, and enhancing service delivery. In healthcare, robotic-assisted surgeries have improved precision and patient outcomes. Autonomous vehicles and robotic warehouse solutions are optimizing supply chains and transportation. Additionally, service robots, such as AI-driven customer assistants, are streamlining consumer interactions.

Moreover, robotics is reshaping employment patterns. While automation replaces certain low-skill jobs, it simultaneously creates new opportunities in robotics engineering, AI development, and maintenance. Educational institutions are increasingly emphasizing STEM (Science, Technology, Engineering, and Mathematics) to prepare the workforce for this evolving landscape.

INDUSTRIAL ROBOTICS AND ECONOMIC CONTRIBUTIONS

The manufacturing sector has seen the most profound impact of robotics, leading to increased productivity, cost reductions, and enhanced output quality. The industrial component of GDP in major economies like the U.S., China, and Japan has been heavily influenced by robotic automation.

United States

The U.S. has witnessed a steady rise in robotics adoption, particularly in automotive and electronics manufacturing. According to the International Federation of Robotics (IFR), the U.S. had approximately 310,000 industrial robots in operation as of 2023, contributing to an annual manufacturing GDP of $2.3 trillion (about 11% of total GDP). Projections indicate that the U.S. industrial robotics market will grow at a compound annual growth rate (CAGR) of 10%, reaching $50 billion by 2030.

China

China leads the world in robotics adoption, with over 1.5 million industrial robots in use as of 2023. Robotics plays a crucial role in China's industrial GDP, which accounts for approximately 28% of the total GDP, amounting to $4.5 trillion. The Chinese government's "Made in China 2025" initiative aims to increase domestic robotics production and reduce reliance on foreign technology. By 2030, China's robotics sector is projected to surpass $100 billion in market value, with annual installations exceeding 400,000 units.

Japan

Japan is a pioneer in robotics, particularly in precision manufacturing, automotive production, and service robotics. The country has over 400,000 industrial robots deployed, and robotics contributes significantly to Japan's manufacturing GDP, which stands at around $1 trillion, accounting for nearly 20% of its total GDP. Japan's robotics industry is expected to grow at a CAGR of 8%, reaching $70 billion by 2030. With an aging population, Japan is also investing heavily in robotic healthcare solutions to support elderly care.

Outlook

The global robotics market is expected to exceed $300 billion by 2030, with continued expansion in AI integration, collaborative robotics (cobots), and autonomous systems. Governments and private sectors are investing in research and development to enhance robotic capabilities and efficiency. While concerns regarding job displacement persist, the economic and societal benefits of robotics in industrial growth, healthcare, and service industries remain undeniable.

Conclusion

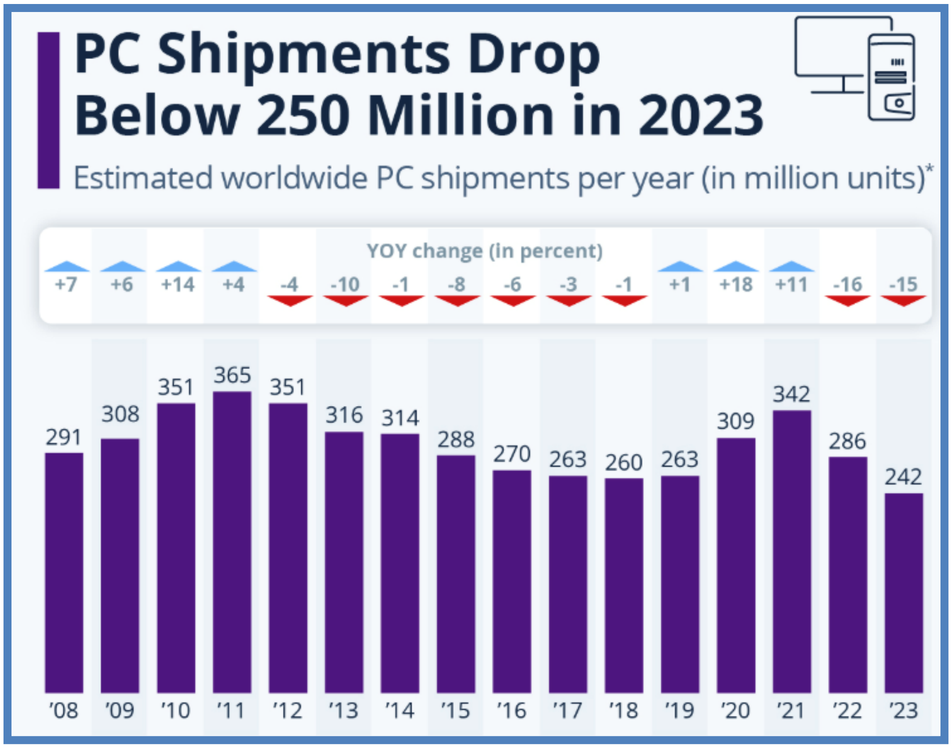
Robotics has become an indispensable part of modern society and industry, driving economic growth and improving quality of life. The industrial contribution to GDP in major economies like the U.S., China, and Japan underscores the vital role robotics play in global competitiveness. As technology continues to advance, nations that embrace and innovate within the robotics sector will position themselves at the forefront of economic and technological progress.

CONSUMER ELECTRONICS AND ENTERTAINMENT

Gaming boomed during the pandemic, further contributing to GPU shortages. Augmented reality/virtual reality (AR/VR) remains niche, but with Apple's Vision Pro and Meta's push, spatial computing may become a serious use case by 2026.

2022–2024: Rebalancing and Strategic Shifts

Post-pandemic, the industry faced: Inventory corrections after overordering in 2021. Softening demand for PCs and smartphones. Rising geopolitical tensions that led to export controls (notably on advanced AI chips to China). In response, governments-initiated semiconductor industrial policy: The U.S. CHIPS and Science Act, Europe's Chips Act, and similar moves in Japan and South Korea are reshaping global supply chains. Major fabs are being built in the U.S., Germany, and India with the intent of reducing reliance on Taiwan and China but frankly in the end companies like TSMC and Samsung are the main ones reaping the benefits of the CHIPS acts. Overall sales of PCs skyrocketed in 2020-21 after years of decline. PCs and internet kept societies connected and operational during the pandemic lockdown.



Source: https://www.statista.com/chart/12578/global-pc-shipments/

Figure ES-19. PC shipments resumed declining after the pandemic induced surge of 2020-22

PC shipments have fallen back to pre-pandemic levels but remain quite stable. PCs remain an extremely flexible device for consumer and business applications as well as for communications. The five leaders controlled 87% of the market in 2024 with shipment results very similar to the ones of 2023. No significant changes are anticipated in the future.

Company	2024 Shipments	2024 Market Share (%)	2023 Shipments	2023 Market Share (%)	2024-2023 Growth (%)
Lenovo	62,498	25.5	59,791	24.7	4.5
HP Inc.	53,028	21.6	52,903	21.8	0.2
Dell	39,448	16.1	40,238	16.6	-2.0
Apple	22,458	9.2	21,532	8.9	4.3
Asus	17,390	7.1	17,161	7.1	1.3
Acer	16,927	6.9	15,889	6.6	6.5
Others	33,612	13.7	34,783	14.4	-3.4
Total	**245,361**	**100.0**	**242,296**	**100.0**	**1.3**

Source: Gartner

Figure ES-20. Shipments of PCs in 2023-24 by leading suppliers

Source: Statista

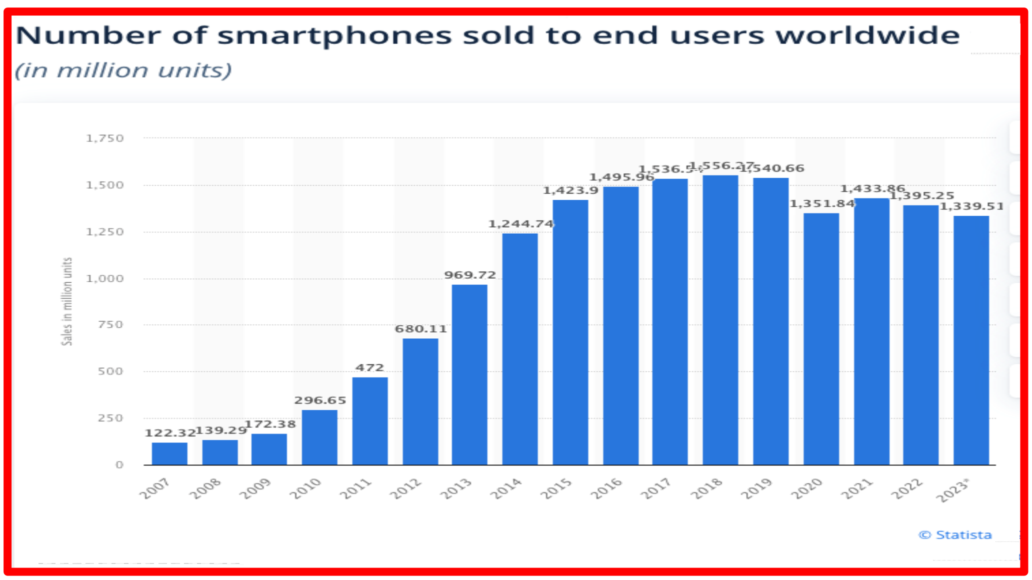
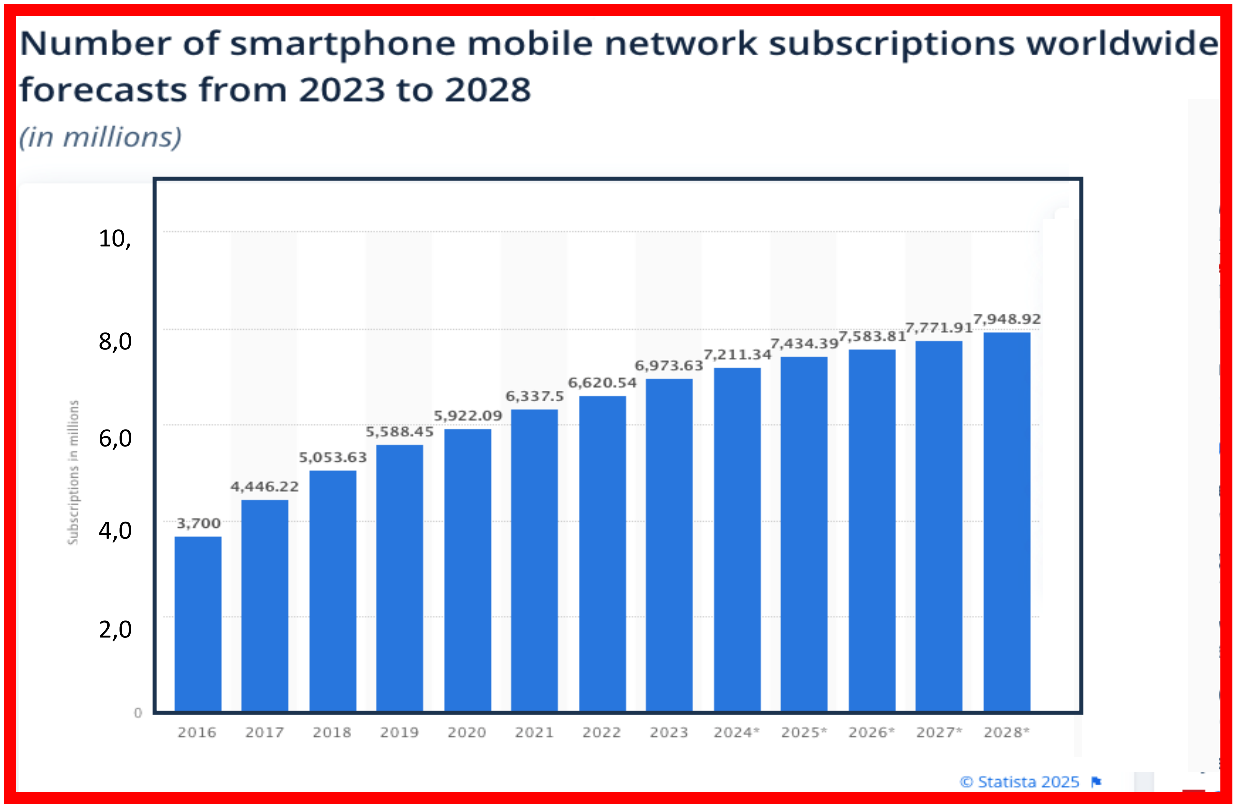


Figure ES-21. Smartphones shipments declining since 2018

Smartphones shipments continued to decline from the peak reached in 2018 but at a very small rate and it seems that shipments will continue to remain above the 1.3B units/year for the foreseeable future. On the other hand, the number of mobile network subscriptions continues to grow unabated.



Source: Statista

Figure ES-22. Mobile subscriptions continue to moderately grow

4. FOCUS—THE GROWTH OF SEMICONDUCTORS: FROM THE PC ERA TO THE AI ERA

4.1. INTRODUCTION

Let's now address the semiconductor contributions in some details. The semiconductor industry is the beating heart of the digital age. It powers every electronic device, every internet connection, and increasingly, every intelligent decision made by a machine. Across the past four decades, semiconductors have evolved not just technologically, but in their role within society. From enabling the rise of personal computers to shrinking entire supercomputers into smartphones—and now giving silicon brains to AI models—the industry has scaled and shifted with each technological epoch.

THE PC ERA (1980S–2005): LAYING OUT THE DIGITAL FOUNDATION

The personal computer revolution marked the first major consumer-driven inflection point in the semiconductor industry.

Key Characteristics

- Dominant chips: CPUs (primarily x86 architecture), dynamic RAM (DRAM), hard disk drive (HDD) controllers, video graphics array (VGA) graphics chips.
- Key players: Intel, AMD, Microsoft (software), IBM, and early Taiwanese and Korean memory makers.
- Moore's Law was in full force: performance doubled every ~18 months, and the PC industry thrived on it.

Semiconductor Growth

Volume grew as millions of households and businesses adopted PCs. Average selling prices (ASPs) were high, especially for CPUs and DRAM. Most semiconductors were general-purpose, optimized for computing, display, and I/O. The PC Era saw Intel's dominance with the Pentium line. However, the 300 mm wafer size conversion saw the demise of the "One Billion Club." During this wafer size conversion, the cost of a new factory rapidly escalated from the hundreds of million to over 1 billion and this put many companies out of the semiconductor manufacturing business but not from the circuit design business. These companies connected in and intermingled in a symbiotic way with foundries (especially Taiwan, Korea, and Japan) and this business arrangement created a winning association that gave rise to the growth of a new supply chain. Semiconductors were essential, but their influence was limited mostly to desktops and enterprise computing.

Furthermore, The MPU reached the power limits in 2004 and from then on multicore architecture with limited operational frequency (~5GHz) became the new norm; MPU performance continued to improve but well below historical rates.

The Smartphone Era (2007–2020): Ubiquity and Mobility

With the iPhone's debut in 2007, a new chapter began. Semiconductors went mobile—and global. Devices became pocket-sized, power-sensitive, and sensor-rich. The rise of the System-on-Chip (SoC): CPUs, GPUs, in-system programming (ISPs), modems, and memory all integrated into one package. Explosion of wireless connectivity chips: Wi-Fi, Bluetooth, GPS, LTE, then 5G.

Summary of Industry Impacts

- Mobile overtook PC shipments by 2011.
- Semiconductors shifted toward high integration and low power.
- New leaders emerged: Qualcomm, Apple (design), MediaTek, ARM.
- Foundries like TSMC and Samsung became central players providing technologies with ever-smaller nodes (7 nm, 5 nm, 3 nm).

Growth Pattern

Unit volume exploded: billions of smartphones sold yearly. Margins compressed compared to the PC era, but total market value expanded. Semiconductors moved from the background to differentiators—camera performance, modem speed, and AI processing became selling points.

Source: Statista

Figure ES-23. Semiconductor unit shipments

THE AI ERA (2020–PRESENT): INTELLIGENCE IN SILICON

Shipment of PC had been declining since 2012 and temporarily went up again to provide a viable communication link during the pandemic, but this was a short-lived resurgence. It was however clear that performance improvements of MPUs were limited since only any parallel operation benefitted from multicore architecture, but inevitably serial executions did not benefit from it. So, the real question to be asked was:" Could a fully parallel architecture be devised?". This question went unanswered, for a long time and finally it took a group of "unbiased minds" to pose this question quite naturally, then seriously search for an answer and finally finding the answer! This is precisely what a group of students at Stanford did in 2009 and they discovered that not only was there indeed a branch of computer science that required a fully parallel architecture called Artificial Intelligence (AI) but discovered also that an IC capable of carrying on this task already existed: the GPU!

However, this discovery would have been remained only as an intellectual exercise had it not been for the subsequent events that took place. NVIDIA the leading maker of GPU for graphic applications like computer games etc. fully understood the relevance of this discovery and hired Bill Dally, director of Stanford computer science department to become the new CTO.

Architectural and prototyping work feverishly continued for the next 10 years until finally a viable solution useable towards making a real AI workhorse was demonstrated. *The IRDS took noticed of this and in 2018 heralded the discovery to the world.*

By then the pandemic crisis exploded and people attention was diverted to more serious existential problems.

FINALLY, AI COMES OF AGE!

The current era is defined not just by computation, but by learning and inference. *Artificial Intelligence (AI)—once a research novelty—is now the core driver of semiconductor innovation.*

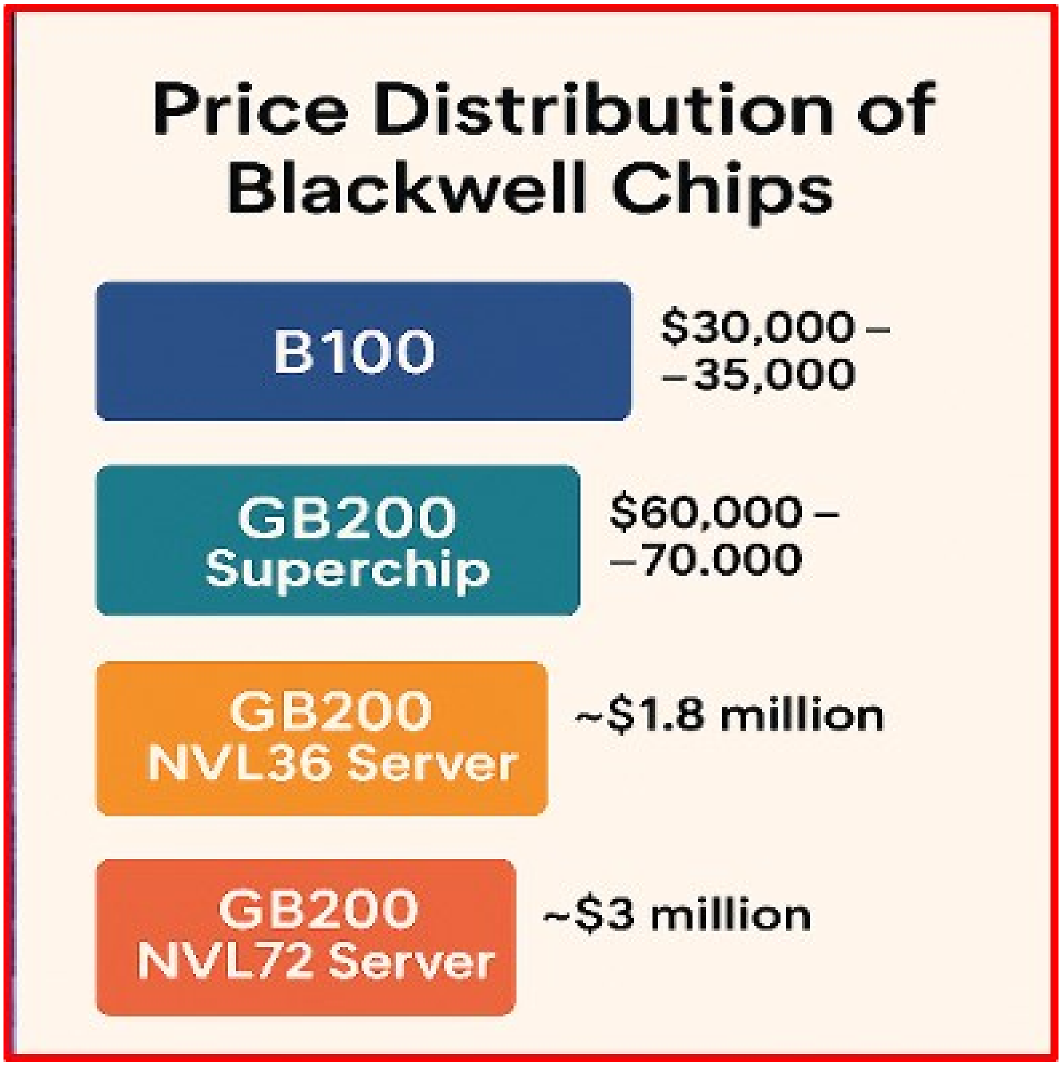
Milestones

- NVIDIA launched the A100 GPU in 2020, primarily for AI training.
- In 2022, the H100 with 80 billion transistor and an 814mm^2 die size debuted on the market scene and gained mainstream recognition with the rise of ChatGPT and other generative AI tools.
- ASICs like Google's TPU and Tesla's full self-driving (FSD) chips represent vertical integration of silicon and software.

But this is not all. The next GPU to be produced named Blackwell opened new avenues to forgotten technologies. Having Blackwell design reached and exceeded the limits of what was possible to realizable with lithography tools it became necessary to "fuse" together two dice to achieve an astonishing 208Billion transistor product. The basic technology to overcome the limits of viable die sizes known as multi-chip module (MCM) had been pioneered in the 60s for supercomputers but then dementated and made easily manufacturable in the 90s. Nevertheless, the technology was too complex and costly at the time and had to wait for the perfect product to be re-introduced again and so Blackwell expeditiously adopted this packaging technology and was introduced in 2024 with even more outstanding results.

Explosion of Semiconductor Demand:

Training state-of-the-art models requires thousands of GPUs, generating multi-billion-dollar hardware clusters. Inference at the edge is also growing via phones, cars, cameras—all now run AI models locally. Governments and hyperscalers are racing to secure chip capacity, and sovereign AI chip initiatives are emerging. However, the technical achievement comes with a bonus. Since the demand for GPUs is outstripping supply the basic laws of economics demand that unusually high prices be paid for these AI GPUs!



Source: NVIDIA

Figure ES-24. Blackwell sale values confirm that sale price is set by market demand and not by cost only in a virtual monopoly

However, many additional chips in addition to the GPUs are required to assemble servers, racks, power distributions and communications and as a result the overall semiconductor industry is benefitting on large scale from the AI race. In summary, the AI race has boosted the sales of semiconductors to new highs.

2024 Rank	2023 Rank	Vendor	2024 Revenue	2024 Market Share (%)	2023 Revenue	2024-2023 Growth (%)
1	2	Samsung Electronics	66,524	10.6	40,942	62.5
2	1	Intel	49,189	7.9	49,117	0.1
3	5	NVIDIA	45,988	7.3	25,053	83.6
4	6	SK hynix	42,824	6.8	23,027	86.0
5	3	Qualcomm	32,358	5.2	29,225	10.7
6	12	Micron Technology	27,843	4.4	16,123	72.7
7	4	Broadcom	27,641	4.4	25,613	7.9
8	7	AMD	23,948	3.8	22,307	7.4
9	8	Apple	18,880	3.0	18,052	4.6
10	9	Infineon Technologies	16,001	2.6	17,022	-6.0
		Others (outside top 10)	274,775	43.9	263,483	4.3
		Total Market	**625,971**	**100.0**	**529,964**	**18.1**

Source: Gartner

Figure ES-25. Ranking of Top Semiconductor Sellers. A fabless company (NVIDIA) climbed to the third position for the first time

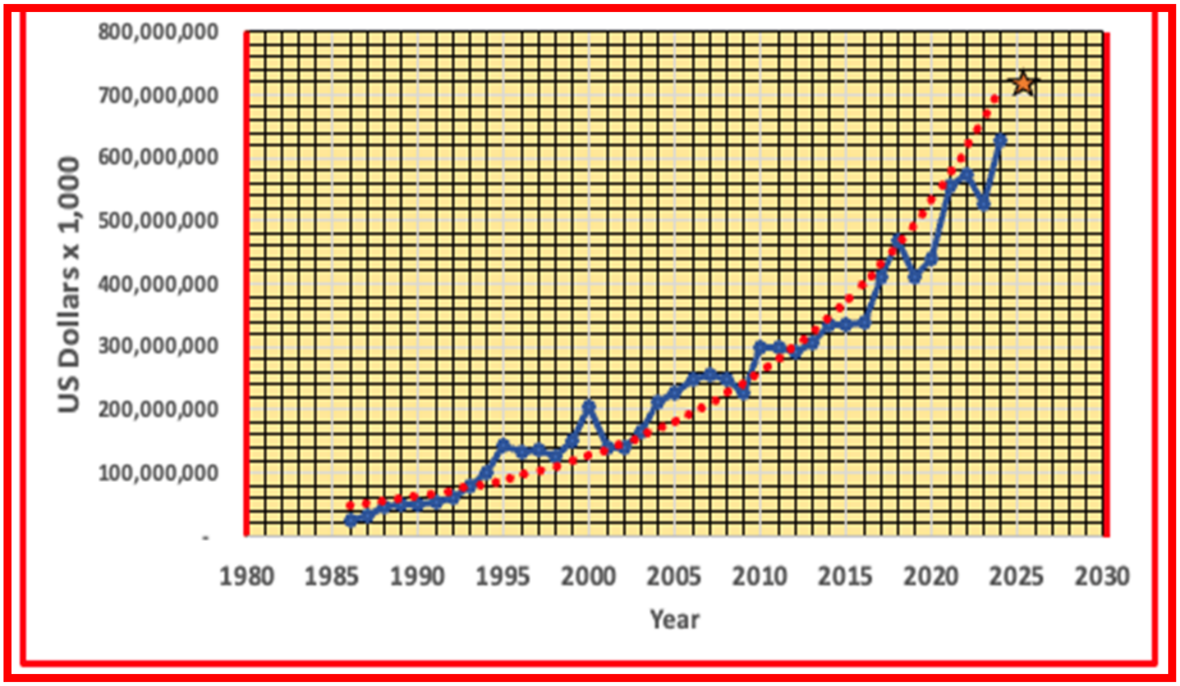
It should be noted that TSMC revenue does not appear in the list in Figure ES-25 since the semiconductor sales are reported by its customers in the table above. However, these are TSMC revenue results for the past three years, as shown in Table ES-1.

Table ES-1. Taiwan Semiconductor Manufacturing Annual Revenue – 2022-2024

Year	Annual Revenue	% Increase from Previous Year
2022	$73.67 B	28.74% increase from 2021
2023	$70.599 B	4.17 % decrease from 2022
2024	$88.268 B	25.03% increase from 2023

The chart in Figure ES-25 shows that 2024 was the year of memory cycle as the three participants in this market (Samsung, SK Hynix, and Micron) moved up in the ranking as compared to 2023. It also shows for the first time the Fabless/Foundry combination of NVIDIA designed products manufactured by TSMC silicon technology climbed to the third position (the highest ranking position for a fabless company) behind well-known semiconductor powerhouses (mostly known for their integrated device manufacturer (IDM) products) as a demonstration of the success of GPUs as the power engines of the AI revolution.

Expect another NVIDIA surprise in 2025.



Source: WSTS

Figure ES-26. Semiconductor Sales 1996–2024. 1 Trillion-dollar sales appear in sight for 2030.

Semiconductor market grew 18.1% in 2024, and it is expected to reach the $700B mark in 2025 but this is not all

- Semiconductors are projected to become a *$1 trillion industry by 2030*, with AI accounting for a significant share.
- AI chips command premium ASPs and require advanced packaging (chiplets, high bandwidth memory (HBM), 2.5D/3D stacking).

The chart in Figure ES-27 identifies and clarifies the roles of the 3 pillars of the semiconductor industry.

Era	Focus	Key Semiconductors	Driving Applications	Impact
PC	Compute (static)	CPU, DRAM, GPU	Desktops, office tools	Personal productivity
Smartphone	Compute (mobile)	SoC, modem, sensor, PMIC	Social media, camera, cloud access	Digital lifestyle
AI	Compute (intelligent)	GPU, NPU, TPU, HBM	Training, inference, real-time data	Machine intelligence, automation

Figure ES-27. The roles of the three sequential pillars of the semiconductor industry

Table ES-2. World-wide semiconductor sales by region and product classifications

Fall 2024	Amounts in US$M			Year on Year Growth in %		
	2023	2024	2025	2023	2024	2025
Americas	134,377	**186,635**	**215,309**	-4.8	38.9	15.4
Europe	55,763	**52,031**	**53,736**	3.5	-6.7	3.3
Japan	46,751	**47,410**	**51,866**	-2.9	1.4	9.4
Asia Pacific	289,994	**340,792**	**376,273**	-12.4	17.5	10.4
Total World - $M	526,885	**626,869**	**697,184**	-8.2	19.0	11.2
Discrete Semiconductors	35,530	**31,546**	**33,377**	4.5	-11.2	5.8
Optoelectronics	43,184	**42,092**	**43,705**	-1.6	-2.5	3.8
Sensors	19,730	**18,732**	**20,034**	-9.4	-5.1	7.0
Integrated Circuits	428,442	**534,499**	**600,069**	-9.7	24.8	12.3
Analog	81,225	79,433	83,157	-8.7	-2.2	4.7
Micro	76,340	79,291	83,723	-3.5	3.9	5.6
Logic	178,589	208,723	243,782	1.1	16.9	16.8
Memory	92,288	167,053	189,407	-28.9	81.0	13.4
Total Products - $M	526,885	**626,869**	**697,184**	-8.2	19.0	11.2

Source: WSTS (World Semiconductor Trade Statistics)

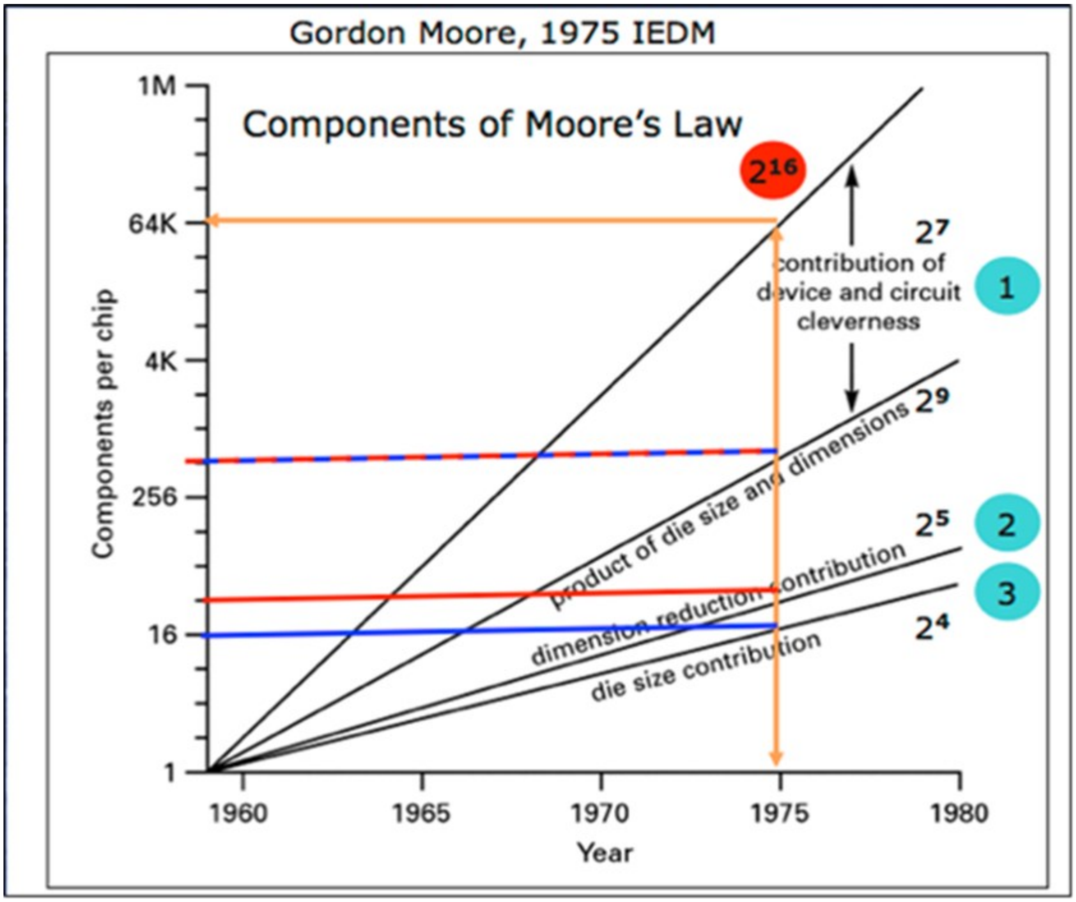
This table summarizes where semiconductors are sold around the world and shows the classifications of semiconductor products by seven categories. See detailed analysis for each region later in Figures ES-33.).

5. LOOKING AHEAD: THE ELECTRONICS INDUSTRY OF THE FUTURE

5.1. KEY TRENDS TO WATCH

The next 5–10 years are set to be transformative, as follows:

- AI everywhere: From phones to datacenters to cars. Expect more custom silicon (Apple Neural Engine, Tesla FSD chip, Google TPU).
- Advanced packaging and chiplets: Moore's Law continues through integration, not scaling. This shouldn't be something new for those that read Gordon Moore 1975 presentation to IEDM (see top right on next figure). (More to say on this subject later.)
- Photonics and quantum electronics: Still early, but potentially revolutionary.
- Sustainability and energy efficiency: Power-per-compute becomes a dominant metric.



Source: IEDM 1975, Gordon Moore presentation

Figure ES-28. Relative contributions of the three components of Moore's Law

The Figure ES-28 points out that (1) "device and circuit cleverness" was the main contribution to demonstrate the validity of the prediction made by Gordon Moore in 1965. Most of the observers of the semiconductor industry concentrated their attention on (2) dimensional reduction of features and only mildly on (3) the die size contribution. However, it is now device and circuit cleverness that still enables Moore's Law via the new 3D IC Fabric and packaging related innovation.

MARKET OUTLOOK

Electronics content per system (especially in automotive and industrial) continues rising. Semiconductor industry is expected to surpass $1 trillion by 2030, driven largely by AI, IoT, and edge compute. Figure ES-29 highlights the forces behind the relentless growth of the semiconductor market forecasted until 2030.

Year	Market Size (USD Trillion)	Key Technological Trend	Geopolitical & Economic Factors	Challenges
2020	0.45	5nm chips mass production begins (TSMC, Samsung)	US-China trade tensions impact supply chains	Global chip shortage due to pandemic
2021	0.53	AI, IoT, and EVs drive demand	US CHIPS Act proposed; EU announces semiconductor strategy	Supply chain disruptions persist
2022	0.58	3nm process development starts	US CHIPS Act signed into law; Europe Chips Act launched	Talent shortages in semiconductor industry
2023	0.63	AI accelerators, advanced packaging growth	China invests heavily in domestic chip industry	Rising costs of semiconductor fabs
2024	0.67	2nm process development underway (Intel, TSMC, Samsung)	US, EU subsidies for semiconductor fabs increase	Geopolitical tensions impact supply chain resilience
2025	0.75 (Projected)	Growth in AI-driven chips, edge computing	Expansion of semiconductor production in US, Europe	Need for more skilled workforce
2026	0.82 (Projected)	Quantum computing chips emerge	Increased government funding for R&D	Energy efficiency concerns in chip manufacturing
2027	0.90 (Projected)	Sub-2nm chips reach commercialization	Stronger competition in AI chip sector	High costs of R&D for next-gen semiconductors
2028	0.95 (Projected)	Neuromorphic and optical computing growth	China further develops domestic chip industry	Environmental impact of semiconductor production
2029	1.02 (Projected)	Advanced AI-specific semiconductors	Global semiconductor industry consolidates	Increasing cost of raw materials
2030	1.10 (Projected)	Post-silicon materials (e.g., graphene) in development	US, EU, and China lead in semiconductor production	Ethical concerns around AI and semiconductor use

Figure ES-29. Semiconductor market growth projection

SUMMARY

The electronics industry is navigating one of the most volatile and exciting decades in its history. From a pandemic-induced PC boom to the silent rise of AI hardware, to geopolitical shifts and electrification of vehicles—it's a sector that not only reflects technological change but drives it. The once-overlooked A100 GPU set the stage for an AI revolution, and the H100 turned it into a headline act when GPT4 attracted the attention of the consumers. As we step into the AI-native era, electronics is not just enabling innovation—it *is* the innovation. The GPUs are the drivers of the growth of the semiconductor market bound for the $1Trillion mark by 2030 (or sooner).

OUTLOOK: THE AI-FIRST SILICON WORLD

We are entering a world where every system will be AI-enhanced, and semiconductors will define the following: "*How fast we can train a language model?" "How quickly a car can react to a pedestrian?," and "How energy-efficiently can a satellite or drone navigate?"*

Future growth areas include:

- Edge AI (smart sensors, AR glasses, robotics)
- Autonomous systems (vehicles, industrial automation)
- Cloud-to-edge AI pipelines
- Quantum-inspired and neuromorphic computing

3D silicon fabric, advanced packaging, custom silicon, and sovereign chip design are set to further fragment and expand the industry landscape, creating both opportunity and geopolitical tension.

CONCLUSION

The journey from PCs to smartphones to AI reflects not just technological progress, but the increasing centrality of semiconductors in modern life. Once invisible components inside beige boxes, chips are now the engines of cognition, intelligence, and autonomy. As we look forward, semiconductors aren't just keeping up with human ambition—they're accelerating it.

6. SEMICONDUCTORS—AN ECONOMIC ENGINE

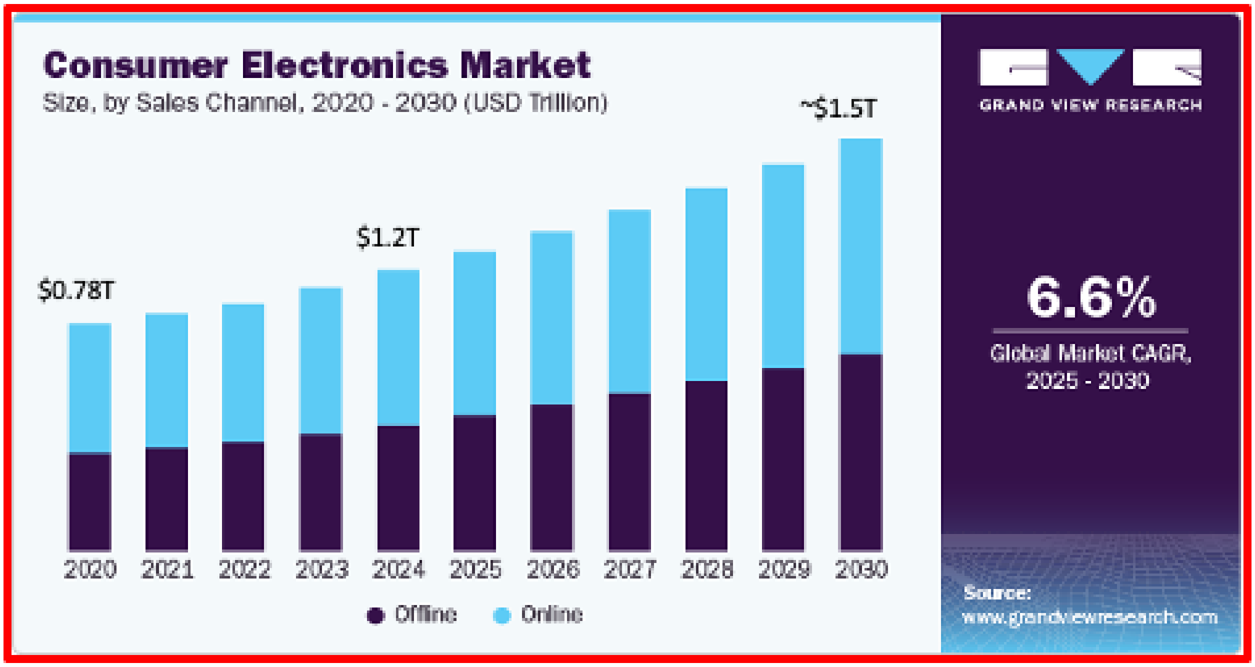
Semiconductor industry health is a bellwether for broader economic resilience. AI advancements (especially generative AI), edge computing, and automotive electrification are fueling a fresh wave of demand. Governments continue to court chipmakers, seeing them as strategic assets for national security and economic competitiveness. In the US, new fabs are under construction or operational particularly in Arizona and Texas. R&D partnerships between industry and academia are expanding. Workforce development is underway to meet the talent demand in advanced manufacturing although more realistic understanding of the needed skills' requirements is finally driving the work force demand towards a more realistic and reduced number of new jobs required. Nevertheless, new factories are also operational and under construction in Japan and China while Europe is closely following this trend.

However, a key element is emerging. While all regions wished and expected domestic suppliers of semiconductors taking the lead in this resurgence this has not occurred and what is happening is quite different. Manufacturing of semiconductors in the regions where semiconductors are an essential contributor to the economy and have therefore led in investing in the chips acts, is indeed substantially increasing but the providers are not domestic suppliers but international leaders that have established new manufacturing plants in US, Japan, China and Europe. However, most of the semiconductor manufacturing of these international leaders remains in their own countries of origin and so we may ask: *"Is the outcome of chips acts around the world nothing more than a <u>limited insurance policy</u> against possible future shortages as opposed to the expected resurgence of manufacturing capacity of domestic suppliers of semiconductors?"* The next few years will provide the final answer to this question.

6.1. OUTLOOK: GDP GROWTH AND THE ROLE OF SEMICONDUCTORS

Looking ahead, GDP growth will be increasingly influenced by digital infrastructure, green transition policies, and AI-driven productivity gains. Key projections and themes:

- Global GDP is expected to grow at a moderate pace (2.5–3.5%) through 2026, barring new shocks.
- Emerging markets like India and Southeast Asia will be growth engines.
- Semiconductors will underpin economic competitiveness, much like oil did in the 20th century.
- Technological sovereignty will remain a priority, prompting more investments in domestic fabs, resilient supply chains, and R&D ecosystems but all of this will remain subjected to the conditions outlined above.



Source: www.grandviewresearch.com

Figure ES-30. Consumer electronics past and future growth

Once upon a time semiconductor electronics in automobiles were limited to the radio. Over the past decade, the semiconductor content in consumer electronics has exhibited notable trends.

Year	Semiconductor Content in Consumer Electronics (%)	Notes
2015	25%	
2021	33.2%	Semiconductor content in electronic systems worldwide.
2023	~35%	Stable semiconductor content in mobile and computing subsegments.

Figure ES-31. Semiconductor content in consumer electronics

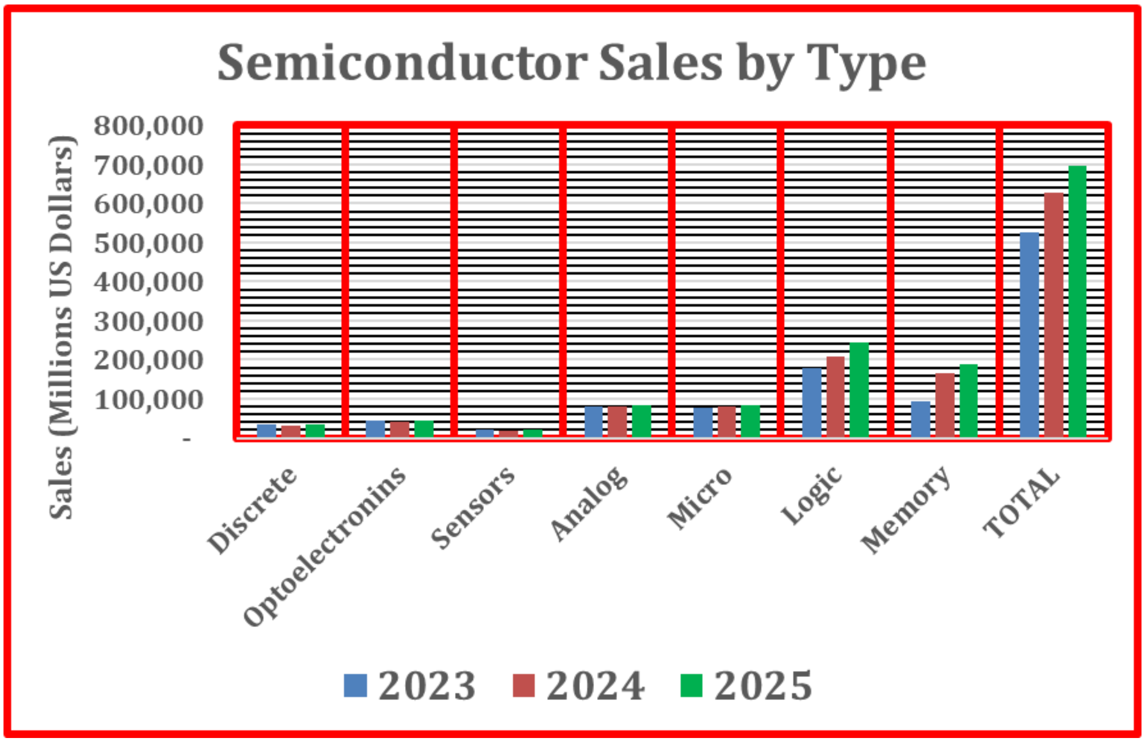
WHERE DO THE SEMICONDUCTORS GO?

- AI and Data Centers: Reports from Deloitte and Gartner emphasize that generative AI and data center upgrades are key drivers behind the dramatic growth of certain segments—especially memory (through HBM innovations) and logic (via advanced microprocessors and GPUs). This trend is expected to have a cascading effect on related categories over the next few years.
- Automotive and Industrial Shifts: Automotive electrification and the move toward software-defined vehicles (SDVs) are reshaping demand, particularly benefiting sensors, analog circuits, and discrete components. This

has been reflected in many industry studies that show strong growth forecasts in these segments as automotive semiconductor content rises.

- Supply Chain and Innovation: While 2023 was marked by supply chain disruptions and inventory corrections, 2024 and 2025 are benefitting from companies refining their supply chains and investing in next-generation fabrication technologies. This dynamic is contributing to rising optimism across nearly every segment, although the magnitude of growth differs according to end-use applications.

Below is a summary of the current and forecast trends for semiconductor sales in several key product segments—Memory, Logic, Micro, Analog, Sensors, Opto, and Discrete—for the years 2023, 2024, and 2025. These insights draw on industry outlooks from sources such as KPMG, Gartner, Deloitte, and Allied Market Research, and they help explain how evolving demand drivers (from AI and data centers to automotive electrification and 5G) are reshaping each segment's performance.



Source: WSTS

Figure ES-32. Semiconductor sales by type

OUTLOOK BY PRODUCT CATEGORY AND DRIVING TRENDS

This segmented outlook below not only provides a sense of how each product category is trending but also contextualizes these trends within broader market drivers.

Memory

The memory category experienced volatility in 2023 as manufacturers adjusted inventories. Even though cyclical challenges persisted, products like DRAM and emerging high-bandwidth memory began to show signs of recovery—driven by data center and AI application requirements. Memory sales rebounded strongly in 2024. According to industry reports, demand was notably spurred by AI workloads and high-performance computing, with HBM leading a fast-growing niche and overall DRAM regaining momentum. The rebound continues into 2025 with robust growth as data center expansion and consumer upgrades further buoy demand. The segment is expected to consolidate gains with higher bit growth and efficient scaling in advanced nodes.

Logic

Overall logic chip sales in 2023 were moderate as supply chain constraints and inventory corrections weighed on performance. However, early signals of increased AI and IoT demand began emerging. In 2024, logic revenues accelerated as increased investments in AI processors, server CPUs, and integrated SoCs provided momentum. Market realignment—prompted by generative AI and cloud computing—spurred product innovation in this space. With generative AI, cloud expansion, and the introduction of next-generation process nodes, logic chip sales are forecast to advance further in 2025, establishing a robust growth path amid renewed global demand.

Microcontrollers

Microcontrollers (and similar embedded processors) showed mixed performance in 2023. While some areas (like traditional consumer electronics) remained flat, sectors such as automotive and industrial automation signaled cautious recovery. In 2024, there was a modest recovery as renewed investments in automotive electronics and IoT accelerated shipments of microcontrollers and embedded processors. By 2025, broader adoption in smart devices, edge computing, and automotive systems is expected to drive continued, though measured, growth across the micro segment.

Analog

The analog segment remained relatively stable in 2023 despite supply pressures. Its steady demand in power management, consumer electronics, and automotive applications underpinned performance during a year of broader semiconductor headwinds. In 2024, as digital transformation in automotive and industrial sectors accelerated, analog chip revenues recorded a moderate recovery. Enhanced integration in mixed-signal and power management solutions contributed to this stability. Continuing into 2025, the analog market is expected to grow steadily—supported by improved design integration and ongoing demand for battery management, sensor interfaces, and power regulatory components.

Sensors

Sensor sales in 2023 were bolstered by increased adoption in automotive ADAS systems and various IoT applications, even as overall market uncertainties (such as economic headwinds) capped the pace of growth. healthcare, and industrial IoT (with improved MEMS and photonic sensors) provided additional impetus for strong performance. deepening integration in smart vehicles, autonomous driving systems, and consumer health-monitoring devices.

Optoelectronics

Optoelectronic components—covering optical transceivers, lasers, and photonic chips—saw modest growth in 2023 as telecommunications and data center investments were still in early stages. In 2024, upgrades in 5G/6G infrastructure and the digital transformation of data centers began to boost opto revenues. Increased demand for high-speed interconnects and optical communication components marked a turn toward stronger growth. In 2025, the opto segment is expected to experience significant upward momentum. Expansion of fiber-optic networks and photonic integration—for example, in AI-driven data centers—should drive further investments in these components.

Discrete

Discrete components—such as diodes, transistors, power modules—witnessed stable, low single-digit growth in 2023. Their performance was largely supported by steady demand in industrial and automotive markets. For 2024, the trend was toward modest improvement as vehicle electrification and renewable energy applications required more robust power semiconductors. The discrete segment is projected to continue its steady, moderate growth into 2025, bolstered by sustained demand in power management and high-voltage applications across sectors like automotive, industrial, and consumer electronics.

SEMICONDUCTOR SALES BY REGION: ASIA'S LEAD AND NORTH AMERICA'S RISE

The global semiconductor industry is a cornerstone of modern technological advancement, powering everything from smartphones to electric vehicles and data centers. Over the past decade, regional dynamics in semiconductor sales have shifted, but Asia has consistently maintained its lead. However, North America has shown notable growth recently, driven by strategic investments and supply chain recalibration.

Asia's Continued Dominance

Asia has long been the epicenter of semiconductor manufacturing and consumption. In 2023, Asia-Pacific, including China, Taiwan, South Korea, and Japan, accounted for more than half of global semiconductor sales. Several factors underpin this dominance:

- Manufacturing Powerhouses: Taiwan and South Korea host the world's leading foundries, such as TSMC and Samsung, which supply chips for global giants like Apple, NVIDIA, and Qualcomm.
- Consumer Electronics Demand: China remains the largest end-market for semiconductors due to its massive consumer electronics, automotive, and industrial sectors.
- Established Supply Chains: Decades of investment in infrastructure, talent, and logistics have made Asia a cost-effective hub for semiconductor production and assembly.

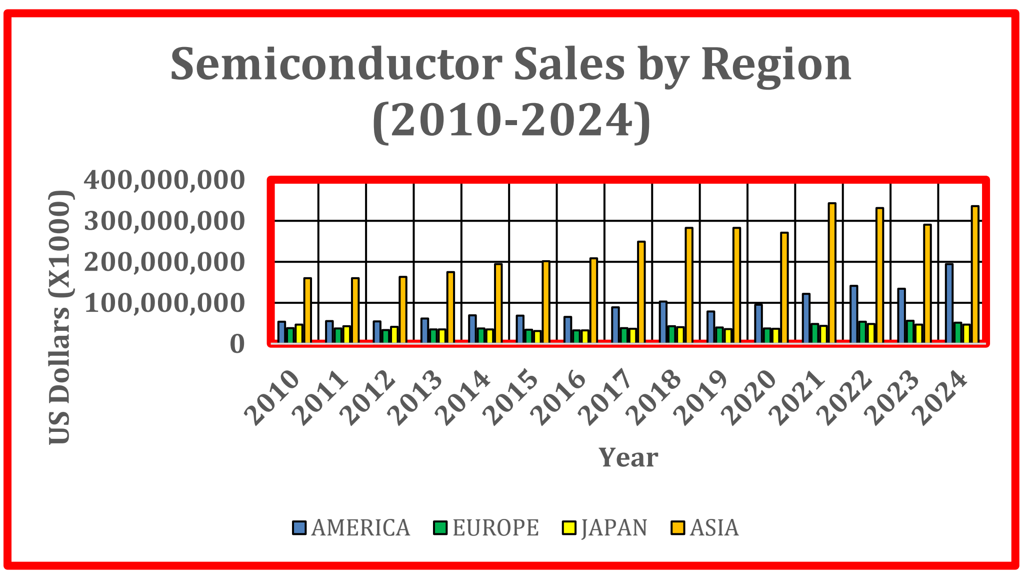
North America's Growth Trajectory

While Asia remains dominant, North America—especially the United States—has been experiencing significant growth in semiconductor sales and capacity. Several drivers explain this trend:

- Government Policy and Incentives: The CHIPS and Science Act (2022) allocated over $50 billion to revitalize semiconductor research, development, and manufacturing in the U.S. This has spurred investments from Intel, TSMC (in Arizona), and other firms.
- Geopolitical Shifts: The trade tensions between the U.S. and China, along with concerns over supply chain disruptions during the COVID-19 pandemic, highlighted the need for domestic production.
- AI and High-Performance Computing (HPC): North America leads in designing cutting-edge chips for AI, data centers, and supercomputing. Companies like NVIDIA, AMD, and Intel are central to this innovation, and demand for these chips has surged.
- Reshoring Trends: Tech companies are increasingly prioritizing resilient supply chains, leading to reshoring efforts that boost local semiconductor production.

CONCLUSION

The semiconductor landscape remains dynamic. Asia's entrenched manufacturing capabilities and market size keep it at the forefront of global sales. However, North America's resurgence—fueled by policy, innovation, and strategic realignment—marks a pivotal shift. As the industry becomes more geopolitically sensitive and technologically complex, the balance of power may gradually even out, with both regions playing critical roles in shaping the semiconductor future.



Source: WSTS

Figure ES-33. Semiconductor sales by region. China continues to lead by America

6.2. CAPITAL INVESTMENTS

According to SEMI, Semiconductor capital expenditures (CapEx) decreased in the first half of 2024 but saw a strong rebound, particularly in the fourth quarter, resulting in 3% annual growth by the end of 2024. Memory-related CapEx continued to lead the growth surging 53% quarter-on-quarter (QoQ) and 56% YoY in Q4 2024.

Installed wafer fab capacity surpassed a record 42 million wafers per quarter worldwide (in 300mm wafer equivalent) in 2024. Foundry and Logic-related capacity continues to show stronger increases, growing 2.3% QoQ in Q4 2024, and the segment is projected to rise 2.1% in Q1 2025 driven by capacity expansion for advanced nodes. Memory capacity increased 1.1% in Q4 2024 and is forecasted to remain at the same level in Q1 2025 driven by strong demand for HBM.

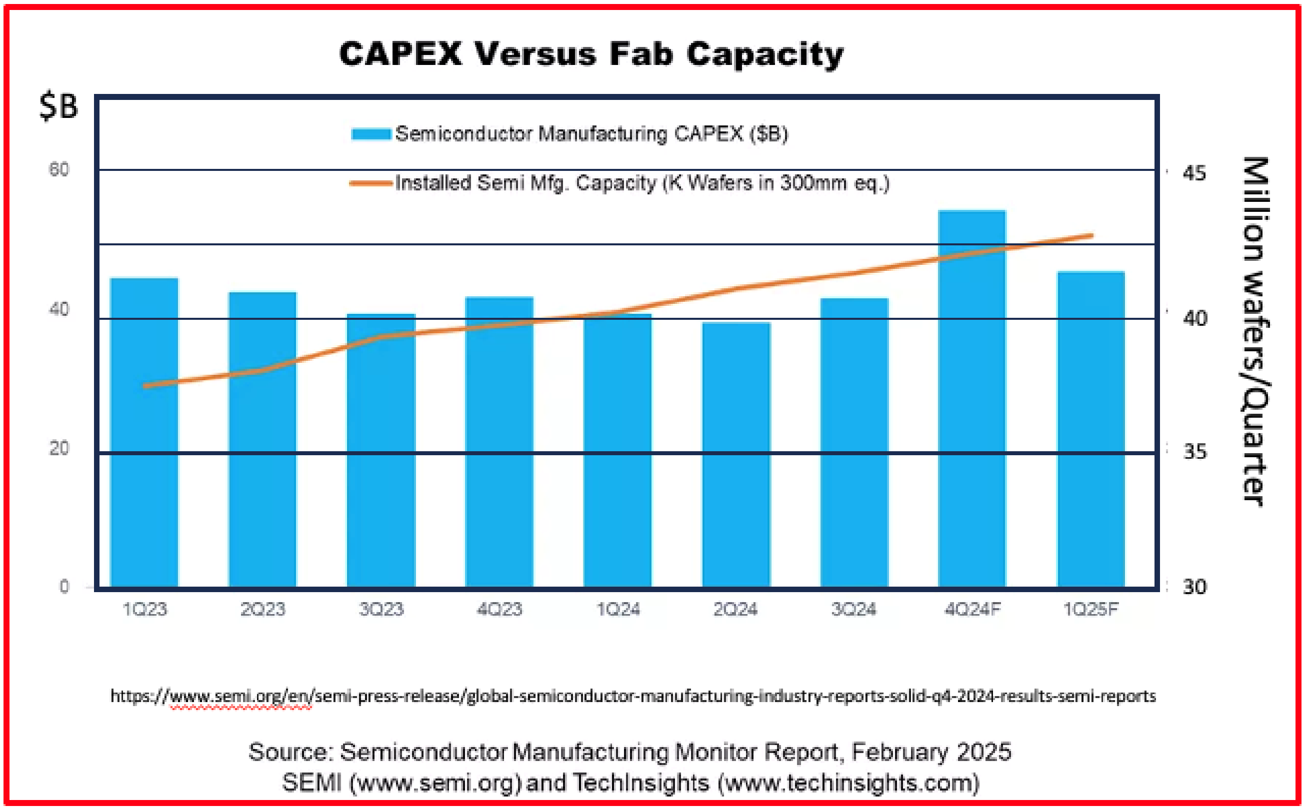


Figure ES-34. CAPEX and 300 mm wafer installed capacity

7. THE EXPANSION AND GROWTH OF DATA CENTERS: SCALING INTO THE EXABYTE ERA

Any report on the health of the electronics industry cannot be completed without talking about the Data Centers. In the digital age, data is the new oil—and data centers are the refineries. As the world global appetite for data-intensive applications continues to grow, data centers have scaled dramatically, both in number and in capacity. Measured in exabytes (1 exabyte = 1 billion gigabytes), today's data infrastructure is reaching unprecedented levels, driven by cloud computing, AI, video streaming, and the IoT. At the forefront of this transformation are technology giants like Amazon, Microsoft, Google, and Meta, each investing heavily to build the digital backbone of the 21st century.

7.1. EXPONENTIAL GROWTH IN DATA STORAGE

Over the past decade, global data center storage has surged from petabyte-scale to exabyte-scale. According to IDC, the total amount of data created and stored worldwide is projected to exceed 180 zettabytes by 2025, up from just 33 zettabytes in 2018. To manage this, hyperscale data centers have expanded in both physical footprint and capacity, with some now housing exabytes of data individually.

This growth is not just about storing more information—it's about enabling high-speed access, AI computation, edge processing, and real-time analytics across industries.

7.2. BIG TECH LEADING THE CHARGE

The world's largest tech companies have been key drivers of this explosive growth, indicated by the following:

- Amazon Web Services (AWS): As the global leader in cloud infrastructure, AWS operates dozens of hyperscale data centers worldwide. Amazon has invested tens of billions of dollars in expanding capacity across North America, Europe, and Asia. Its infrastructure supports not only retail and logistics operations but also countless third-party businesses.
- Microsoft Azure: Microsoft has announced plans to build up to 50–100 new data centers per year, with significant footprints in the U.S., Europe, and Africa. It has committed to scaling responsibly, with a focus on sustainability and carbon-neutral operations by 2030.
- Google Cloud: Google's focus on AI and data-intensive services has led it to invest over $10 billion annually in its global data center network. Its facilities support massive storage and real-time processing for services like YouTube, Search, and Google Photos—each consuming exabytes of data storage monthly.
- Meta (Facebook): With the rise of the metaverse, virtual reality, and social media content, Meta is constructing some of the world's largest data centers. Its infrastructure supports petabytes of video and user-generated content daily, with projected storage needs in the multi-exabyte range in the next few years.

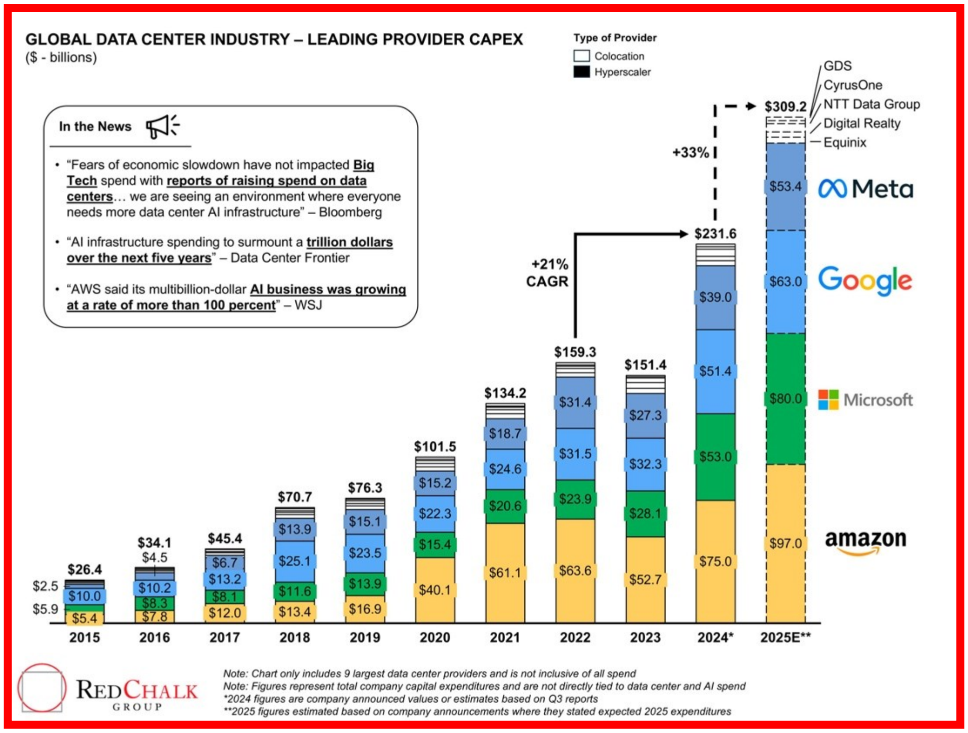
WHY THIS GROWTH MATTERS

The expansion of data centers has wide-ranging implications, as follows:

- AI and Machine Learning: Exabyte-scale data is essential for training large language models (like GPT) and computer vision systems. Data centers are becoming AI factories, with specialized hardware like GPUs and TPUs.
- Global Connectivity: The cloud powers nearly all digital services today—video conferencing, banking, healthcare, logistics, and more. Reliable data center infrastructure ensures uptime and low latency.
- Economic Impact: Data center construction has become a key area of investment, bringing jobs, infrastructure, and economic growth to local regions, especially in states like Virginia, Texas, and Oregon in the U.S.

SUMMARY

The growth of data centers into the exabyte era marks a defining feature of our digital civilization. What once fit in server rooms now spans football-field-sized campuses of blinking racks and humming machines. The leadership of Amazon, Microsoft, Google, and Meta is pushing the boundaries of what's possible, enabling the cloud-first world we now live in. As emerging technologies like AI, AR/VR, and 5G continue to mature, data centers will remain at the heart of global innovation—bigger, faster, and smarter than ever.



Source: RedChalk

Figure ES-35. Data centers investment growth 2015-2025

8. CONCLUSION

Since 2019, global GDP has experienced a dramatic arc: from steady growth to historic contraction, followed by a rapid rebound and an ongoing adjustment to new realities. Among the most consequential developments has been the revaluation of semiconductors as strategic economic assets. The pandemic exposed systemic vulnerabilities but also accelerated transformation. As an example, PC proliferated at unprecedented levels as people locked in their homes resorted to this means of communication with relatives and with the outside world in general; home deliveries became the norm and so on. Consumer electronics powered by semiconductors (more than 35% content) reached the $1.2 Trillion mark in 2024 and will grow at a CAGR of 6.6% until 2030; semiconductors sales will reach the $1Trillion mark by 2030. Most of all after more than 7 decades form inception AI architecture finally found the technology of choice in GPU and demonstrate the feasibility of this revolutionary architecture. By 2018 the basic GPU had evolved and had been customized to handle all the aspects of AI. However, the world, occupied with the challenges of the pandemic, did not noticed this until late in 2022 when ChatGPT attracted 100million users in 2 months. The most dramatic societal transformation has just initiated. Data centers investment powered by GPUs and MPUs is projected to grow 33% in 2025.

With targeted policies like the CHIPS Acts and other investments in semiconductors and electronics, nations are striving not just for recovery, but for resilient, innovation-driven futures. *As we step further into the digital age, the evolution of GDP will be increasingly tied to how economies navigate these complex, tech-centered landscapes.*

9. 2024 INTERNATIONAL FOCUS TEAMS' SUMMARIES

This section covers a summary of the 2024 IRDS Chapters. In depth assessments and analysis of needs, challenges, and potential solutions for the next 15 years can be found in the individual chapters, provided on the IRDS website:

1. Systems and Architectures (SA)
2. More Moore (MM)
3. Outside System Connectivity (OSC)
4. Beyond CMOS (BC)
5. Lithography (L)
6. Yield Enhancement (YE)
7. Metrology (M)
8. Environment, Safety, Health, and Sustainability (ESH/S) as Environmental Sustainability of Semiconductor Facilities (ESSF)
9. Autonomous Machine Computing

- This section is structured with respect to Computer Architecture/Systems (such as Systems and Architectures, Outside System Connectivity, and Beyond CMOS, and Autonomous Machine Computing) and
- Device Technologies and manufacturing considerations (such as More Moore, Lithography, Metrology, Yield Enhancement and Environmental Sustainability of Semiconductor Facilities).

Computer Architectures/Systems

9.1. SYSTEM AND ARCHITECTURES

INTRODUCTION

The Systems and Architectures section of the roadmap analyzes a broad range of applications of computing, electronics, and photonics. This chapter identifies key challenges and technology requirements by observing the patterns in play in the systems and architectures anticipated during the period under study.

In previous editions of the roadmap, we have focused upon aspects of reliability, availability, serviceability and security of individual systems, in line with the vertically integrated and turnkey nature of many solutions within the industry. Current thinking in this space has evolved rapidly, however, and it is important to recognize the intrinsically fractal nature of the challenges found in this problem domain. As a result, it has become obvious that the aspects of reliability, availability, serviceability and security should be considered as holistic components of maintaining the supply chain.

The SA chapter identifies four main areas of applications, as follows:

1) Internet-of-Things networks perform important services in a range of applications

2) Cyber-physical systems (CPS) provide essential services

3) Personal augmentation devices number in the billions worldwide—regardless of whether they are operated privately or for public consumption

4) Cloud systems are engendering new programming languages and methodologies and cloud-native computing

It must be added that close association among any of these categories occurs under practical operating conditions since all categories of systems considered are intrinsically linked by the information lifecycle of AI and this has implications for the optimization of any given type.

Present and Future Drivers

Artificial intelligence continues to be a primary driver for the demand for growth in capabilities both in its training phases where vast amounts of data are analyzed to create models and in its inference phases where models are evaluated against new data. Increasing levels of AI-driven computation are being provisioned in the cloud, but many advantages exist to performing decision-making tasks in real time at the edge, so we expect personal augmentation systems, IoT edge devices, and cyber-physical systems to all include significant AI components.

It is still expected that AR/VR will emerge as an important application area, however there are significant challenges in developing suitably compact interface devices that will begin to drive social acceptance of this degree of obvious augmentation in social spaces. Early enthusiasm for full VR headset devices failed to translate into sustained business, but work continues both immersive VR and lightweight AR equipment.

A primary challenge in this domain remains the need to synthesize data streams captured locally with geographically and contextually related live and pre-distributed data streams, within the perceptual limitations of the users, which places speed-of-light limitations for low latency computational turnaround. In the meantime, a core driver for personal augmentation systems is instrumentation, with health monitoring, safety alerting and sensory enhancement all being significant value-added features to devices in this category.

From 2D to 3D

The semiconductor industry is evolving from 2D to 3D packaging technologies and this migration represents an inflection point in how a system is organized that creates an opportunity for an explosion of new architectures that exploit the potential of stacking combinations of processing, memory, acceleration and I/O in denser and more efficient forms. However, this new flexibility on how a system is organized makes it difficult to project many elements of the roadmap forwards as we move from a period of familiar 'general purpose computing' devices to a proliferation of exploration into new ways to address current challenges.

It is anticipated that there will be a period of searching for commercial advantage through the application of specialized architectures and it is likely that there will be a significant product differentiation before modular IP within devices will become as common as the current 'modular devices on board' model.

From Written Software to AI

This shift is driven by the combination of open-source software frameworks and the rise of AI machine learning frameworks capable of the creation of very effective models based on statistical inference. Unsupervised learning techniques can trawl huge volumes of structured and unstructured data to find correlations independently of expert blind spots. Intelligence craves data and artificial intelligence is no exception.

The release of ChatGPT in November 2023 was a watershed event where the efficacy of large language models cross significant levels of accuracy across many knowledge work activities, not the least of which is the task of code generation itself. What was not predicted at the time of the original hypothesis was that this would demand all of the high quality publicly available data on the internet. As large language models move from their initial powerful demonstrations of potential to adoption in regulated industry and enterprise, the pursuit of increasing quality and fitness for particular use will bifurcate the market between those continuing to pursue larger models, which will demand quadratically increasing data (and synthetic data may be required) and computational power, and those assembling smaller language models trained on data of proven quality and provenance which play specialized roles.

Furthermore, the open-source development and collaboration model has proven incredibly effective in software, not only in the complexity of systems that can be delivered, but also in the diversity of those who are enabled to participate. This creates the virtuous cycle where internationalization and localization occur as primary efforts coincident with innovation rather than after the fact, creating greater diversity of representation that again fuels greater inclusion in the economic and social benefits of innovation. But, most of all the same guiding principles of open source software development are being extended down the stack.

From Data Centers to Data Everywhere

Today, 90% of information that the enterprise (public or private) cares about is housed in a data center. The advent of so many rich, high-definition sensors housed in the ever-proliferating number of personal augmentation devices, that ratio is expected to shift to as much as 75% of enterprise information will *never* being housed in a data center. This since that data will grow exponentially and disproportionately at the edge, in distributed social infrastructure, in edge devices (personal, public, and private - in other words, in all things intelligent).

System Categories: CLOUD

The term *cloud* refers to the engineering of data center scale computing operations—compute, storage, networking engineered for scale and for continuous resource redeployment and reconfiguration via APIs. Cloud infrastructure has undergone several waves of optimization from its initial deployment of industry standard rack servers, storage and compute at data center scale.

Initially, scientific computation traditionally emphasized numerical algorithms, whereas cloud applications, in contrast, emphasized streaming for multimedia and transactions for commerce and other database applications. Now, with AI/machine learning (ML) integrated into so many applications, the demand for accelerated floating point computation is more universal and all applications are being dominated by operational and capital costs of data movements at scale. In all cases, the general trend is also to utilize this as-a-service consumption model to foster independence of the user from not only a particular piece of hardware infrastructure but from one architectural approach.

The power consumption of processors used for high-end systems such as cloud, servers, and high-performance computing ranges from tens of watt to a few hundreds of watts depending on the performance requirement of each processor chip and its cooling capacity. As these systems are composed of a cluster of multiple compute nodes, total system performance is usually a crucial metric rather than a single processor performance. Since the total system performance depends on the number of installed nodes which is mainly limited by total power and/or cooling budget of a data center facility, power efficiency, sometimes measured by performance per watt, is used to be a critical factor of the processors. This trend is outlined in the figure below.

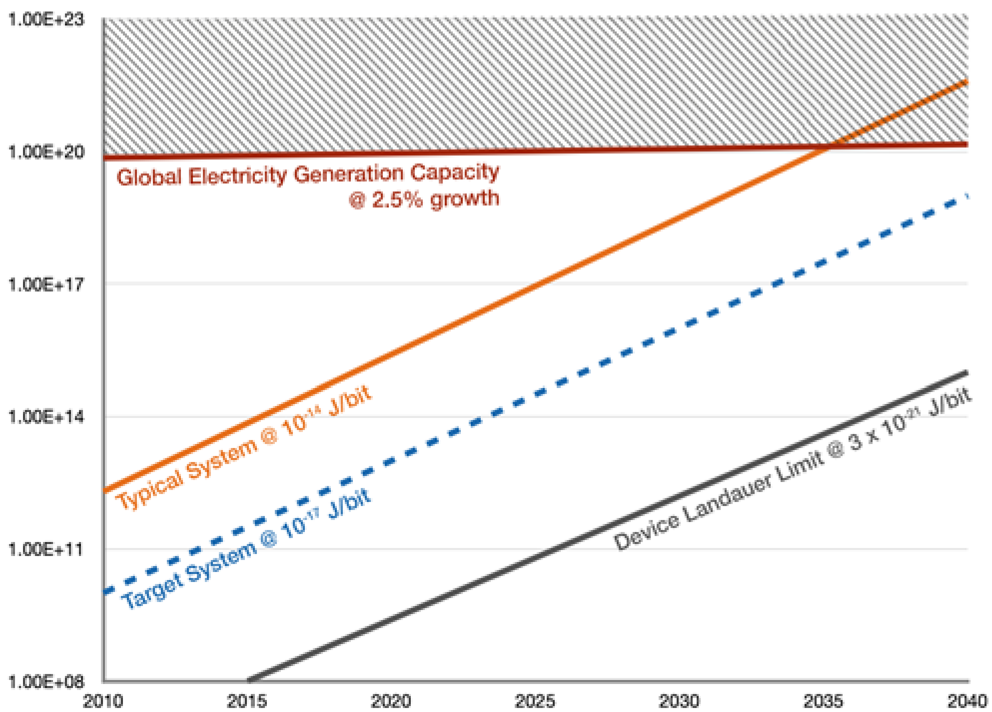


Figure ES-36. Power consumption trends for cloud operations

Table ES-3. Overall Roadmap Systems Characteristics

2024 IRDS ORSC									
YEAR OF INTRODUCTION	***2024***	***2025***	***2027***	***2029***	***2031***	***2033***	***2035***	***2037***	***2039***
Cloud Computing (CC) Latency Sensitive Processors									
Number of MPU Cores per socket (max)	96	96	128	256	384	512	640	800	900
Processor Base Frequency (for multiple cores together) [2]	2.8-3.4	2.8-3.4	3.0-3.5	3.2-3.6	3.4-3.7	3.4-3.7	3.4-3.7	3.4-3.7	3.4-3.7
L1 Data Cache Size (in KB) [3]	42	42	44	44	44	44	44	44	44
L1 Instruction Cache Size (in KB) [4]	128	128	160	160	160	160	160	160	160
DDR bandwidth (TB/s)	6.6	6.6	6.6	6.6	6.6	6.6	6.6	6.6	6.6
Number of DDR channels	12	12	12	16	16	16	16	16	16
	DDR5	DDR5	DDR5	DDR5	DDR6	DDR6	DDR6	DDR6	DDR6
Data Rate/link (Gbit/s)	800	800	1600	1600	3200	3200	3200	3200	6400
Socket TDP (Watts)	400	400	450	600	600	700	700	800	800
SA Augmented Data Systems (ADS) Table - Focus Drivers Line Items									
# CPU cores	18	18	25	28	28	30	30	30	30
# GPU cores	64	64	128	256	256	512	512	512	512
CPU frequency (GHz) [2]	3.10	3.39	3.51	2.94	4.17	3.93	4.18	3.99	3.82
Maximum Mobile Data Rate (Gb/s) [6]	10	10	10	20	20	50	50	50	50
# Sensors	12	12	12	16	16	16	16	16	16
Board Power (mW)	6504	6829	7529	8301	9152	10090	11000	12000	13000
SA IoT Table - Focus Drivers Line Items									
CPUs per device [7]	4	4	6	6	8	8	8	8	8
Max CPU Frequency (MHz)	320	325	335	346	357	369	380	390	400
Energy Source (B = battery, H = energy harvesting)	B+H	B+H	B+H	B+H	B+H	B+H	B+H	B+H	B+H
Sensors per device	12	12	16	16	16	16	17	19	21
SA CPS Table - Focus Drivers Line Items									
Number of Devices [7]	128	128	256	256	512	512	512	512	512
CPUs per Device [7]	8	12	12	16	16	16	16	16	16

Table ES-3 Notes

[1] Required for specintrate scaling

[2] CC Frequency increase slowing down, but increases because of better cooling (allowing higher TDP); ADS frequency slowed from past forecasts

[3] Load-to-use latency dictates constant or limited growth in L1 data cache size

[4] Instruction footprint for cloud apps going up (refer Google data warehouse paper)

[5] Personal Augmentation was defined as Mobile in the 2021 IRDS

[6] Max mobile cellular data rate includes 4G LTE, 5G, 6G (beyond 2031), and beyond mobile cellular data rates. 6G will start to be introduced in 2033; maximum data rate is projected to be 50 Gbits/sec extended out through 2039.

[7]. For clarification purposes, the name "devices" refers to chip- level VLSI chips and CPU architectures which may contain many CPU's integrated into a single chip; with many VLSI Devices and many CPU's integrated onto Mobile phones and IoT's and flexible or solid circuit boards, and many circuit boards onto, for example, server and switch "blades" ; which themselves are numerous in "racks," boxes, rows, etc.

HPC/AI TECHNOLOGY HIGHLIGHTS

The performance of HPC/DC systems is increasingly dependent on accelerators, such as GPUs, Tensor Processing Units (TPUs) and AI processors, rather than general-purpose CPUs. It is therefore necessary to project a roadmap of accelerators with their characteristics and performance, in order to project architectures of future systems in the HPC and DC categories. The table below highlights some of the key indicators.

Table ES-4. Single-socket Performance Trends of High-End GPUs with HBM or Advanced Future Memory

	2024	2026	2028	2030	2032	2034	2036	2038	2040
FP64 vector [TFLOPS]	45	53	62	73	86	100	118	140	165
FP32 vector [TFLOPS]	90	106	124	146	172	200	236	280	330
FP16/BF16 matrix [PFLOPS AI]	5	8	12.5	20	30	45	65	95	140
FP8 matrix [PFLOPS AI]	10	16	25	40	60	90	130	190	280
INT8 matrix [POPS AI]	10	16	25	40	60	90	130	190	280
Memory bandwidth per socket (TB/s)	6	7	10	20	40	60	66	72	80
Memory type	HBM 3	HBM3e	HBM3e	HBM4 or equiv.	HBM4e or equiv.	3D Stacked	3D Stacked	3D Stacked	3D Stacked
Total number of memory modules (stacks for HBM)	8	8	12	12	16	16	16	16	16
TDP per socket	700	1000	1400	1800	2200	2400	2500	2600	2700

PACKAGING HIGHLIGHTS

Monolithic/Package Integration, aimed at high-speed processing of generative AI data, is rapidly evolving. The IRDS More Moore (MM) Chapter discusses backside functions as a roadmap for Monolithic Integration. The introduction of Backside Via and Backside Power Delivery Network (BSPDN) is set to begin in 2025, requiring circuit driving designs based on DTCO. Subsequently, IO, Clock, and Bus will be introduced, transforming backside functions. Full-scale application of Direct Contact is expected to commence in 2027, and by 2029, active devices such as IO and IVR (Integrated Voltage Regulator), along with intrinsic Decaps (Decoupling Capacitors), will be mounted on the backside.

In current mainstream packaging, discrete passive components are mounted on printed circuit boards equipped with CPUs and GPUs, with Decaps being used extensively. In next-generation Packaging Integration, these components will be positioned closer to the processors. Decaps mounted on interposers are referred to as HDMIM Cap (High Density Metal-Insulator-Metal Cap), DTC (Deep Trench Cap), ISC (Integrated Stacked Cap), among others.

SYSTEM CATEGORY: PERSONAL AUGMENTATION

Personal augmentation devices integrate computation, communication, storage, capture and display, and sensing. These systems are highly constrained in both form factor and energy consumption. As a result, their internal architectures tend to be heterogeneous. Typical examples are: telephony and video telephony; multimedia viewing; photography and videography; email and electronic communication; positioning and mapping, authenticated financial transactions, health and fitness monitoring, personal safety and environmental warning. Current and upcoming market drivers include gaming and video applications; productivity applications; social networking; augmented reality and context-aware applications, and mobile commerce. Personal augmentation devices already make use of AI technologies such as personal assistants. Deployment of AI on and through personal augmentation devices will accelerate. The processors used for personal augmentation systems need high processing performance, but cannot be equipped with cooling systems like heat sinks or cooling fans. The performance is restricted by power consumption limits due to thermal dissipation. Heat, generated by chips, must be radiated to ambient air via conduction through packaging. The power consumption of this type of processor must be in the range from 100mW to 10W. Key performance indicators are included in the table below.

Table ES-5. Personal Augmentation KPIs

	2024	2026	2028	2030	2032	2034	2035	2036	2037	2038	2039
Core performance (DMIPS/MHz/Core)	20.8	23.2	25.5	27.8	30.1	32.4	33.6	34.8	35.9	37.1	38.3
Number of Cores	16	16	16	32	32	32	48	48	48	48	64
Maximum frequency (GHz)	3	3.2	3.2	3.4	3.4	3.4	3.6	3.6	3.6	3.6	3.6
Performance (kDMIPS)	1,000	1,186	1,304	3,025	3,277	3,530	5,807	6,008	6,209	6,409	8,813
Process node (nm)	3	2.1	1.5	1.5	1.0 eq	0.7 eq	0.7 eq	0.7 eq	0.5 eq	0.5 eq	0.5 eq

SYSTEM CATEGORY: INTERNET-OF-THINGS EDGE DEVICES

An IoT device is a wireless device with computation, sensing, communication, and possibly storage. The device may include one or more CPUs, memory, non-volatile storage, communication, security, and power management. It may be line powered, battery powered or utilize energy harvesting.

IoT devices must satisfy several stringent requirements. They must consume small amounts of energy for sensing, computation, security, and communication. They must be designed to operate with strong limits on their available bandwidth to the cloud.

Many IoT devices will include AI capabilities; these capabilities may or may not include online supervision or unsupervised learning. These AI capabilities must be provided at very low energy levels. A variety of AI-enabled products have been introduced. Several AI technologies may contribute to the growth of AI in IoTe devices—convolutional neural networks; neuromorphic learning; stochastic computing.

IoT edge devices must be designed to be secure, safe, and provide privacy for their operations. The processors in this category are classified into two types: standard systems with power consumption of 10 mW or less, and low power systems with power consumption of 1 mW or less.

A wireless sensor node mainly consists of a power-supply unit, sensor unit, micro controller unit (MCU) unit, and wireless unit. A battery and photovoltaic power-generation device and/or energy harvesting device are mounted on the power-supply unit together with a power-supply stabilizing circuit. In cases where the performance of the installed battery or energy harvester is insufficient, the range of IoTe utilization is limited.

The primary benefit of processing IoT data at the edge is the significant reduction in data traffic to the cloud. Beyond this traditional view of data reduction, there is growing focus on the benefits of reducing computational loads and preventing temperature increases in cloud GPU servers, spurred by the explosive growth of generative AI. As the hardware and operational costs for sensors and machine-to-machine (MtM) devices continue to justify the reduction in cloud computing loads, the adoption of IoT at the edge is expected to increase. Furthermore, the development of smartphones for generative AI and the creation of new edge devices may be propelled by the combined benefits of reduced computational processing and data storage at the edge. The IoT edge is characterized by a heterogeneous integration of devices and chips. The adoption of technologies such as 3D stacking and 2.5D interposers will facilitate this heterogeneous integration within IoT environments. Technologies initially developed for advanced processors, which benefit from substantial development funds, are expected to be adapted for IoT use with a typical delay of two to three years. Some key indicators for IoT devices are reported in the table below.

Table ES-6. Key Indicators for IoT Devices

	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
CPUs per device	2	4	4	4	4	6	6	6	8	8	8	8	8
Maximum CPU frequency (MHz)	310	315	320	325	330	335	340	346	351	360	363	369	375
Energy source (battery, harvesting)	B+H	B+H	B+H	B+H	B+H	B+H	B+H	B+H	B+H	B+H	B+H	B+H	B+H
Tx/Rx power/bit (µW/bit)	0.09	0.06	0.04	0.03	0.02	0.01	0.008	0.01	0.003	0.002	0.002	0.002	0.002
Battery operation lifetime (months)	9	9	9	12	12	12	12	18	18	18	18	18	18
Deep suspend current (nA)	32	27	23	20	17	14	12	10	9	8	7	7	7
Sensors per device	8	8	12	12	12	16	16	16	16	16	16	16	16

SYSTEM CATEGORY: CYBER-PHYSICAL SYSTEMS

Cyber-physical systems are networked control systems. These distributed computing systems perform real-time computations to sense, control, and actuate a physical system. Many cyber-physical systems are safety-critical. They interface to the systems they control via both standard and proprietary interconnects broadly known as operational technology (OT), where ruggedness, extended environmental capabilities, low-cost have historically been paramount over considerations such as security and attestation. As these systems are increasingly connected edge-to-cloud, this will present an increasing attack surface, either for data theft, false signal injection, systems commandeering, or as a back door into the IT domain.

Cyber-physical systems must be highly reliable at all levels of the design hierarchy. Physical security and isolation have traditionally been part of the design of these systems, but that is becoming a greater challenge as edge to cloud connected design becomes the dominant methodology.

Some cyber-physical systems, such as vehicles, are powered by generators. In these systems, available power for the computational engine is determined by the capabilities of the generator and the electrical load presented by the physical plant. Cyber-physical systems typically perform roles that are of a critical nature and therefore also require uninterruptible power supplies that may constrain total power budget. Many cyber-physical systems present extreme temperature, electromagnetic and vibrational environments in which the electronics must operate. The power consumption must be in the range from 100mW to 10W. Some of the key indicators are reported in the table below.

Table ES-7. Cyber-physical Systems Key Indicators

	2024	2026	2028	2030	2032	2034	2035	2036	2037	2038	2039
Core performance (DMIPS/MHz/Core)	15.1	17.3	19.7	22	24.3	26.6	27.8	29	30.1	31.3	32.4
Number of Cores	8	8	8	16	16	32	24	24	24	24	24

Maximum frequency (GHz)	1	1	1.2	1.2	1.2	1.2	1.4	1.4	1.4	1.6	1.6
Performance (kDMIPS)	121	139	189	422	467	1,023	934	973	1,012	1,201	1,246
Process node (nm)	7	5	3	2.1	2.1	1.5	1.5	1.0 eq	1.0 eq	1.0 eq	0.7 eq

For mobile applications, such as autonomous vehicles, there is a trade-off between ML-capability, range and payload capacity. In this envelope, it is necessary to reduce the volume and mass of the processing capability, whilst also reducing power usage. Depending upon the velocity and operating environment of the vehicle, heat dissipation may be a lesser or greater issue. This challenge becomes more significant in mobile robotic systems intended to function at human scale or smaller. It can be assumed that external factors such as mechanical vibration and electromagnetic interference will be higher in this category.

To enhance CPS cycles, technology for high-speed, large-capacity data transfer between edge and cloud environments is advancing. In particular, the role of silicon photonics, which brings optical technology close to processors, is becoming increasingly crucial. Silicon photonics facilitates photonic-electronic integration and spans various applications from chip-level to next-generation optical interposers and optical interconnects. The choice of substrate—whether PCB, silicon, glass, or plate—will depend on its suitability for the specific application. In CPS, silicon photonics is utilized in high-speed receivers that enable rapid generative AI training across multiple cloud servers.

COMPUTER ARCHITECTURES AND SYSTEMS CONCLUSIONS, RECOMMENDATIONS AND FUTURE PLANS

Despite having top supply chain processes, unexpected global events have caused significant problems, affecting all downstream industries. It would be unwise to view this experience as a transient event followed by a return to business as usual, however. As we move further into the 'vertical' phase of Moore's law growth, rapidly increasing change and volatility are to be fully expected, with the associated socio-political change that new technologies always bring. Consideration should be given to volatility as a regular and ongoing problem.

Personal augmentation systems have emerged globally as a new driver for consumer technology, with uptake progressing beyond the original communications functions of mobile phones to encompass a wide range of enhancements to human capabilities, including augmented reality, the extension of vision into infrared, the provision of accurate contactless temperature and distance measurement, 3D scanning, sensing of multiple health parameters, fall detection and alerting etc. Provision of secure financial transactions continues as an important emerging application for personal augmentation system.

Cyber-physical systems will continue to proliferate in scenarios where they are required to control physical systems, the reliability of these systems will increasingly fall into the purview of government regulation. As this regulation is currently being driven by demands very different from the concerns of the industry, attention should be given to ensuring that incompatible regulation does not become problematic for some markets.

AI/ML, IoT and CPS all present drivers for massive increases in data flow from edge to cloud. As this grows, the problems of information transfer bottlenecks within systems will transition from the HPC domain down through all the design envelopes. Utilizing the performance available in new devices will become harder as increasingly complex software approaches are necessary to face these challenges so it will be necessary to disseminate appropriate knowledge and tooling to allow customers to take advantage of these capabilities.

Looking forward beyond the 2024 edition, the Systems and Architecture team are planning a major update which investigate the following themes:

Accelerators are a cross-cutting technology impacting all system categories and design envelopes—IoTe, CPS, cloud, personal devices. Despite the continuous increase in design, validation, and manufacturing costs associated with leading edge processes nodes, the ability of accelerators to solve for size, weight, power constraints and yield sufficient return on investment when accounting for capital and operational cost is increasing across all system categories. These are no longer application specific functions in standards CMOS process, an increasingly wide variety of accelerators – in-memory, analog, quantum, and photonic approaches are being explored alongside GPU, FPGA, and data processing unit (DPU).

9.2. DEVICE TECHNOLOGIES AND MANUFACTURING CONSIDERATIONS -- MORE MOORE

INTRODUCTION

The More Moore IFT provides physical, electrical, and reliability requirements for logic and memory technologies to sustain power, performance, area, cost (PPAC) scaling for edge and cloud applications. This is done over a time horizon of 15 years for mainstream/high-volume manufacturing.

A major portion of semiconductor device production is devoted to digital logic that needs to support two technology platforms: 1) high-performance logic, and 2) low-power/high-density logic. Key considerations for these technology platforms are speed, power, density, cost, capacity, and time-to-market.

The goal of the semiconductor industry is to be able to continue to develop new technology generations to improve overall performance at reduced power and cost. The performance of the components and of the final chip can be measured in many ways: higher speed, higher density, lower power, form factor reduction, bill-of-material reduction, more functionality, etc. Traditionally, dimensional scaling was adequate to bring about these performance merits, but it is no longer the case. Processing modules, tools, material properties, etc., are presenting difficult challenges to continue scaling.

Transistor Technology Ground Rules Scaling

The More Moore roadmap focuses on effective solutions to sustain the performance and power scaling at scaled dimensions and scaled supply voltage. Ground rule scaling drives die-cost reduction. However, this scaling increases the portion of parasitics in the total loading and brings diminishing returns of scale in performance and power scaling. Therefore, it is necessary to focus on technology scaling solutions that also scale the parasitics of device and interconnect. Ground-rule scaling needs to also enable design-technology co-optimization (DTCO constructs that accommodate the area reduction as well as tighten the critical design rules that limit the area scaling. Due to the rising costs and process complexity of multiple patterning, EUV is used as a remedy to pattern-tight ground rules in fewer process steps. The projected roadmap of ground rules as well as device architectures is shown in Table ES-8. Evolution of ground rules in shown Figure ES-37. There is not yet a consensus on the node naming across different foundries and integrated device manufacturers (IDMs); however, the projected rules give an indication of technology capabilities in line with the PPAC requirements. Key parameters in the ground rules are the gate pitch, metal pitch, fin pitch, gate length, and 3D tier stacking capability, which are important factors in core logic area scaling.

This implied that two different paths exists and any product design must carefully select the best approach however, typically a combination of these two paths provides the best solution.

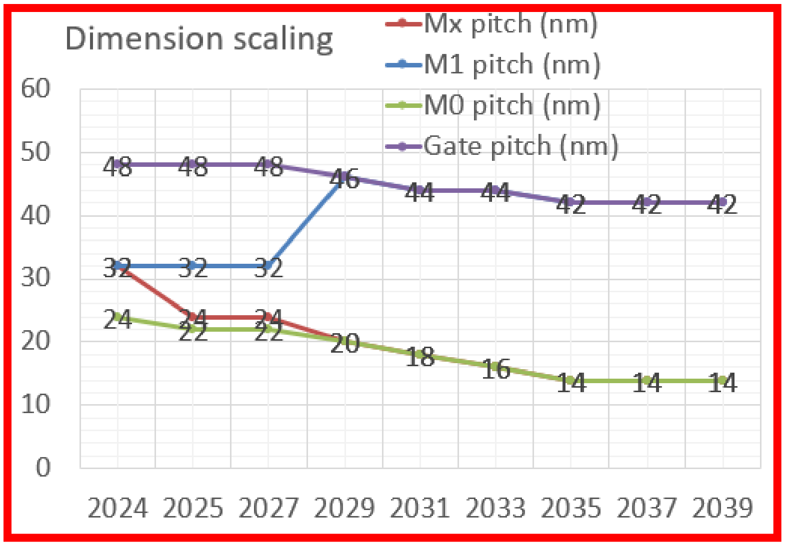


Figure ES-37. Evolution of ground rules

Technology Evolution

Transition to new transistor architectures

The FinFET transistor architecture introduced in 2011 will remain approach until 2025 as anticipated in the 2028IRDS. In 2025 a transition to lateral GAA devices is needed to overcome the limits of single fin geometry as fin-width down scaling required scaling to sustain the electrostatics control is reaching physical limits. Parasitic capacitance mitigation, effective drive width (Weff), and replacement metal gate (RMG) integration continue to be the challenge to sustain GAA adoption across multiple nodes. Projected evolution of device architectures is shown in Figure ES-38. It is projected that complementary FET (CFET) will be the subsequent evolution of lateral GAAs in the 3D form, where N devices will be stacked over P devices. Table ES-8 describes the detailed evolution in time of the transistor's structure.

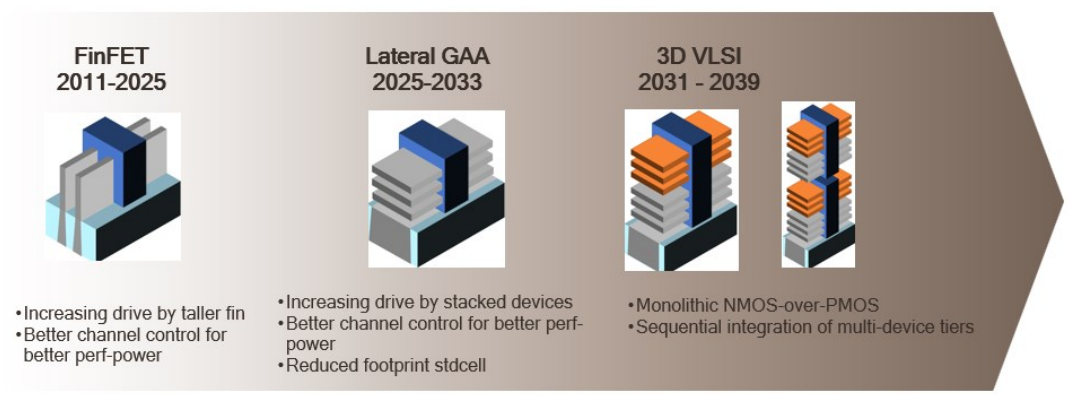
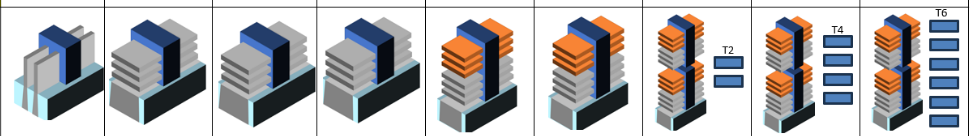


Figure ES-38. Projected evolution of device architectures

Table ES-8. Detailed Evolution In Time of the Transistor's Structure

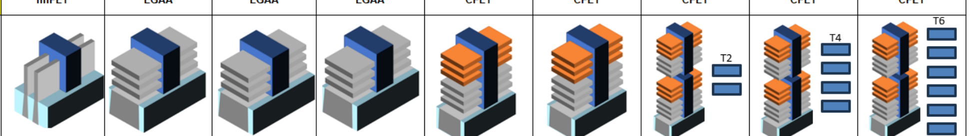
YEAR OF PRODUCTION	2024	2025	2027	2029	2031	2033	2035	2037	2039
	G48M24	G48M22	G48M22	G46M20	G44M18	G44M16	G42M14/T2	G42M14/T4	G42M14/T6
Logic industry "Node Range" Labeling	"3nm" Enhanced	"2nm"	"1.4nm"	"A10 eq"	"A7 eq"	"A5 eq"	"A3.5 eq"	"A2.5 eq"	"A1.8 eq"
Fine-pitch 3D integration scheme	Stacking	Stacking	Stacking	Stacking	CFET	CFET	3DVLSI	3DVLSI	3DVLSI
Logic device structure options	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET-3D	CFET-3D	CFET-3D
Backside structure options	-	Backside via	Direct contact	Decap+ESD	Active devices	Active devices	Active devices	Active devices	Active devices
Platform device for logic	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET	CFET	CFET
	 T2 T4 T6								
LOGIC TECHNOLOGY ANCHORS									
Device technology inflection	Taller fin	LGAA	LGAA	LGAA	CFET	CFET	3DVLSI	3DVLSI	3DVLSI
Patterning technology inflection for Mx interconnect	193i, EUV DP	193i, EUV DP	193i, High-NA EUV	193i, High-NA EUV	193i, High-NA EUV	193i, High-NA EUV	193i, High-NA EUV	193i, High-NA EUV	193i, High-NA EUV
Specialty transistors add-ons	-	-	-	2D, IGZO	2D, IGZO	2D, IGZO	2D, IGZO	2D, IGZO	2D, IGZO
Local interconnect inflection	Self-aligned vias	Backside via	Backside contact Substractive metal	Backside contact Substractive metal	Backside contact Active backside	Tier-to-tier via Active interconnect	Tier-to-tier via Active interconnect	Tier-to-tier via Active interconnect	Tier-to-tier via Active interconnect
Process technology inflection	Channel, RMG	Lateral/AtomicEtch	Low-temp contact	Low-temp contact	CFET	CFET	3DVLSI	3DVLSI	3DVLSI
Heterogenous stacking generation inflection	2.5D Chiplet, Mem-on-Logic	Backside Power	Backside Fabrics	Backside Fabrics	CFET	CFET	3DVLSI	3DVLSI	3DVLSI

This structural transition does not come without problems and challenges. Parasitic capacitance between gate and source/drain terminal of the device is expected to increase with technology scaling. In fact, this component is getting more important than channel-capacitance-related loading whenever the standard cell context is considered and even more elevated in the GAA structures as a result of unused space between stacked devices. There is a need to focus on low-κ spacer materials and integration of inner spacer schemes in the GAA device is necessary to block the gate dielectric interface facing the source/drain at the unused space of GAA device. Those schemes still need to provide good reliability and etch selectivity for S/D contact formation.

Performance-Power-Area (PPA) Scaling

The projected roadmap for the electrical specifications of logic core device is listed in Table ES-9. This edition of More Moore roadmap includes both logic and analog specifications of More Moore platform device. Analog specifications are derived from the device targets of a logic device but this would potentially require relaxation of contacted poly pitch on the same wafer to allow longer channel lengths. There would also be considerations such as reliability and matching where performance targets need to be derated in an effort to meet those concurrent goals, e.g. increasing overdrive voltage by stacking devices.

Table ES-9. Roadmap for the Electrical Specifications of Logic Core Devices

YEAR OF PRODUCTION	2024	2025	2027	2029	2031	2033	2035	2037	2039
	G48M24	G48M22	G48M22	G46M20	G44M18	G44M16	G42M14/T2	G42M14/T4	G42M14/T6
Logic industry "Node Range" Labeling	"3nm" Enhanced	"2nm"	"1.4nm"	"A10 eq"	"A7 eq"	"A5 eq"	"A3.5 eq"	"A2.5 eq"	"A1.8 eq"
Fine-pitch 3D integration scheme	Stacking	Stacking	Stacking	Stacking	CFET	CFET	3DVLSI	3DVLSI	3DVLSI
Logic device structure options	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET-3D	CFET-3D	CFET-3D
Backside structure options	-	Backside via	Direct contact	Decap+ESD	Active devices	Active devices	Active devices	Active devices	Active devices
Platform device for logic	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET	CFET	CFET
	 T2 T4 T6								
LOGIC DEVICE ELECTRICAL SPECS									
Power Supply Voltage - Vdd (V)	0.70	0.65	0.60	0.60	0.60	0.60	0.60	0.60	0.60
Subthreshold slope (mV/dec) - HP (mV/dec)	82	72	70	70	70	70	70	70	70
Subthreshold slope (mV/dec) - HD (mV/dec)	75	67	67	65	65	65	65	65	65
Capacitive equivalent thickness (CET) (nm) [2]	1.00	1.00	0.90	0.90	0.90	0.90	0.90	0.90	0.90
Vt,sat at Ioff=10nA/um - HP (mV)	156	165	164	152	164	164	158	151	146
Vt,sat (mV) at Ioff=100pA/um - HD (mV) [3][4]	288	271	276	260	266	265	261	257	257
Average effective mobility of N/P-MOS (cm2/V.s)	125	100	80	80	60	60	60	60	60
Rsd (Ohms.um) [5]	271	257	245	232	221	210	199	189	180
Ballisticity.Injection velocity (cm/s)	9.00E+06	9.00E+06	9.00E+06	9.00E+06	9.00E+06	9.00E+06	9.00E+06	9.00E+06	9.00E+06
Vdsat (V) - HP	0.092	0.101	0.108	0.108	0.144	0.120	0.120	0.120	0.120
Vdsat (V) - HD	0.104	0.101	0.108	0.108	0.144	0.120	0.120	0.120	0.120
Average Ion (uA/Perimeter Weff-um) of N/P-MOS at Ioff=10nA/um - HP [6]	874	787	759	797	768	800	827	858	885
Average Ion (uA/sheet or fin) of N/P-MOS at Ioff=10nA/um - HP [7]	88	170	158	115	161	139	119	98	85
Average Ion (uA/Perimeter Weff-um) of N/P MOS at Ioff=100pA/um - HD [8]	644	602	546	589	565	595	614	635	646
Average Ion (uA/sheet or fin) of N/P MOS at Ioff=100pA/um - HD [9]	65	130	114	85	119	104	88	72	62
Cch,total (fF/um2) - HP/HD [8]	34.52	34.52	38.35	38.35	38.35	38.35	38.35	38.35	38.35
Gate height over fin (nm)	20	15	10	10	10	10	10	10	10
Cch (fF/um) - HP [8]	0.44	0.39	0.37	0.37	0.37	0.31	0.31	0.31	0.31
Cch (fF/um) - HD [8]	0.50	0.39	0.37	0.37	0.37	0.31	0.31	0.31	0.31
CV/I (ps) - FO3 load, HP [9]	1.06	0.96	0.87	0.83	0.86	0.69	0.67	0.64	0.62
I/(CV) (1/ps) - FO3 load, HP [10]	0.94	1.04	1.15	1.20	1.16	1.45	1.50	1.55	1.60
Energy per switching [CV2] (fj/switch) - FO3 load, HP	0.65	0.49	0.40	0.40	0.40	0.33	0.33	0.33	0.33

System-On-Chip (SoC) PPA Metrics

SOC has become the fundamental building block of most of the new electronics devices. Due to standard cell and bitcell density improving on a node-to-node basis, it is possible to integrate more functions in a given SoC footprint.

The footprint if SoC for mobile applications is increasing across generations because of newly added functions exceeding the scaling reduction that must be compensated by increasing the die size. Therefore, the amount of memory as well as graphical processing unit (GPU) processors and neural processing units (NPU) follow the density scaling of SRAM and standard cell, respectively, as the trend for more parallel architectures continues. However, thanks to advances in DTCO, lateral nanosheets, followed by device-over-device stacking (e.g., P-over-N) and 3D VLSI, SoC footprint scaling factor for the same function can still be maintained as an alternative to indefinitely increasing die size. In this roadmap edition we also added more details on machine learning blocks such as the MAC (multiply-accumulate) block clock frequency and number of MACs in SoC. Integration capacity of logic technology is shown in Figure ES-39 (amount of NAND2-equivalent standard cell density as well as bitcell density). The overall Near-Term and Long-Term Challenges have been summarized in the following Tables ES-10 and ES-11 below.

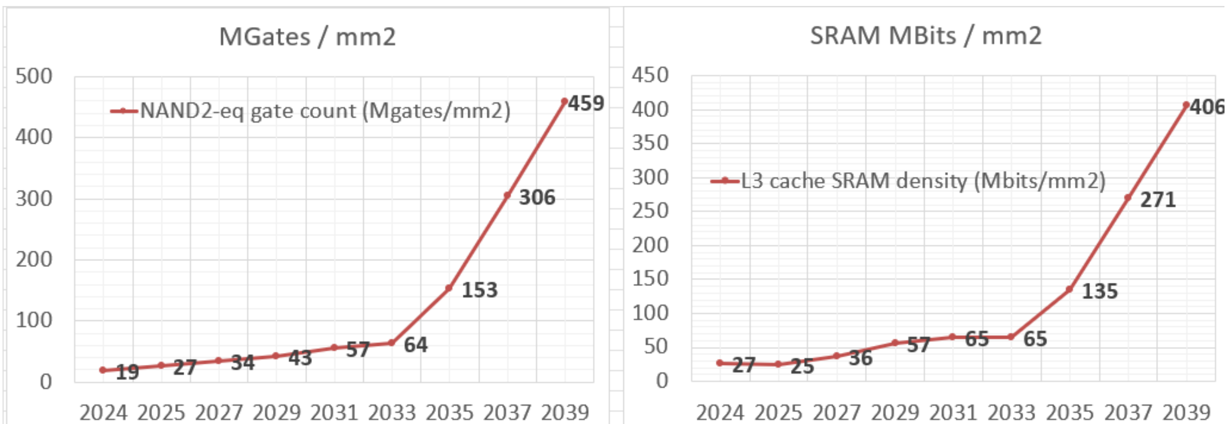


Figure ES-39. Number of MACs in SoC. Integration capacity of logic technology

Table ES-10. More Moore Near-term Challenges

Near-Term Challenges: 2024-2029	**Description**
Power scaling	Voltage and capacitance scaling slow down and there is a lack of solutions for power reduction. Introduction of gate-all-around (GAA) devices is a remedy to reduce the supply voltage, but not in a sustained manner that allows continuous scaling. Power scaling is also limited because of slow-down of loading capacitance scaling, impacted by increasing portion of parasitic components with scaling. Therefore, low-κ materials, design-technology-co-optimization (DTCO), backside wafer processing for better power delivery and contact access schemes, and novel power landing vias, are needed.
Parasitics scaling	Maintaining control of increased parasitics in stacked devices. Stacked devices require high-aspect ratio contacts to access the bottom contact. This will increase both the contact resistance as well as the fringe capacitance between the gate and drain/source. Interface resistance will also require new silicidation schemes that conformally wrap the source/drain.
Cost reduction	Cost-effective area scaling through High-NA EUV, DTCO, and 3D stacking. Throughput and yield challenges necessitate a careful selection of ground rules that optimize the die cost as a significant part of cost is determined by the middle-of-line (MOL) and BEOL stack. Therefore, new process-enhanced design constructs that tighten the secondary design rules such as tip-to-tip and the P-N separation rule are necessary to allow a further shrink of the standard cell and bitcell area on top of ground rule scaling for low-cost die. Process integration of those design constructs might

Near-Term Challenges: 2024-2029	**Description**
	require new materials to allow better etch selectivity and self-deposition. Slow-down in SRAM density scaling brings disintegration of last-level-cache and/or buffer memories as a stacked die on logic.
Integration enablement for SRAM-cache applications	Bitcell scaling is slowing down because of the slow-down of the device pitch and gate pitch. New device schemes such as N-over-P stacked device (CFET) bring an opportunity to significantly reduce the SRAM area. This is enabled because of optimized layouts that eliminate the critical design rules impacting the area. Option of embedded NVM in high-performance logic. Being able to integrate most of emerging memories (e.g., MRAM) at the interconnect stack also brings an opportunity for high-density memories. However, the stack as well as the materials should be compatible with BEOL.
Interconnect scalability	Maintaining control of interconnect resistance and electromigration (EM) and time-dependent dielectric breakdown TDDB limits. Interconnect resistance has now entered an exponential increase regime because of non-ideal scaling of the barrier for Cu and increased scattering at the surface and grain-boundary interfaces. Therefore, there is a need for new barrier materials and high-aspect ratio (AR) Cu alternative solutions. In addition to resistance scalability, TDDB is putting a limit on the minimum space between the adjacent lines for a given low-κ dielectric.

Table ES-11. More Moore Long-term Challenges

Long-Term Challenges: 2029-2039	**Description**
Power scaling	Power scaling — no solutions left to reduce the supply voltage because of lack of steep-subthreshold devices. Therefore, new power delivery schemes such as backside power to further scale down the power supply voltage by reducing the supply noise. New architectures are necessary to attain performance through parallelization and increased memory capacity, and bandwidth are required to reduce the amount of less energy-efficient external memory accesses.
Thermal issue due to increased power density	Thermal challenges of 3DVLSI and 3D stacking. Gate-all-around (GAA) and CFET (complementary FET) devices have limited heat conductance due to confined architecture. Increased pin density due to aggressive standard cell height scaling and increased drive by stacked devices put a significant pressure on the power density.
Cost reduction with 3D integration	Managing cost, yield, and process complexity of 3D integration. Using vertical devices separated by the interconnect significantly increases the wafer cost and the number of masks (i.e., process complexity) adding pressure to the defectivity (e.g., D0) control. Architecture needs to be refined for reducing the interconnect complexity between tiers as well as simplified integration and function per tier (e.g., I/O in one tier, SRAM in another tier, etc.).
Integration of non-Cu metallization to replace Cu	Adoption of high-AR non-Cu interconnects for low-resistance, meeting EM/TDDB, and temperature budget compatibility with devices used in 3D integration. Introduction of backside wafer processing for power distribution will require careful assessment of material and thermal considerations as this metallization will be near logic devices.

Interconnect Technology Requirements

Transistor performance has always been at the center of the attention of the technology observers as new shorter channel transistor equated to faster products, but it was the invention introducing the interconnections among transistors that made the Integrated Circuit possible. As more transistors are integrated in a single die more interconnections are required to make the IC functional.

After the replecement of aluminum with copper and interconnections and after the replacement of silicon dioxide with low k dielectric materials the most difficult challenge for interconnects nowadays is the introduction of yet new materials that meet the wire conductivity requirements, reduce dielectric permittivity, and meet reliability requirements. As for wire conductivity, the impact of cross section reduction absolutely necessary to meet higher density requrements increases the total line resistance increasing signal propagation delay and this effect must be mitigated. New technology are needed!

Table ES-12. More Moore—Interconnect Roadmap

YEAR OF PRODUCTION	2024	2025	2027	2029	2031	2033	2035	2037	2039
	G48M24	G48M22	G48M22	G46M20	G44M18	G44M16	G42M14/T2	G42M14/T4	G42M14/T6
Logic industry "Node Range" Labeling	"3nm" Enhanced	"2nm"	"1.4nm"	"A10 eq"	"A7 eq"	"A5 eq"	"A3.5 eq"	"A2.5 eq"	"A1.8 eq"
Fine-pitch 3D integration scheme	Stacking	Stacking	Stacking	Stacking	CFET	CFET	3DVLSI	3DVLSI	3DVLSI
Logic device structure options	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET-3D	CFET-3D	CFET-3D
Backside structure options	-	Backside via	Direct contact	Decap+ESD	Active devices	Active devices	Active devices	Active devices	Active devices
Platform device for logic	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET	CFET	CFET
									
FRONTSIDE INTERCONNECT TECHNOLOGY									
Number of Mx layers	2	2	2	2	2	2	2	2	2
Number of P40 layers	2	2	2	2	2	2	2	2	2
Number of P80 and <P720 layers	8	9	10	11	12	12	12	12	12
Number of P720 layers	2	2	2	2	2	2	2	2	2
Number of wiring layers - M1+Mx+ >P40	15	16	17	18	19	19	19	19	19
Mx - tight-pitch interconnect resistance (Ohms/um)	300	475	475	475	475	475	475	475	475
Mx - tight-pitch interconnect capacitance (aF/um)	270	270	270	270	270	270	270	270	270
Vx - tight-pitch interconnect via resistance (Ohms/via)	50.0	53.0	64.0	64.0	64.0	64.0	64.0	64.0	64.0
MP80 - 80nm pitch interconnect resistance (Ohms/um)	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.3
MP80 - 80nm pitch interconnect capacitance (aF/um)	198	198	198	198	198	198	198	198	198
VP80 - 80nm pitch interconnect via resistance (Ohms/via)	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
Aspect ratio - M0, M1, Mx, MP80, MP720	1.5-2.5	1.5-2.5	1.5-4.0	1.5-6.0	1.5-6.0	1.5-6.0	1.5-6.0	1.5-6.0	1.5-6.0
Power rail layer	M0	M0	Backside	Backside	Backside	Backside	Backside	Backside	Backside
Power rail material	Co, W, Ru	W, Ru	W, Ru	W, Ru	W, Ru	W, Ru	W, Ru	W, Ru	W, Ru
Metallization - M0	Co, W, Ru	Co, W, Ru	Co, W, Ru	Co, W, Ru	Co, W, Ru	Co, W, Ru	Co, W, Ru	Co, W, Ru	Co, W, Ru
Barrier - M0	0.5nm TaNCo	0.5nm TaNCo	Barrierless	Barrierless	Barrierless	Barrierless	Barrierless	Barrierless	Barrierless
Metallization - M1, Mx	Cu	Cu	Cu	Cu, Ru, Mo	Cu, Ru, Mo	Cu, Ru, Mo	Cu, Ru, Mo	Cu, Ru, Mo	Cu, Ru, Mo
Barrier metal - M1,Mx	1.5nm TaNCo	1.0nm TaNCo	0.5nm TaNCo	Barrierless	Barrierless	Barrierless	Barrierless	Barrierless	Barrierless
Di-electrics k value - M0,M1,Mx	SiCOH (2.70-3.20)	SiCOH (2.70-3.20)	SiCOH (2.70-3.20)	SiCOH (2.70-3.20)	SiCOH (2.20-2.55) Airgap (1.0)	SiCOH (2.20-2.55) Airgap (1.0)	SiCOH (2.20-2.55) Airgap (1.0)	SiCOH (2.20-2.55) Airgap (1.0)	SiCOH (2.20-2.55) Airgap (1.0)
Metallization - >MP40	Cu	Cu	Cu	Cu	Cu	Cu	Cu	Cu	Cu
Di-electrics k value - >MP40	SiCOH (2.40-2.55)	SiCOH (2.40-2.55)	SiCOH (2.40-2.55)	SiCOH (2.40-2.55)	SiCOH (2.40-2.55)	SiCOH (2.40-2.55)	SiCOH (2.40-2.55)	SiCOH (2.40-2.55)	SiCOH (2.40-2.55)

Like in the transistor case materials solutions are hard to come and structural solutions are required to continue making progress. Therefore, it is necessary to pursue 3D integration routes such as device-over-device stacking, fine-pitch layer transfer, and/or monolithic 3D (or sequential integration). These pursuits will maintain system performance and power gains while potentially maintaining the cost advantages such as treating expensive non-scaled components somewhere else and using the best technology fit per tier functionality. 3D stacking routes should factor in know-good-die sorting and test methodologies to improve the stacking yield where wafer-to-wafer stacking would require a very yielding process on each of the stacked wafers because of testing and wafer sorting challenges.

Pursuing heterogeneity in die stacking will require careful planning how tiers are partitioned, for instance having a smaller I/O die on top logic die would require a lot of 2D routing in the logic die to fan-in the connections from the corresponding logic block on the logic tier to the I/Os in the IO tier above. However, this routing would be more complex and would introduce an area penalty in the logic tier. The overall trade-off should also include the assembly/stacking yield and additional wafer process steps such as TSV, wafer thinning, Cu pad/µBump processing.

3DVLSI can be routed either at gate or transistor levels. 3DVLSI offers the possibility to stack tiers enabling high-density contacts at the tier level (up to several million vias per mm²). The partitioning at the gate level allows IC performance gain due to wire length reduction while partitioning at the transistor level by stacking nFET over pFET (or the opposite), enabling the independent optimization of both types of transistors while enabling reduced process complexity compared to a planar co-integration, for instance the stacking of III-V NMOS field effect transistors (nFETs) above SiGe PMOS field effect transistors (pFETs). These high mobility transistors are well suited for 3DVLSI because their process temperatures are intrinsically low. There is a significant momentum on integrating device-on-device stacking (e.g., N device over P) to decouple the channel engineering (e.g., Ge channel for PMOS) for better performance. Better performance of higher tiers achieved by a freedom to select a better substrate should however factor in the potential degradation of performance due to processing them at lower temperature budget compared to the devices at the most bottom tier.

In addition, the relocation of multiple interconnects layers dedicated to power distribution to the backside of the wafer where transistors are fabricated has been reported as a viable solution to improve density and performance.

Table ES-13. 3D Stacking and Backside Metallization Technology Ground Rules

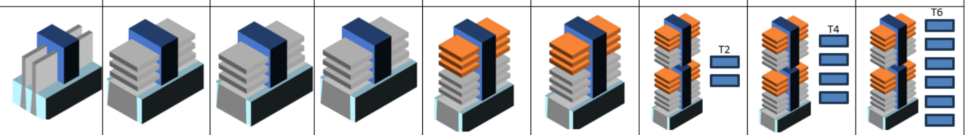
YEAR OF PRODUCTION	2024	2025	2027	2029	2031	2033	2035	2037	2039
	G48M24	G48M22	G48M22	G46M20	G44M18	G44M16	G42M14/T2	G42M14/T4	G42M14/T6
Logic industry "Node Range" Labeling	"3nm" Enhanced	"2nm"	"1.4nm"	"A10 eq"	"A7 eq"	"A5 eq"	"A3.5 eq"	"A2.5 eq"	"A1.8 eq"
Fine-pitch 3D integration scheme	Stacking	Stacking	Stacking	Stacking	CFET	CFET	3DVLSI	3DVLSI	3DVLSI
Logic device structure options	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET-3D	CFET-3D	CFET-3D
Backside structure options	-	Backside via	Direct contact	Decap+ESD	Active devices	Active devices	Active devices	Active devices	Active devices
Platform device for logic	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET	CFET	CFET
									
STACKING, BACKSIDE INTERCONNECT and 3DVLSI									
Die-to-water hybrid bonding pad pitch (um)	10	10	7	5	3	3	3	3	3
Die-to-wafer uBump pitch (um)	40	25	25	15	15	15	15	15	15
Stacking wafer thickness (um)	40	40	30	30	20	20	20	20	20
3DVLSI (layer transfer or monolithic) tier-to-tier via pitch (nm)	-	-	-	80	63	56	49	49	49
Number of backside metals	-	4	4	5	5	6	6	6	6
Backside/via contact pitch (nm)	-	120	32	27	21	19	16	16	16
First backside metal pitch (nm)	-	192	32	27	21	19	16	16	16
Backside functions	-	PDN, IO	PDN, IO, Clock	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus

Table ES-14. Critical Challenges for Interconnects

Critical Challenges	**Summary of Issues**
Materials—Mitigate impact of size effects in interconnect structures	Line and via sidewall roughness, intersection of porous low-κ voids with sidewall, barrier roughness, and copper surface roughness will all adversely affect electron scattering in copper lines and cause increases in resistivity.
Metrology—Three-dimensional control of interconnect features (with its associated metrology) will be required	Line edge roughness, trench depth and profile, via shape, etch bias, thinning due to cleaning, CMP effects. The multiplicity of levels, combined with new materials, reduced feature size and pattern dependent processes, use of alternative memories, optical and RF interconnect, continues to challenge.
Process—Patterning, cleaning, and filling at nano-dimensions	As features shrink, etching, cleaning, and filling high aspect ratio structures will be challenging, especially for low-κ dual damascene metal structures and DRAM at nano-dimensions.
Complexity in Integration—Integration of new processes and structures, including interconnects for emerging devices	Combinations of materials and processes used to fabricate new structures create integration complexity. The increased number of levels exacerbates thermomechanical effects. Novel/active devices may be incorporated into the interconnect.
Practical Approach for 3D—Identify solutions that address 3D interconnect structures and other packaging issues	Three-dimensional chip stacking circumvents the deficiencies of traditional interconnect scaling by providing enhanced functional diversity. Engineering manufacturable solutions that meet cost targets for this technology is a key interconnect challenge.
Growing gap between area-efficient signaling and power distribution in the local interconnects	Increasing trade-off between area-efficient wiring within the cell versus low-resistance power delivery to the standard cell, necessitating decoupling of power rails from the signal routing such as introduction of buried power rails.

MORE MOORE CONCLUSIONS

The semiconductor industry has evolved from IC made of planar CMOS transistors to FinFET in 2011 and it is now transitioning in 2025 to 3D GAA nanosheet transistors with vertically stacked NMOS and PMOS transistors. Looking further out stacking of multiple NMOS/PMOS couples on top of each other (CFET) is anticipated beyond 2030. New interconnects approaches gravitate around backside power and clock delivery by means of hybrid stacking. Table ES-15 shows the Overall Roadmap Technology Characteristics that presents projected attributes over the next 15 years.

Table ES-15. Overall Roadmap Technology Characteristics

2024 IRDS ORTC										
YEAR OF PRODUCTION	***2024***	***2025***	***2026***	***2027***	***2029***	***2031***	***2033***	***2035***	***2037***	***2039***
Logic device technology naming note definition [1a]	G48M24	G48M22	G48M22	G48M22	G46M20	G44M18	G44M16	G42M14/T2	G42M14/T4	G42M14/T6
Logic industry "Node Range" Labeling [2]	"3nm" Enhanced	"2nm"	"2nm+"	"1.4"	"A10-eq"	"A7-eq"	"A5-eq"	"A3.5-eq"	"A2.5-eq"	"A1.8-eq"
Platform device for logic [1b]	finFET	LGAA	LGAA	LGAA	LGAA	CFET	CFET	CFET	CFET	CFET
Patterning technology inflection for Mx interconnect	193i, EUV DP	193i, EUV DP	193i, EUV DP	193i, High-NA EUV	193i, High-NA EUV	193i, High-NA EUV	193i, High-NA EUV	193i, High-NA EUV	193i, High-NA EUV	193i, High-NA EUV
Backside structure options	Self-aligned vias	Backside via	Backside via	Direct contact		Active devices	Active devices	Active devices	Active devices	Active devices
LOGIC CELL AND FUNCTIONAL FABRIC TARGETS										
Digital block area scaling	0.74	0.74	0.74	0.74	0.55	0.26	0.26	0.13	0.08	0.08
DEFECTIVITY TARGETS										
Defectivity Poisson D0 target (defects/inch^2)	0.290	0.270	0.270	0.260	0.250	0.240	0.300	0.250	0.210	0.210
Process complexity exponent	28.0	30.2	30.2	31.4	32.5	33.2	33.6	52.1	95.2	136.3
Litho Field Size for Litho Field 26x16.5 NA>0.5 (4 chips, @ 100 mm2)	429	429	429	429	429/858	429/858	429/858	429/858	429/858	429/858
Litho Field Litho Field 26x33 NA=.33 (4 chips, @ 200mm2) [Note 15]	858	858	858	858	858	858	858	858	858	858
LOGIC DEVICE GROUND RULES										
MPU/SoC M0 1/2 Pitch (nm) [3]	12	11	11	11	10	9	8	7	7	7
Number of Backside Metals [Data from Table MM14]		4	4	4	5	5	6	6	6	6
First Backplane Contact (Via) Pitch (nm) [Data from Table MM 3_4 Excel]		120	120	32	27	21	19	16	16	16
Backplane intereconnect Pitch (nm) [Data from Table MM 3_4 Excel]		192	192	32	27	21	19	16	16	16
Backplane Functions [Data from Table MM14] [need Note>]		PDN, IO	PDN, IO	PDN, IO, Clock	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus
Gate length (nm) [4]	16	14	12	12	12	12	10	10	10	10
Number of stacked tiers [5]	1	1	1	1	1	1	1	2	4	6
Number of stacked nanosheets in one device [5]			4	4	4	3	3	3	3	3
Number of stacked nanosheets in one device (PMOS/NMOS) [5]		3	3	4	4	3	3	3	3	3
Cell height (nm) - HD	160	120	120	96	80	63	56	49	49	49
CPU frequency (GHz)	3.10	3.39	3.39	3.51	2.94	4.17	3.93	4.18	3.99	3.82
LOGIC DEVICE Electrical										
Vdd (V)	0.70	0.65	0.65	0.60	0.60	0.60	0.60	0.60	0.60	0.60
DRAM TECHNOLOGY										
DRAM Min half pitch (nm)	16.6	14	14	14.6	12.4	10.5	8	8	7	7
DRAM cell size (μm^2)	0.00165	0.00118	0.00118	0.00085	0.00062	0.00044	0.00044	0.00044	0.00035	0.00035
DRAM storage node cell capacitor voltage (V) [9]	0.45	0.45	0.45	0.43	0.40	0.4	0.37	0.37	0.33	0.33
NAND Flash										
Flash 2D NAND Flash uncontacted poly 1/2 pitch – F (nm) 2D [10a,b]	15	15	15	15	15	15	15	15	15	15
Product highest density (3D) (commerciallized) [11]	1.3	2.6T	2.6T	4T	6T	8T	8T	12T	12T	24T
Flash Product Maximum bits/cell (2D_3D) [12]	4	5	5	6	6	6	6	6	6	6
Flash 3D NAND Maximum Number of Memory Layers [13]	128-192	256-384	256-384	384-576	576-768	576-768	576-768	1024-1536	1024-1536	2048-3072
Maximum chip size (mm^2) [14]	140	140	140	140	140	140	140	140	140	140

Notes for the Overall Roadmap Technology Characteristics Table

[1a] GxxMxxTx notation refers to Gxx: contacted gate pitch, Mxx: tightest metal pitch in nm, Tx: number of tiers. This notation illustrates the technology pitch scaling capability. On top of pitch scaling there are other elements such as cell height, fin depopulation, DTCO constructs, 3D integration, etc. that define the target area scaling (gates/mm2). [Note from MM Chapter - Table MM-7]

[1b] Acronyms used in the table (in order of appearance): LGAA—lateral gate-all-around-device (GAA), CFET (Complementary Field Effect Transistor), 3DVLSI—fine-pitch 3D logic sequential integration. [Note from MM Chapter - Table MM-7]

[2] Industry labeling convention (x0.7 with respect to earlier node) showing a full PPA (performance-power-area) gain from node cycle to another where SoC area typically scales with a 0.55-0.70x factor.

[3] Horizontal local interconnect (M0) pitch. This is in most of cases the tightest metal pitch in a SoC enabling the scaling of standard cell height. M0 pitch follows a scaling factor of 0.70-0.85x. When this scaling factor combined with the other scaling factors, which are coming from the contacted poly pitch and design area scaling, and this combination provides a full PPA (performance-power-area) gain to the next node where SoC area typically scales with a factor of 0.55-0.70x.

[4] Gate Length Defined as distance between metallurgical source/drain junctions

[5] 3D fine-pitch stacking is done in two depth hierarchies. First depth hierarchy is done by the number of vertically stacked nanosheets in one device, where devices can be a NMOS or a PMOS device; or can be CFET (complimentary FET) device where PMOS and NMOS are stacked on vertically. Number of vertically stacked nanosheets and the width of nanosheet are determined to give the best performance, power, and area (PPA) scaling node-to-node. Second depth hierarchy is the number of tiers that are vertically stacked. Each tier can be formed from a single device layer

together with the interconnect layer(s) used for connecting devices to each other. For example, first tier can be computation logic, second tier can be a memory layer, third tier can be an I/O layer.

[6]. VDD is the minimum chip-level operating voltage; and in 2021, devices were dealing with higher than anticipated leakage of FinFET transistors and necessitated VDD to be at 0.75 volts, instead of the 2020 operating voltage levels of 0.70 volts

[7] The definition of DRAM Half pitch has been changed from this edition. Because of 6F2 DRAM cell, BL pitch is no more critical dimension. Calculated half pitch is use the following equation "Calculated half pitch=(Cell Area/ Call size factor)^0.5." Critical dimension for process development, the Minimum half pitch is also introduced. Currently Active area (long rectangle island shape) half pitch is the critical dimension of 6F2 DRAM.

[8] The DRAM cell size is the results of minimizing the cost based on technology.

[9] The DRAM storage node capacitor voltage must be low enough that the resulting electric field in the dielectric is within acceptable limits. Array area efficiency is the ratio of cell array area to total chip area. Hence, array area efficiency = 1 / (1 + [peripheral circuit area]/NaF2), where N is the DRAM capacity (number of bits per chip), F is the DRAM ½ pitch, and a is the cell size factor. For a = 8 (past years) , array area efficiency is estimated to be 0.63. For a=6 (Current), the array area efficiency of 0.55. For a=4 (after 2018), area efficiency will be 0.50. These assumptions are based on the same relative peripheral circuit area. But the case of 4F2 cell, the technology is not open to public at this moment.

"[10a] 2D NAND strings consist of closely packed polysilicon control gates (the Word Lines) that separate the source and drain of devices with no internal contact within the cell. Up to now this uncontacted word line pitch is still the tightest in all technologies. However, going forward, with the introduction of EUV lithography, metal half pitch may shrink below that of 2D NAND. Furthermore, 2D NAND is now a legacy technology with no further development thus the table shows constant half-pitch since 2019.

[10b] In order to gain high data bandwidth 3D NAND packs two bit lines within each cell x-y print space. These metal bit lines contact the array vertical channels in a complex and staggered fashion, resulting in tight metal (usually metal-2) pitch. Currently this tight contacted pitch is smaller than DRAM and Logic contacted metal pitches. But since 3D NAND x-y footprint is likely to stay constant, other technologies may surpass 3D NAND in metal pitch in the future."

[11] Product highest density is commonly used to gauge a technology but can be misleading. First, this density is not truly a density (e.g., how many bits per cm^2) but rather, how many bits in a product chip (e.g., 128Gb). Thus this term really means the highest number of bits one can squeeze into one commercially viable chip. Second, due to yield and form factor concerns, manufacturers often do not maximize the die size. Typically, optimized products often have die sizes 50-70% of the potential maximum possible. Thus "commercialized" is added to this category to distinguish from potentially maximal number of bits possible of a maximum size die (which is seldom commercialized).

[12] Multi-level cell (MLC) with 4 logic levels (2-bit/cell) and 8 logic levels (3-bit/cell) became routine for 2D NAND. Because of the larger device dimensions for 3D NAND, 4-bit/cell (QLC) products become possible. Although still challenging, there is now optimism about 5-bit/cell and 6-bit/cell products in the next several years. Increasing the number of storage bits in a cell is desirable because it does not cause higher manufacturing cost or area overhead. Yet, it is both technically challenging and requires difficult performance tradeoffs, such as sacrificing write endurance and access speed. Yet, it still may provide low-cost storage for many read-intensive applications.

[13] The number of 3D layers is not a unique function, as it depends on the cell 1/2 pitch, number of bits per cell, and 3D NAND technology architecture chosen. Lower number of 3D layers generally has lower bit cost, but other factors such as decoding method, speed performance, and yield also need to be considered. The number of 3D layers spans a range since the same density product may be achieved by either using a smaller 1/2 pitch, more bits per cell, and fewer layers: or larger 1/2 pitch, fewer bits per cell, and more layers. The number of layers for all 3D NAND producers have increased by approximately 2X every 3 years in the past several years. Going forward, this trend may slow down due to diminishing returns because of higher manufacturing cost and increased layer contacting overhead. Practically speaking, increasing the number of storage bits per cell, although technically just as challenging, seems to circumvent such issues.

[14] The maximum potential chip size has remained almost constant from node to node, since the x-y footprint is fixed for the dominant architecture. However, for practical products, the die size may be substantially smaller in order to get high yield. However, performance speed is also a concern. Survey of recent products seem to indicate that most commercial products have die sizes roughly 50% of the potential maximum.

[15] New Litho Field Size Line-Item Comment: Field-to-field stitching will be required for producing chips larger than the maximum field size.

9.3. LITHOGRAPHY

Even though lithography and patterning technologies have continued to advance still many challenges lay ahead. EUV exposure tools with a numerical aperture (NA) of 0.55 have been available for 2 years but production use of such tools will not start until 2027. The availability of both multiple patterning and high NA EUV will meet resolution requirements through 2029. Resolution should be extendable to meet requirements through 2033, where minimum metal half-pitch is expected to be 7 nm, and the minimum via pitch is expected to be 31 nm. However, improvements in control of defects and line edge roughness are and will remain substantial challenges. All the basics support elements ranging from light sources, in mask design, in mask materials, in photoresists, and underlayers materials will be needed. An issue with high NA EUV is the field size, which is half that of the 0.33 NA tools currently in production. To produce larger chips than the field size would allow it will be necessary to make two or more exposures for different parts of a chip and subsequently stitched them together. At present however, the industry is successfully using "fusing" chips in the package. Packaging connection pitches will need to shrink to provide improved connectivity. As an alternative, larger format EUV masks are being considered for enabling the patterning of large chips with single exposures.

Furthermore, starting in 2025, there will be metal lines patterned and used on the back of the chip to enable smaller logic unit cells. This will start with four backside layers of metal using 193 nm lithography that provide power distribution and I/O. In 2027 the backside metal will start being patterned with EUV and include more chip functions. 3D logic devices with one or more layers of devices stacked on top of each other are projected to be introduced in 2035 and after that there is little change in minimum chip dimensions. Table ES-16 summarizes most of the requirements of lithograph to 2039.

Table ES-16. Lithography Technology Requirements

Parameter	2024	2025	2027	2029	2031	2033	2035	2037	2039
MPU / Logic									
Logic device technology naming [F]	G48M24	G48M22	G48M22	G46M20	G44M18	G44M16	G42M14/T2	G42M14/T4	G42M14/T6
Logic Industry "Node Range" Labeling	"3nm" Enhanced	"2nm"	"1.4nm"	"A10 eq"	"A7 eq"	"A5 eq"	"A3.5 eq"	"A2.5 eq"	"A1.8 eq"
Logic device structure options	finFET	LGAA	LGAA	LGAA	CFET	CFET	CFET-3D	CFET-3D	CFET-3D
Backside structure options	-	Backside via	Direct contact	Decap+ESD	Active devices	Active devices	Active devices	Active devices	Active devices
MPU/ASIC Minimum Metal ½ pitch (nm)	11.5	11.0	11.0	10.0	9.0	8.0	7.0	7.0	7.0
Metal CD control (3 sigma) (nm) [B]	1.7	1.7	1.7	1.5	1.4	1.2	1.1	1.1	1.1
Contacted poly half pitch (nm)	24	24	24	23	22	22	21	21	21
Physical Gate Length for HP Logic (nm)	16	14	12	12	12	10	10	10	10
Gate LWR (nm) [C]		1.4	1.2	1.2	1.2	1.0	1.0	1.0	1.0
Gate CD control (3 sigma) (nm) [B]	1.6	1.4	1.2	1.2	1.2	1.0	1.0	1.0	1.0
Overlay (3 sigma) (nm) [A]	2.3	2.2	2.2	2.0	1.8	1.6	1.4	1.4	1.4
Metal CDU (nm)	1.8	1.7	1.7	1.5	1.4	1.2	1.1	1.1	1.1
Metal LER (nm) [C]	1.3	1.2	1.2	1.1	1.0	0.8	0.7	0.7	0.7
MPU/ASIC finFET fin minimum 1/2 pitch (nm)	12								
FinFET Fin width (nm)	5.0								
Fin CD control (3 sigma) (nm) [B]	0.50								
FIN LER (nm) [C]	0.35								
Lateral finFET/Gate All Around (LGAA) pitch	24.0	26.0	24.0	24.0	22.0	22.0	22.0	22.0	22.0
LGAA minimum width	7.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
Fin/GAA/CFET CDU (nm) - 10%	0.5	0.7	0.6	0.6	0.6	0.6	0.6	0.6	0.6
GAA LWR (nm) [C]	0.5	0.7	0.6	0.6	0.6	0.6	0.6	0.6	0.6
MPU/ASIC minimum contact hole pitch (nm)	48	48	48	46	44	44	42	42	42
Contact CD (nm) - finFET, LGAA	20	22	26	24	24	26	24	24	24
MPU/ASIC minimum Via pitch (nm) [E]	47	45	45	45	40	36	31	31	31
MPU/ASIC Via half pitch	23	22	22	22	20	18	16	16	16
MPU/ASIC minimum contact hole or via CDU (nm)	3.0	3.3	3.3	3.4	3.0	2.7	2.3	2.3	2.3
Backside Lithography									
Number of backside metals	-	4	4	5	5	6	6	6	6
Backside/via contact pitch (nm)		120	32	27	21	19	16	16	16
First backside metal pitch (nm)		192	32	27	21	19	16	16	16
Backside functions		PDN, IO	PDN, IO, Clock	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus	PDN, IO, Clock, Bus
DRAM									
Dram Minimum 1/2 pitch (nm)	15.5	13.0	14.0	11.5	10.0	7.0	7.0	7.0	7.0
CD control (3 sigma) (nm) [B]	1.6	1.3	1.4	1.2	1.0	0.7	0.7	0.7	0.7
Mininum contact/via after etch (nm) [D]	16	13	14	11.5	10.0	7.0	7.0	7.0	7.0
Minimum contact/via pitch(nm)[D]	47	39	42	35	30	21	21	21	21
Overlay (3 sigma) (nm) [A]	3.1	2.6	2.8	2.3	2.0	1.4	1.4	1.4	1.4
Other Memory									
Other memory ciritical dimensions and patterning specifications are large compared to logic or DRAM									
Chip size (mm ^{2})									
Maximum exposure field width (mm) [E]		26	26	26	26	26	26	26	26
Maximum exposure field length, i.e. scanning direction (mm) [E]		16.5	16.5	16.5/33	16.5/33	16.5/33	16.5/33	16.5/33	16.5/33
Maximum field area printed by exposure tool (mm ^{2})		429	429	429/858	429/858	429/858	429/858	429/858	429/858

LITHOGRAPHY AND MULTI-CHIPS

Throughout the history of ICs there has never been a direct connection between lithography and packaging. The former is utilized 40-60 times during a manufacturing process in a super-cleanroom. Packaging is utilized once when all the processing steps have been completed, and the wafers are sent to completely different location from the wafer manufacturing and quite often in a different continent where labor cost is more beneficial. However, in the past 5 years all of this has completely changed.

The current IRDS roadmaps shows that increases in system performance will now be driven not only by shrinking of dimensions and more compact unit cell designs, but also event more so by delivering modules that integrate multiple chips or chiplets in a single package. The continuation of the trend of increasing transistor count for GPUs will require to "fuse" multiple chips, interconnected with 2.5D or 3D integration, to perform the computation. By this means by 2030 a multichip GPU will have more than 1 trillion transistors and may be made of up to 8 dice operating as one GPU.

Integration of many chips can be accomplished using interposers. Interposers are used by TSMC (CoWoS) and others to integrate high-performance chips for AI and other applications. Today, such interposers consist of 3.3 reticle fields. However, large-format interposers 5.5 and even 9.5 reticle size will soon appear and eventually System on Wafer interposer will cover the full wafer[1.] Interconnection density between chiplets will also improve as time goes on. Adaptable patterning will enable tighter pitches and elimination of bonding pads, bringing benefits in chip-chip communication bandwidth and energy efficiency. Increases in device density in logic chips, increases in required data rates and decreases in minimum line pitches will drive proportional increases in packaging bump density and redistribution line density. Current large field projection exposure systems are specified to print down to 2 mm lines and spaces or a little less and have throughputs on the order of 60 wph. The flatness of the substrates will affect the suitability of these tools. Silicon wafers are certainly flat enough and it is unlikely that other types of substrates may work as well. Furthermore, it is expected that more circuits will be included in the interposer so silicon remails the best choice.

See the 2024 Executive Packaging Tutorials (EPT) on the IRDS website and the "Love and Hate" section below in this document for more details.

9.4. COMPUTER ARCHITECTURE AND SEMICONDUCTOR TECHNOLOGY: A PERFECT LOVE/HATE RELATIONSHIP

THE RISE AND FALL OF BIPOLAR TRANSISTORS IN HIGH-PERFORMANCE COMPUTING

Bipolar transistors dominated high-speed computing due to their superior electron mobility and faster switching capabilities. This advantage made them the preferred choice for high-performance computing systems, including IBM's mainframes and supercomputers. However, the major drawback of bipolar transistors was their significant power consumption, primarily due to their static power dissipation and higher leakage currents compared to CMOS.

By the mid-1990s, the power consumption of bipolar-based processors had reached unsustainable levels. For instance, IBM's 3090 mainframe, released in the late 1980s, consumed hundreds of kilowatts, requiring complex cooling systems. The power density of bipolar processors approached 30-50 W/cm², leading to severe thermal management challenges. Beyond this range, further increases in clock speed became impractical due to excessive heat generation and power dissipation. This marked the tipping point where large computer manufacturers had to abandon bipolar technology in favor of CMOS, which offered lower power consumption while still improving performance through advances in process scaling and microarchitecture.

The Transition to CMOS and Its Success in PCs

Unlike bipolar transistors, CMOS technology exhibited significantly lower power consumption because it only drew power during switching events. This efficiency made CMOS ideal for personal computers, where power and heat dissipation were more constrained. Throughout the 1980s and early 1990s, PCs leveraged CMOS microprocessors, which allowed for steady performance improvements while maintaining reasonable power envelopes. CMOS technology became the standard for both PCs and large computing systems by the late 1990s, with companies such as Intel and AMD focusing on increasing clock speeds to enhance performance.

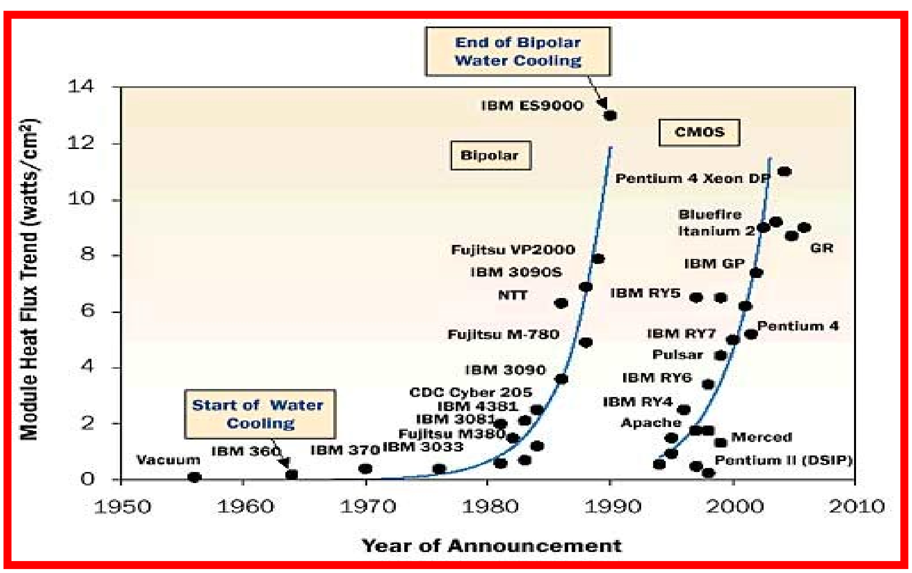


Figure ES-40. Transition from bipolar to CMOS technology

The PC Era had benefitted from the close association between Intel hardware and Microsoft software. As the OS became larger the MPU operated at progressively higher frequencies that masked the increased and raising complexity of software. Transistor speed was the name of the game in the technology world.

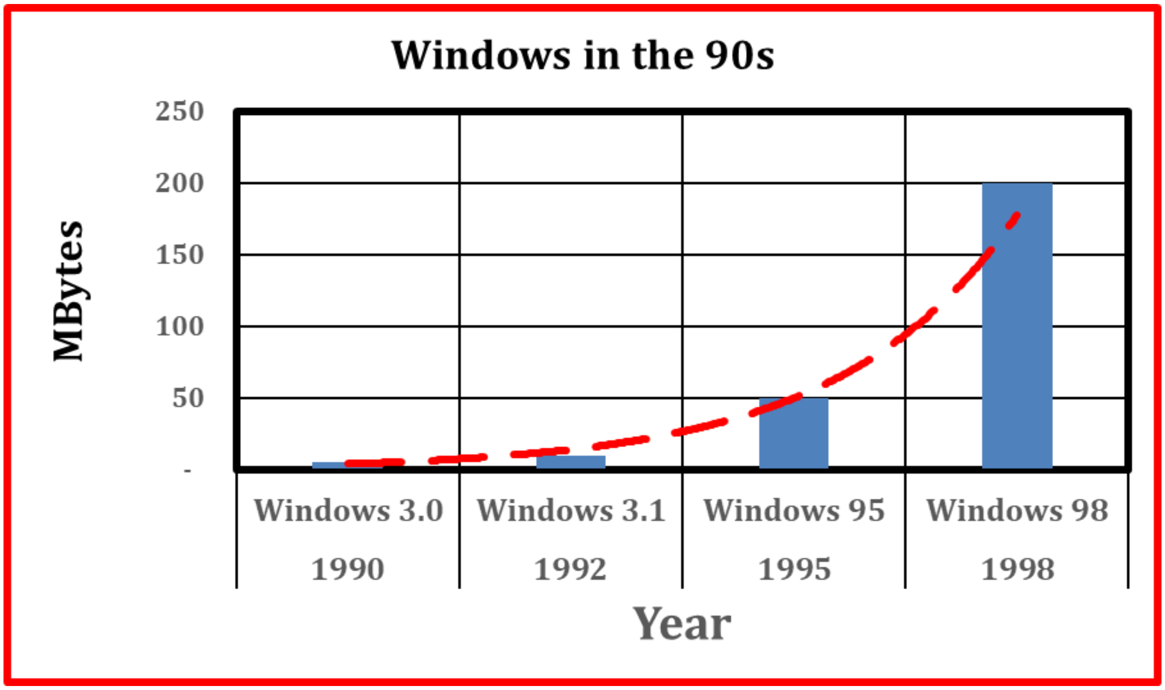


Figure ES-41. Software size increasing exponentially

Software size increased exponentially in the 90s. Frequency of MPUs increased rapidly to keep pace with the increases in software complexity.

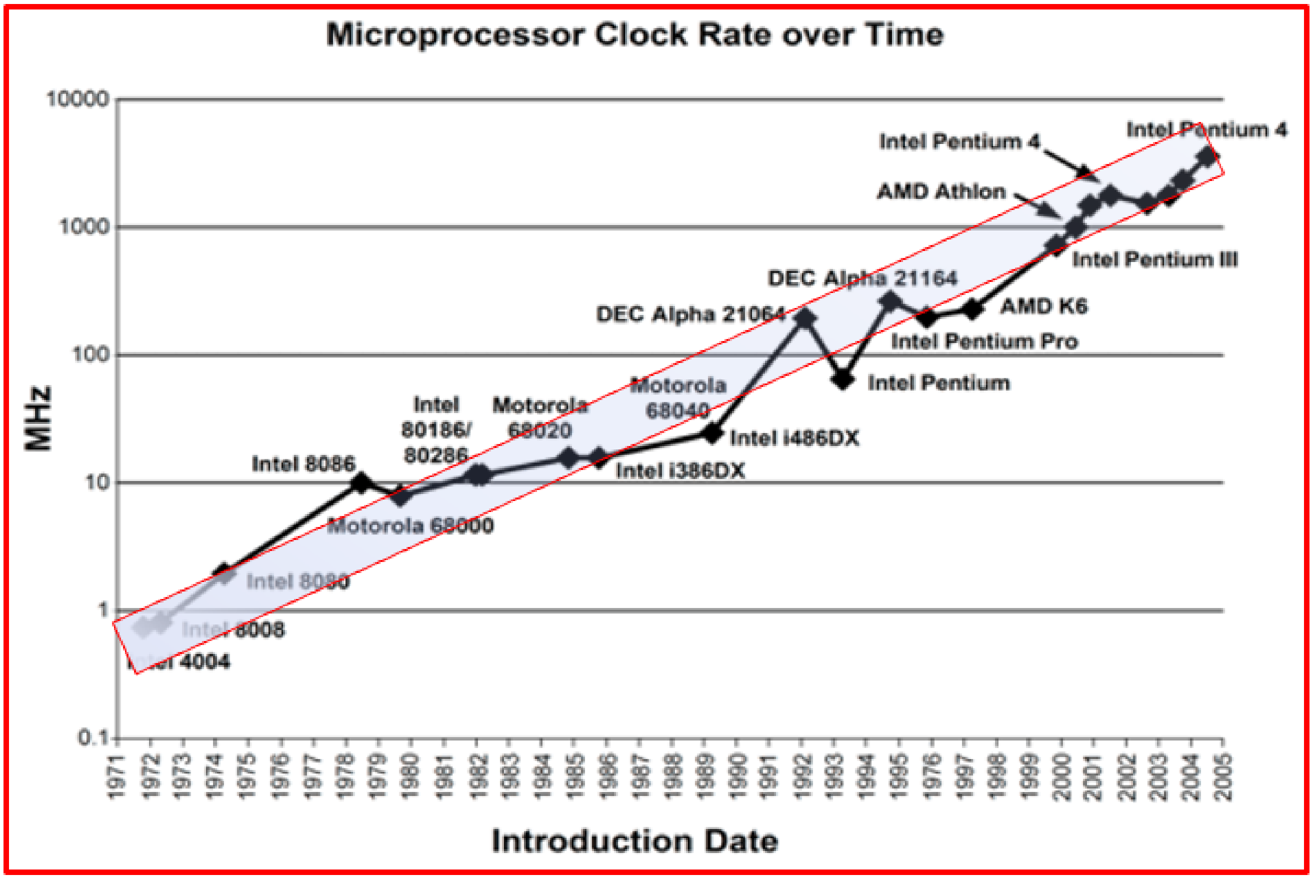


Figure ES-42. MPU frequency trend

This came to an end when Prescott reached the 115W power consumption and from then on, the frequency had to be limited to the 5-6GHz maximum range. Under these conditions performance improvements became to degrade below historical rate. There was however an architectural change that had been predicted as far back as 1989 that ameliorated the performance debacle. This consisted in partitioning the single core architecture of the MPU into multiple cores that could process information in parallel. This approach however suffered from a fundamental limitation since generic algorithms require both serial and parallel processing of the date and in this respect, nothing could be done to improve the serial portion of the data processing. Let's review this situation in detail.

Which one is true: Is Moore's Law dead or is Van Neumann's Architecture dead?

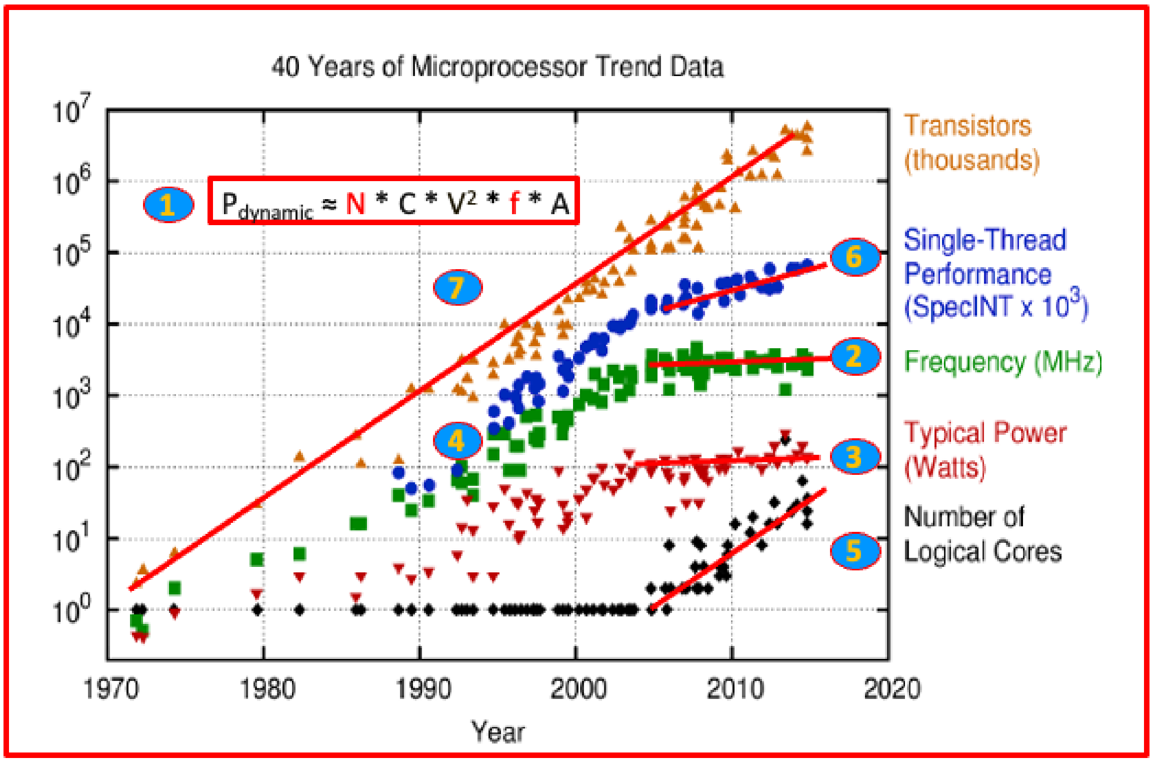
From the 70s on for 30 years MPUs were built accordingly to Van Neumann architecture and MPU performance continued to steadily improve even though software size was increasing but how could this happen? The explanation can be found by considering that the speed transistor was also increasing and so was the frequency of the MPUs: "More software, faster MPUs". Technology was like a black box for the design community that continue to design more complex system with only one penalty: power dissipation.

However, power consumption and heat dissipation began to escalate with increasing clock speeds, particularly as supply voltages remained relatively high. By the early 2000s, desktop CPUs such as Intel's Pentium 4 Prescott (2004) reached power densities of around 100 W/cm², leading to significant cooling challenges. At this point, further frequency scaling became impractical, as increasing the clock speed beyond 3-4 GHz resulted in exponential increases in power dissipation due to dynamic power ($P = CV^2f$) and leakage currents.

The Shift to Multicore Architectures

With single-core frequency scaling hitting a power wall around the mid-2000s, semiconductor manufacturers adopted multicore architectures to continue improving performance. Instead of increasing clock speeds, manufacturers began integrating multiple processing cores onto a single chip. This shift allowed for parallel processing, where multiple tasks could be executed simultaneously, albeit with diminishing returns compared to previous gains from frequency reaching higher values with each technology generations. Multicore processors, starting with dual-core and quad-core architectures, helped mitigate power density issues by distributing computational workloads across multiple lower-frequency cores. However, this approach introduced new challenges, such as software parallelization and diminishing per-core performance improvements. Despite these challenges, multicore architectures have remained the dominant strategy for improving computing performance, with modern processors featuring dozens of cores.

By the middle of the second decade of this century a new brand of computer architects had completely forgotten why frequency had been limited to the 5GHz a decade before to prevent disruption of the IC and began voicing the unfounded opinion that the decline in computing performance had to be attributed to transistors being unable to deliver switching speed at historical rate. This belief led to the next statement: "Moore's Law is dead!". This was systematically repeated in many computer-oriented conferences and panel discussions. Furthermore, the main reference was constituted by a very famous graph published in 2014 (Horowitz, Stanford) that was capturing the evolution in time of several indicators of computer performance but was this graph was looked at with "eyes wide shut."



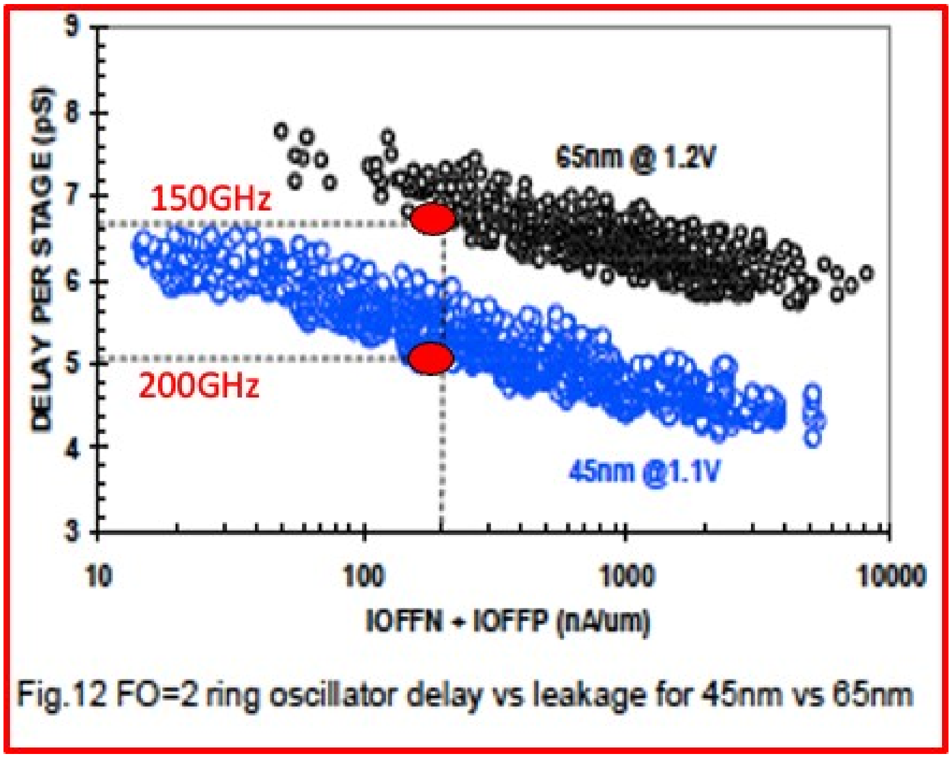
Source: Horowitz, Stanford, 2014

Figure ES-43. A comprehensive graphical overview of MPU performance indicators

It is useful to explain the elements of this graph in Figure ES-43 in some details

1. This equation connects all the contributors to dynamic power dissipation in an IC.
   a. N=Number of transistors
   b. CV^2= Energy necessary to switch a transistor's state
   c. F= Frequency of operation
   d. A=% of time transistors are in the Active state dissipating power
2. Frequency stabilized around 5GHz
3. Power limited to less than 130W
4. Historical single thread performance of MPUs
5. Number of logic cores
6. Sigle thread performance
7. Number of transistors in an IC as function of time

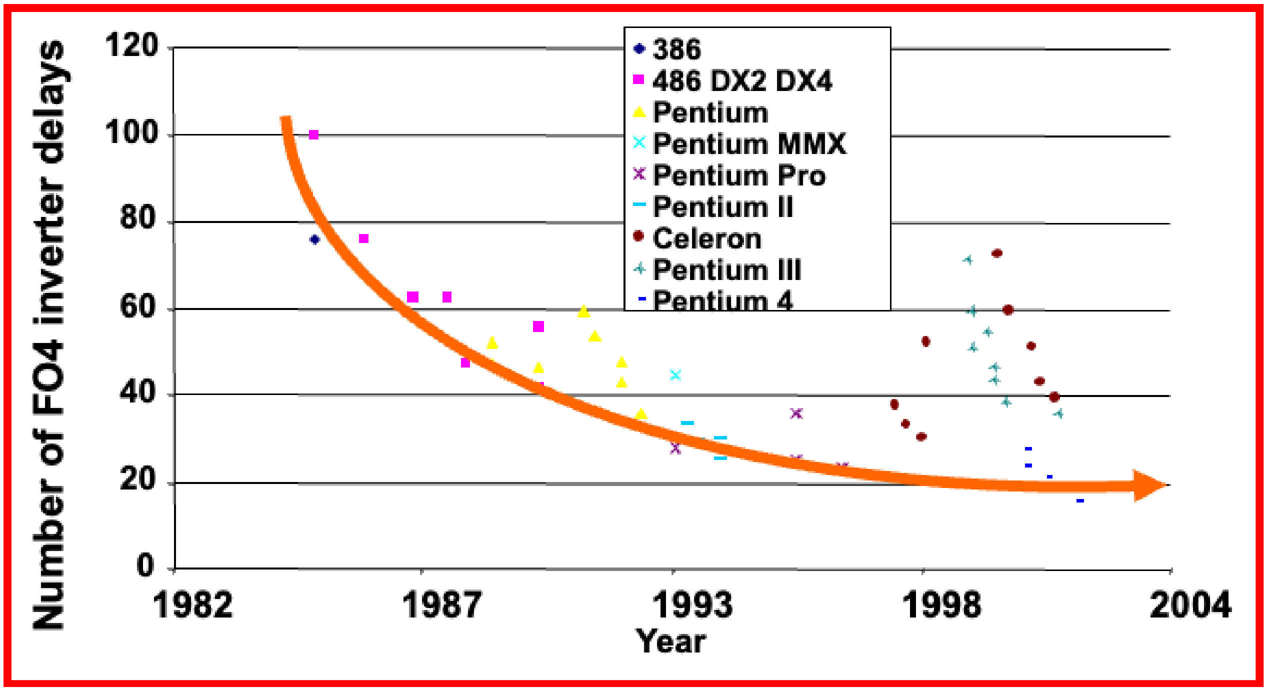
It is indeed clear from this graph that right after the year 2000 frequency of operation remained constant at around 5GHz, and power dissipation remained approximately constant also below 130W. On the other hand, the MPU performance curve deviated from a straight line established in the 80s and 90s and began trending at a rate below the historical rate. It is also however true that line number 7 representing the number of transistors continued unbated to follow the same rate of increase already established since the 70s meaning: no change! But what does this line represent? It represents nothing else, but Moore's Law so how can it be said that Moore's Law is dead? May be than it is the performance of individual transistors that indeed deteriorated?



Source: Intel

Figure ES-44. Delay time and corresponding operating frequency of transistor in 2005 and 2007

This graph shows however that typical transistor switching speed for technologies introduced in 2005 (i.e., 65nm) and 2007 (i.e., 45nm) continued to improve at historical rate well after the 2004 power crisis that had forced operating frequency to be kept constant at 5GHz. The above transistors' speeds can be translated into operational MPU frequencies by considering the standard number of logic operations to be performed in a typical MPU.



Source: Intel

Figure ES-45. Number of inverters necessary to fully operate a logic instruction leading to a result

As a result, by dividing the individual transistor speed by the number of inverters 'operations the logic signal had to go through it resulted that MPU frequencies of 7.5GHz (150GHz/20) and 10GHz (200GHz/ 20) could have been achieved in perfect continuation of past trends hadn't this caused the destruction of the IC. So, what was the real limitation preventing continuation of past trends in MPU performance?

If only a fully parallel architecture could be devised all the computational problems could go away!

BUT THE EMPEROR HAS NO CLOTHES!

The famous tale from Hand Christian Anderson tells the story of an emperor who believed to have acquired clothes from two swindlers that presumedly made his body invisible even though he was not wearing any clothes at all, but all his subjects pretended he was indeed hiding his naked body in invisibility. So, the king went parading down the streets and everybody clapped for his surprising show of invisibility-clothes until a little innocent and unbiased child exclaimed, "The emperor has no clothes!"

From 2000 on the search for a transistor that would consume less power than an MOS transistor franticly continued since with this discovery the world of computer architects could have resumed the ways of the past without having to make any adjustments, but the search was without success. Observing the figure (Horowitz) it can be noted that multicores architecture was indeed contributing to improve MPU performance since the demise of single core architecture. It was also clear that these improvements were since any parallel operation benefited from this type of multicore architecture, but inevitably serial executions did not benefit from it. So, the real question to be asked was**:"** *Could a fully parallel architecture be devised?*" This question went unanswered, and it took a group of "unbiased minds" to quite naturally pose this question and then seriously search for an answer.

The Shift

AI architecture demand processing data on an infinite number of parallel lines and so in 2009 student at Stanford discovered what appeared obvious (as usual) in retrospective. GPU were created to process fast moving graphics images by processing as many lines as possible in parallel and the AI/GPU symbiosis was so created. AI workloads demand massive parallelism, memory bandwidth, and data throughput. This was a great beginning that demonstrated that general-purpose CPUs were no longer enough. GPUs, NPUs,Tensor Processing Units (TPUs), and custom AI accelerators are now the new compute frontier.

This is precisely what a group of students at Stanford did in 2009. They discovered that not only was there indeed a branch of computer science that required a fully parallel architecture but also discovered that an IC capable of carrying on this task already existed!

2009

Large-scale Deep Unsupervised Learning using Graphics Processors

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7. Discussion

Graphics processors are able to exploit finer-grained parallelism than current multicore architectures or distributed clusters. They are designed to maintain thousands of active threads at any time, and to schedule the threads on hundreds of cores with very low scheduling overhead. The map-reduce framework (Dean & Ghe-

Figure ES-46. Students at Stanford demonstrated GPU ability to handle AI algorithms

The association between AI and GPU was proposed by students at Stanford. The breakthrough went largely unnoticed by the multitudes but in the following five years multiple companies quietly began experimenting with this new architecture that became known under the name of Artificial Intelligence or how everybody nowadays calls it "A.I."

Looking back at what happened in 2004 when the last single core MPU was produced clearly does mark an historical date when a more attentive crowd could have said:

"Von Neumann architecture is dead!"

The following figure, Figure ES-47, presented at the 2018 VLSI symposium synthetically includes on the right how AI and GPU had now finally become the new ideal architecture/technology match of the century.

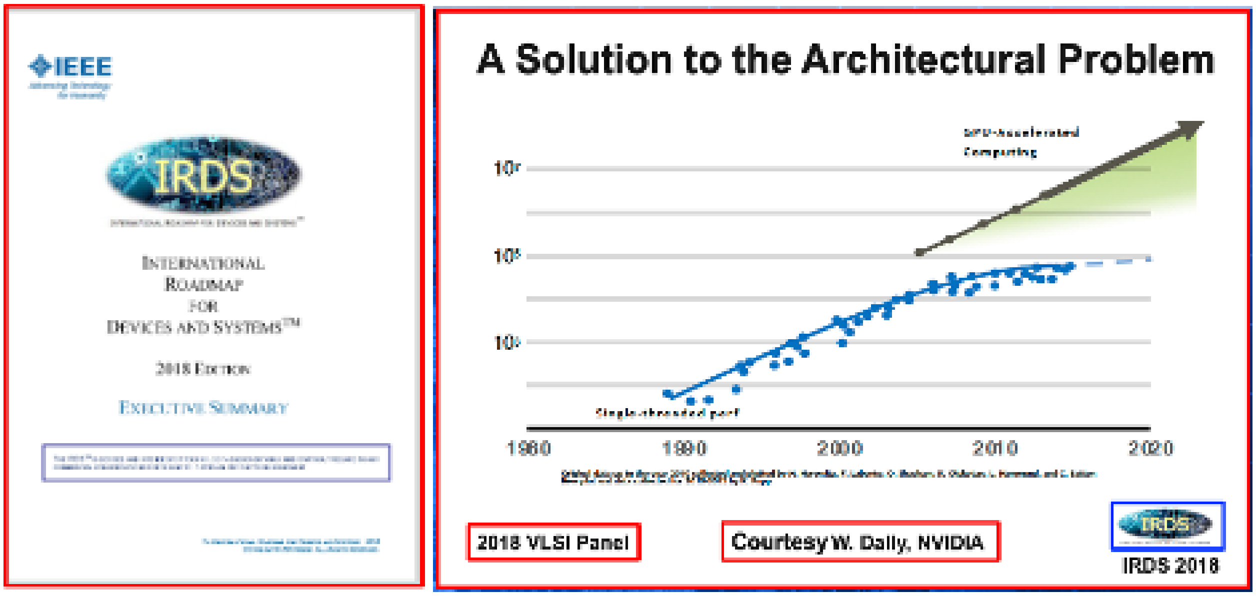


Figure ES-47. IRDS reported that GPU accelerators surpassed MPU performance

Performance of GPU accelerators resumed performance trends established by MPUs in the 90s. After an additional four years of experimentation finally a product embodying all the desired features of the AI and GPU match appeared in the market and its performance was breathtaking.

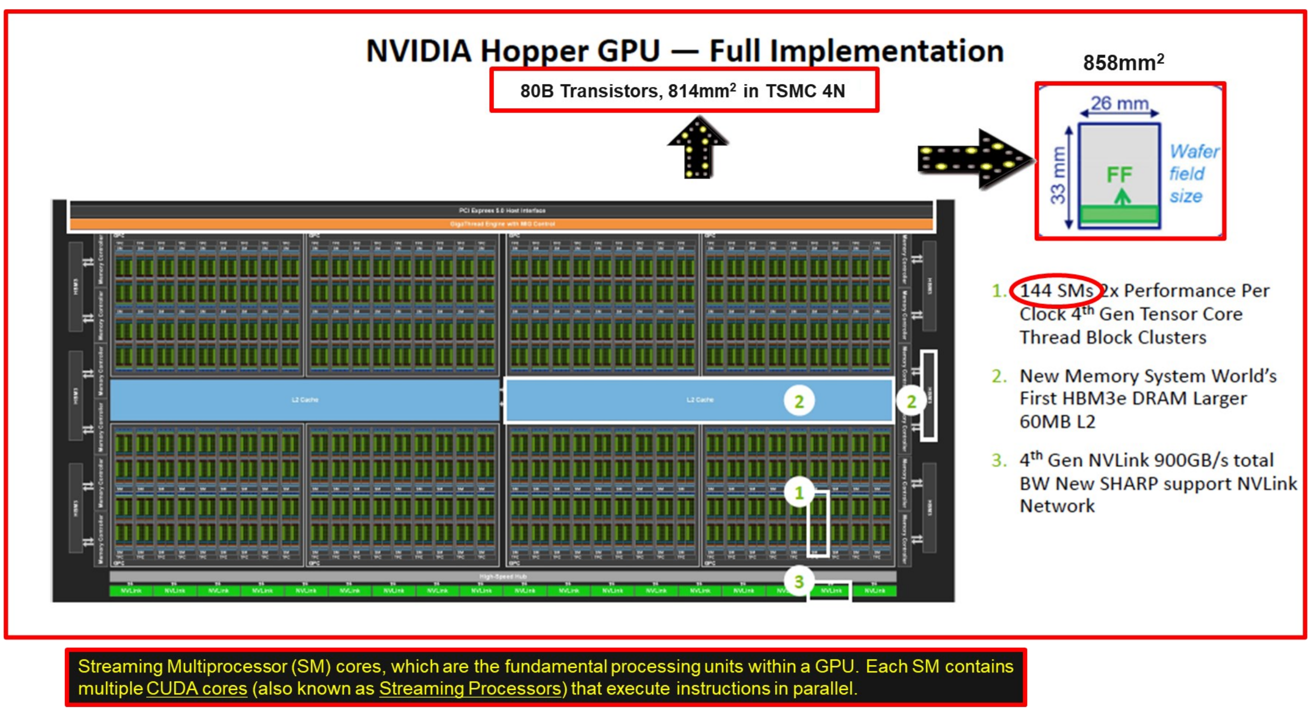
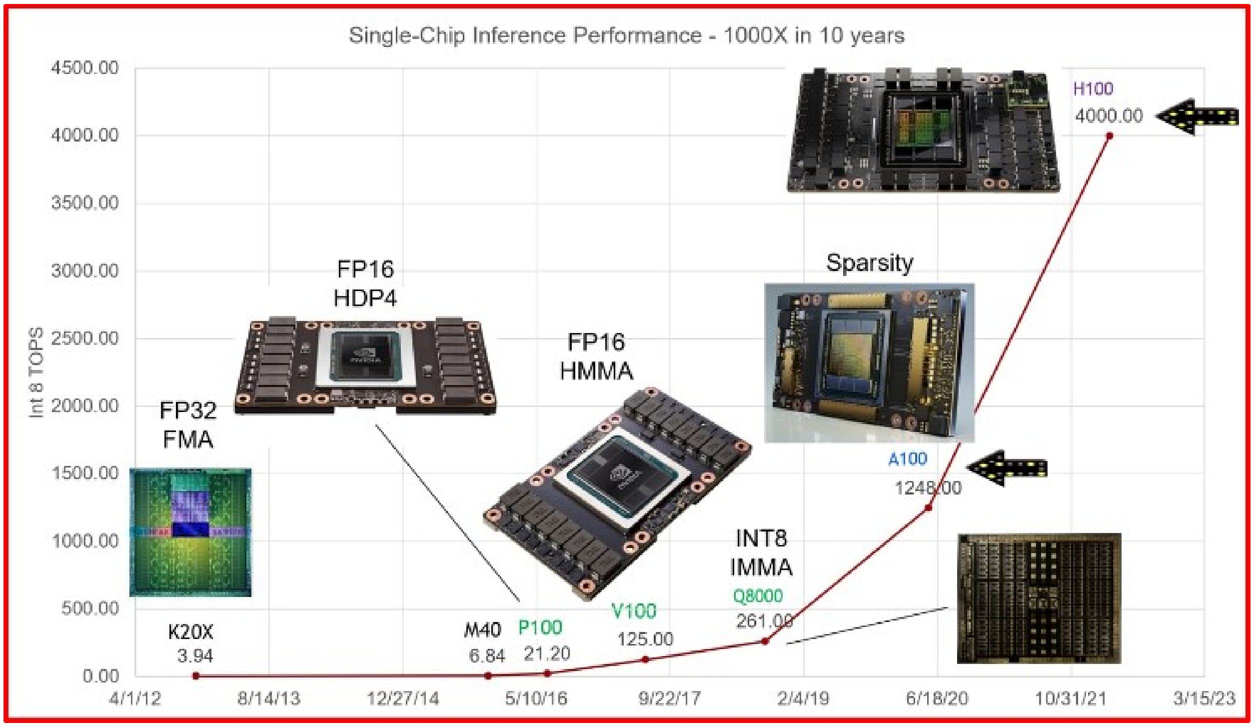


Figure ES-48. Hopper die-size came close the limits of the lithography exposure field of 858mm²



Source: NVIDIA

Figure ES-48B. H100 performance reached the 4,000 TFLOPS level!

But this is not all, as only two years later, Blackwell was introduced with performance reaching the 20,000 TFLOPS.

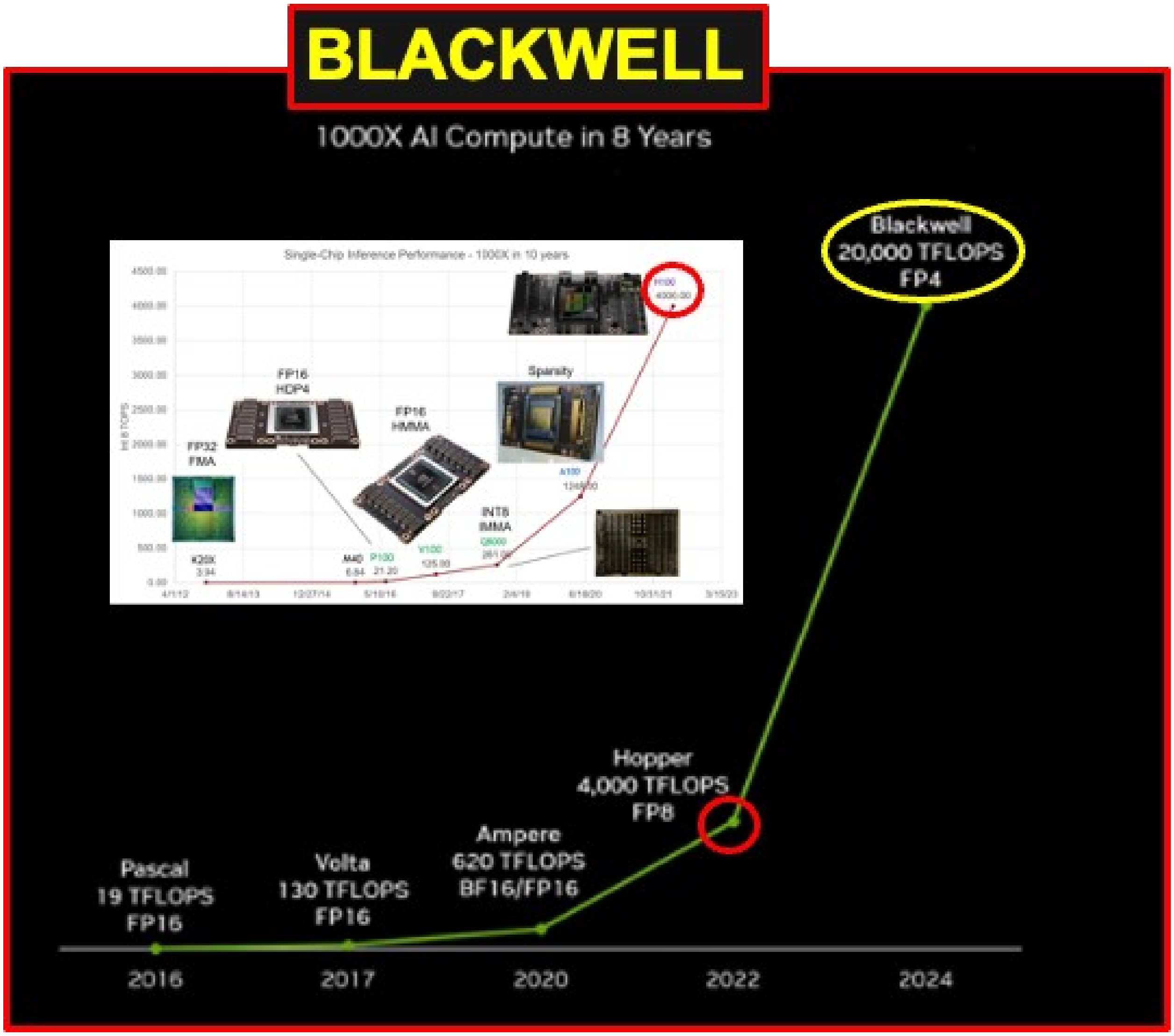
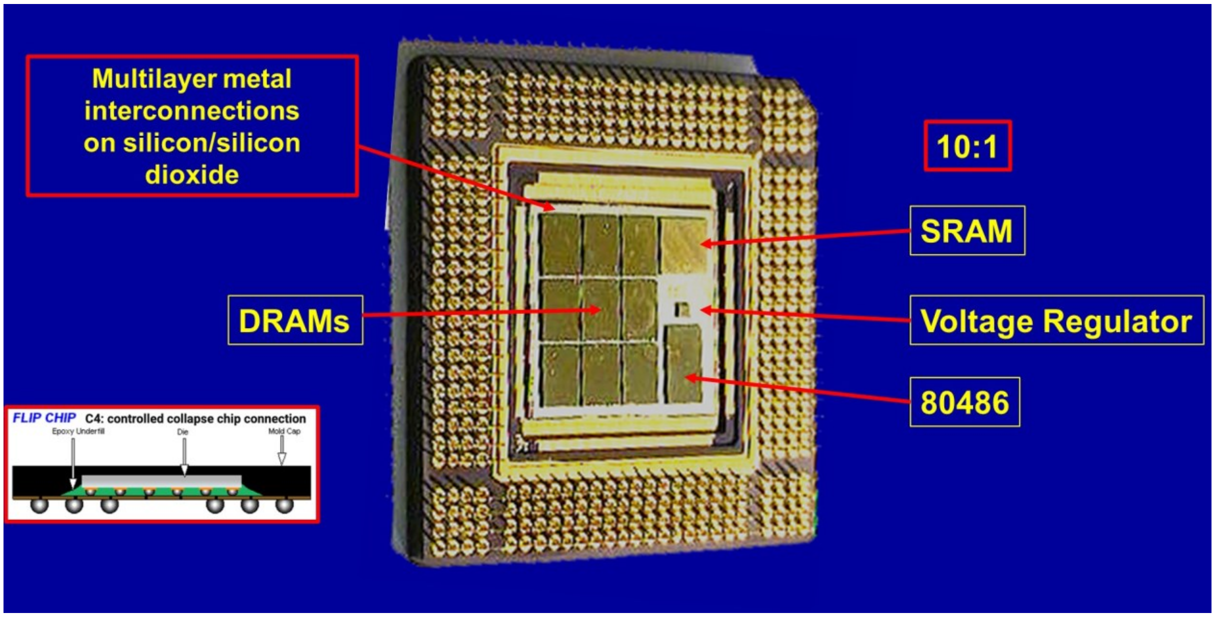


Figure ES-49. Blackwell performance reaching 20,000 TFLOPS level

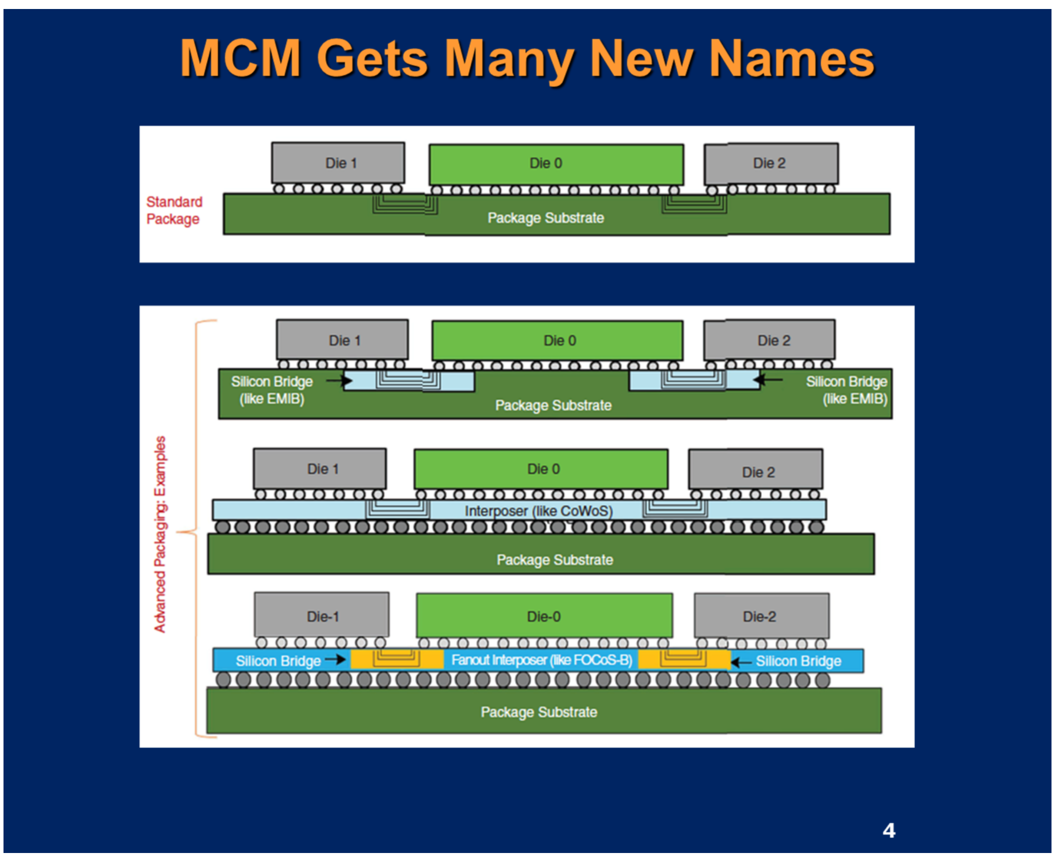
How Lithography Lost the Leadership Position in Semiconductor Technology

The progress of the semiconductor industry can be characterized by a single law enunciated by Gordon Moore in 1975 that has proven to be true since then: *"The number of transistors in an IC will double every 2-years."* So, it was to be expected since proven trends that Blackwell needed to at least double the number of transistors of Hopper. In the past similar goals had always been achieved by means of two main contributions: feature reduction from one technology to the next and increased die size. In fact, feature scaling alone was never sufficient by itself to realize the goal and a contribution from a larger die size was required. However, Hopper 80 Billion transistors had already pushed the die size ($814mm^2$) close to the lithography limits so a new technological element, beyond lithography, was absolutely require meeting the goal of doubling the number of transistors. Fortunately, such technology existed and was readily available as it had been invented in the 60s and demonstrated for MPUs in the 90s.



Source: Intel

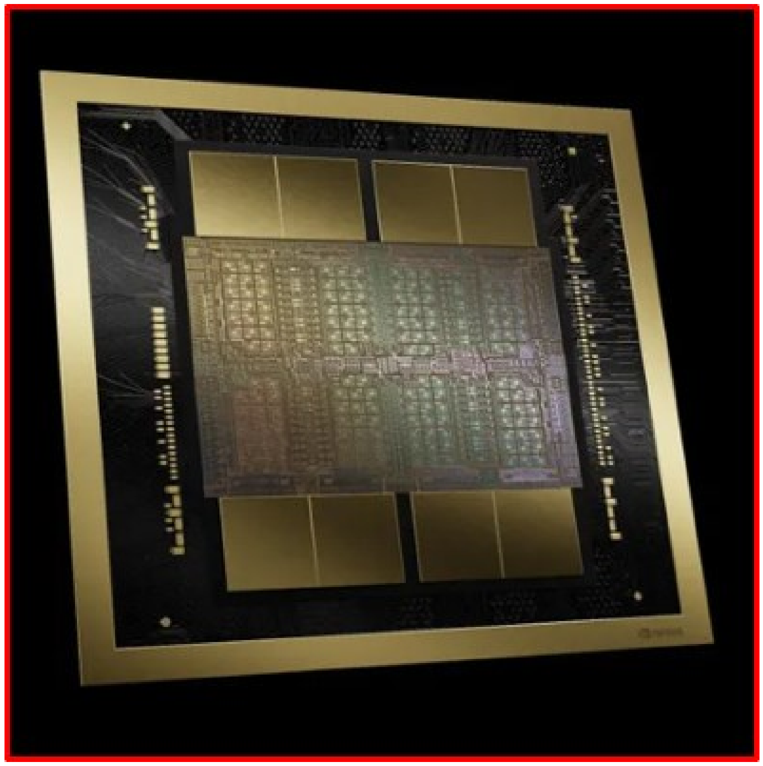
Figure ES-50. 1992-1994. Early demonstration of Multi Chip Module (MCM)



Source: Spectrum, November 10, 2022

Figure ES-51. Multiple variations of MCM at the top of the illustration

For Blackwell to reach (and exceed) the number of transistors goal it was necessary to adopt multi-chip packaging technology that was readily available. See Figure ES-56.



Source: NVIDIA

Figure ES-52. Blackwell two-die implementation

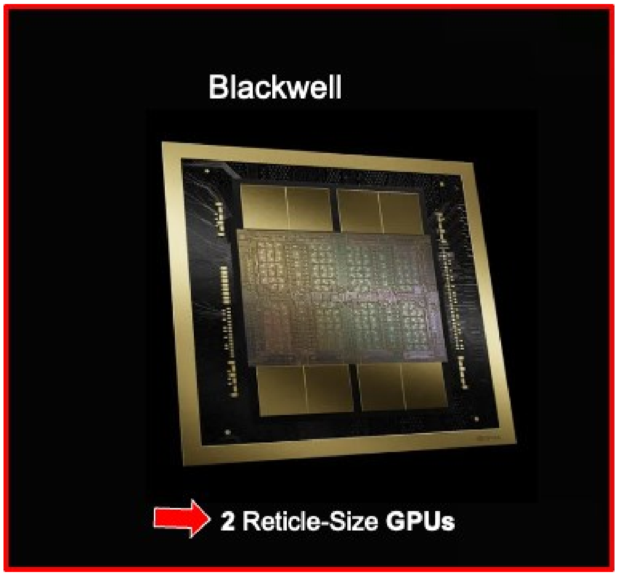
Source: NVIDIA

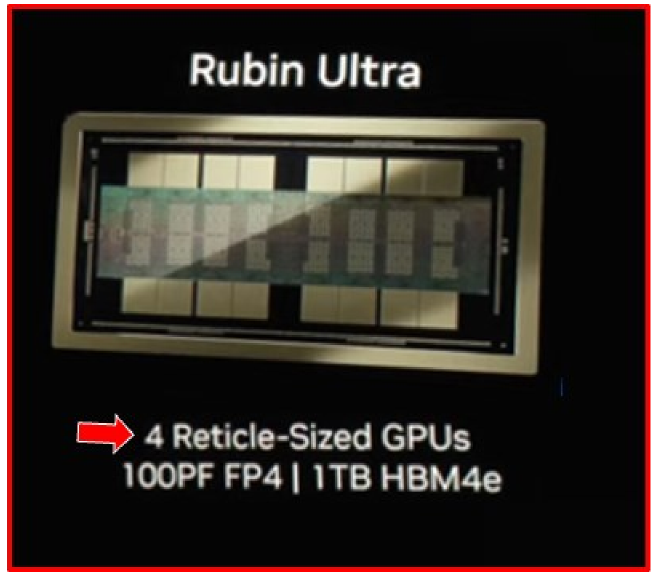
Figure ES-53. Blackwell reaches the 208 Billion transistor mark!

A Tectonic Change: Multidice Replaces Lithography at the Main Technology Enabler of Moore's Law

In the past, packaging had covered the historical role of mechanically protecting the frail die and its interconnection to the package from damage. It had then evolved as an accelerator of new technology generations. For instance, when Prescott failed the power test in 2004, and it became necessary to switch the MPU architecture to two dice configuration the task was easily archived by placing two dice in a single package while the design community was feverishly designing the single die dual core MPU solution. Once this was accomplished the single die solution replaced the two-die emergency solution as a more efficient and cost-effective solution. It was then said that packaging would accelerate new technology introduction by half-generation.

However, this is no longer the case. Present lithography capabilities cannot deliver a feature size reduction or a sufficiently large die size to enable doubling the number of transistors every two years This is not going to change in the future since this is the new norm. In fact, NVIDIA already announced that Rubin Ultra to be introduced in 2027 will consist of four dice. It is therefore expected that this trend will continues and the number of transistors connected by multidice technology will continue to grow according to this formula.





Source: NVIDIA

Figure ES-54. Reticle-Sized GPUs number increases from two for Blackwell to four for Rubin Ultra

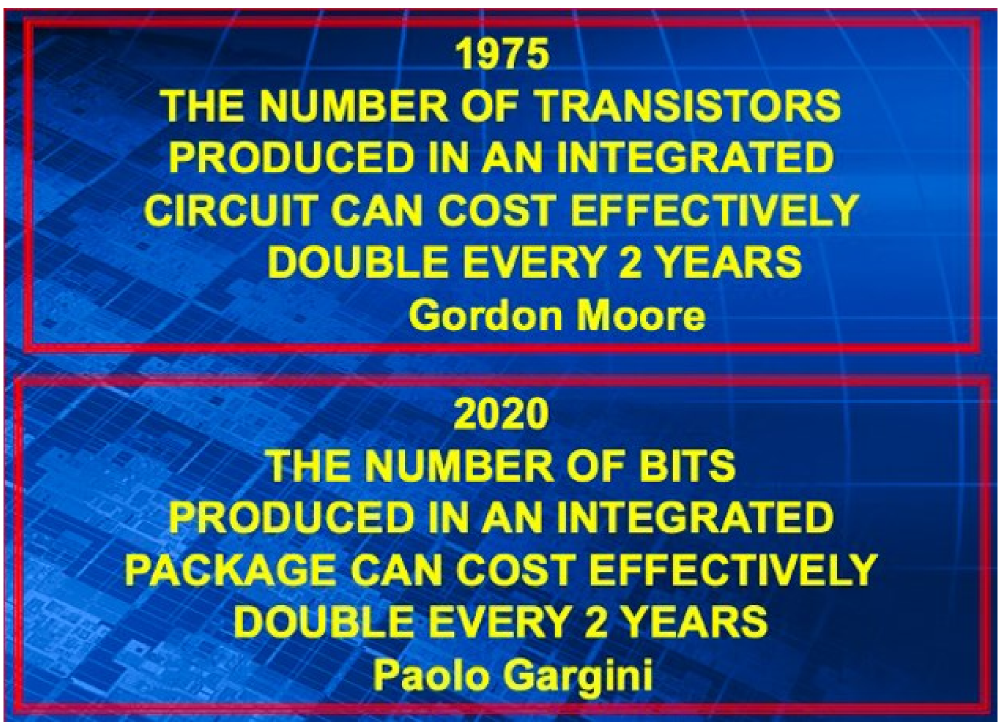


Figure ES-55. Number of transistors continues to double every two years from SOC to SIP

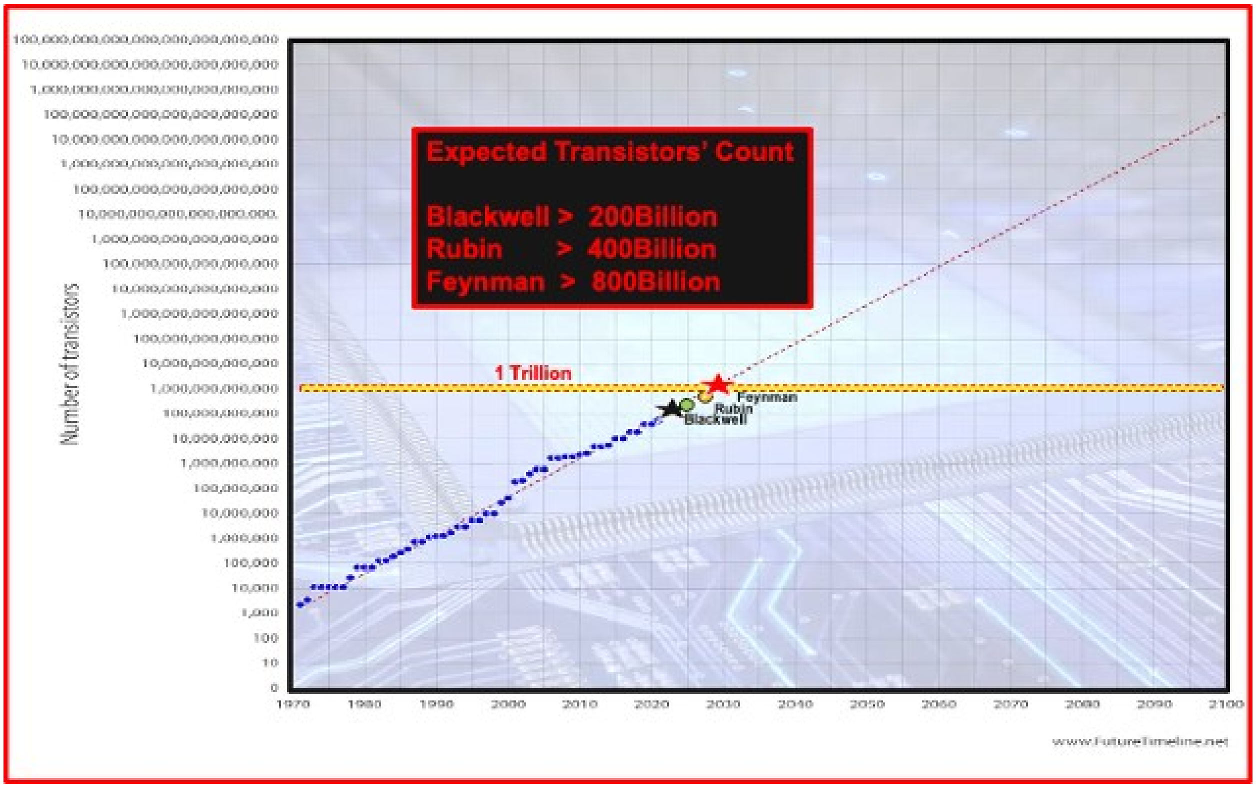


Figure ES-56. Projected number of transistors from Rubin and Feynman GPUs

In fact, it is possible to go on the limb and make the following assumption

Figure ES-57. Projected rate of increase in number of Reticle-Sizes GPUs

CONCLUSION FOR MORE MOORE

Lithography led implementation of Gordon Moore' Law by scaling down device features from one generation to the next generation and by supporting introduction of larger dice. This combination enabled doubling the number of transistors every two years. In addition, by over-scaling by means of extreme lithography the channel length of CMOS transistors made it possible to compensate for an exponential growth in software complexity by running the MPU to continuously higher frequencies. But, after Prescott reaching the 115W power dissipation it required that frequency had to be capped at about 5GHz and reducing transistor channel length was no longer viable. Die size reached the lithographic limit with Hopper and then only by combining two dice became possible to timely reach the 208 Billion transistor goal. The next corollary to enabling Moore's Law consists in doubling the number of dice every two years as supported by Blackwell and Rubin ultra. Would you guess that Feynman ultra in 2029 will have eight dice?

9.5. COMMUNICATIONS (OUTSIDE SYSTEM CONNECTIVITY (OSC))

The Internet of Everything (IoE) is continuing to expand in applications that demand larger volumes of data with faster communication. The IoE was initially defined as a wide range of Internet of Things (IoT) devices communicating with cloud computing that store data and which was analyzed with applications and actions communicated. As IoE was used for a broader range of applications, some applications had unacceptably slow performance due to the latency of communicating with the cloud. To overcome this latency limitation, some applications added local storage and processing close to the IoT devices and network, which is referred to as "Edge" computing.

Most applications will employ RF/microwave wireless communication to connect to the internet which will then connect through high-performance backhaul or fiber optical interconnects to a cloud data center. In the future, millimeter waves (mmWaves), massive multiple input multiple output (MIMO) or other 5G media will be implemented for high-speed connection to terminal devices, while low-power wide area network (LPWAN) communications, such as long-range WAN (LoRaWAN), SIGFOX, LTE Cat 0 and narrow band-IoT (NB-IoT), will be utilized to connect and provide enormous data to the cloud and/or edge computing system from IoT-edge sensor devices. Within the data center, communication to servers is through fiber optical interconnects with signals being routed through multiple routers. Upon arrival at a router, the optical signals are converted to electrical, routed and then converted back to optical signals, which adds to energy consumption and latency. The requested data is then routed out of the data center and returned to the requesting IoT devices through a path like the request path.

Applications that required fast communication and decision making, such as autonomous vehicles and traffic control, are adopting edge computing. In this model fast communications are made between the edge and vehicles and traffic flow controls and filtered data is communicated between the edge computing and the supporting cloud computing capabilities.

With the rapid growth of AI applications, high data rate internet applications and IoT, communication rates in data centers need to grow to support high speed access and provide high speed low latency communication between servers, special AI processors and memory or other servers. With increased use of high-resolution video, virtual reality, and augmented reality applications, ever higher data rate communication with lower latency is required in data centers. Communication between racks and switches is carried by optical interconnects; however, the capacity of switches is increasing faster than the capacity of fiber interconnects. Also, the power consumed by the switches in data centers is ~approximately 30% of the power consumed in the data center. To reduce switching power and overcome the fiber capacity gap, companies are working to integrate silicon photonic I/O into the switch package. There are also efforts to extend fiber into the server into packages and thus reduce power. Currently, short range (<100m) fiber communication uses VCSELs with multi-mode fiber, which has a significant cost and power advantage over single mode communication. There is an interest in employing single mode fiber communication throughout data centers. However, there are significant technical challenges with developing and implementing single mode fiber with low loss and cost for these applications.

Mobile Devices

Currently, mobile devices are supported by 4G LTE and 5G networks with theoretical download data rates of up to 10 gigabits per second (Gbps). Maximum actual 5G download data rates in South Korea were measured to be 432.5Mbps in June 2023 with data rates being lower in other countries. IoT devices communicate over a wide range of networks including multiple LPWAN, Bluetooth, Wi-Fi, and higher data rate devices over cellular 5G networks. The IoT devices communicating to Bluetooth and LPWAN networks typically communicate at data rates less than 2 Mbps, or less than ~100 Kbps in LoWA. Many of the IoT sensors and transducers operate on battery power, so they send data in burst and go into standby mode when not transmitting data. Some IoT devices such as wireless headphones must receive data continuously while operating. For mobile cellular devices, the drivers are the maximum data rate and the number of antennas to support multiple wireless technologies. The maximum theoretical data rate for cellular is currently 7Gbps and the number of antennas is 13 to support GPS, Bluetooth, Wi-Fi, 3G, and 4G LTE, 5G and other needs. Cellular data rates are targeted to increase to 10 Gbps in 2024 with the introduction of improved 5.5G wireless technology. Power amplifiers and A/D converters need to have improved energy efficiency at multi-GHz and mmWave frequencies to support 5G-6G to enable improved battery life for mobile phones at high data rates.

Outlook

Cellular communication devices will be able to automatically connect to data sources seamlessly through terrestrial and non-terrestrial networks with the required speed for the applications being used. Autonomous vehicles will be able to have continuously updated maps of traffic issues, obstacles and hazards as well as communicating with other vehicles to understand their intentions to change lanes or directions. Mobile IoT devices and systems will be able to connect to their host application on the cloud through a variety of networks without allowing security breaches of themselves, the host application or the internet or cloud. Non terrestrial networks (NTN) are being developed to enable communication from mobile phones in underserved areas with voice and the internet. These NTN's consist of either low earth orbit satellites or high-flying drones.

9.6. DATA CENTERS COMMUNICATIONS

Within the data centers, most are using fiber optic active optical cables (AOC) that take electrical data input, convert it to optical data with lasers, transmit it over fiber and then convert it to electrical output at the other end of the cable. Current AOC communicate at 400 Gbps; however, once a data center is "wired" with AOC, the maximum data rate is limited by the cabling. To connect every server in a data center, this requires a server to communicate with a switch(router) and within a large data center, data must go through approximately 6–8 switches before reaching its destination. For data centers, the drivers for communication are the data rate per server unit, which is currently 800 Gbps, and power consumption. Data centers typically upgrade servers approximately every three years and would like to have the data rates increase when they upgrade the servers; however, this would require upgrading the AOCs with every server upgrade. Power consumed in communication and switching is becoming significant, 30% of data center power in 2013, so Data Centers are requesting that optical I/O be integrated into the switch package by the 51.2 Tbps switch. The Consortium for On Board Optics (COBO) is driving to have the photonic transmitter and receiver on the board and connecting this to single mode fiber in the data center. This would enable upgrading the data center communication data rate when the servers are upgraded. At the same time, the switches (routers) in data centers are adding communication capacity and ports that consume considerable power, so the industry is seeking ways to reduce router thermal density and improve energy efficiency.

Outlook

Data centers will have high data rate communication that is upgraded when servers are upgraded without recabling the data center. Latency of communication within the data center will be reduced by all optical switching and routing; however, if latency could be dramatically reduced new data center architectures may be enabled. Power consumption of communication and routing will be reduced by a factor of 10 even though the data rate has increased by a factor of over 10.

9.7. AUTONOMOUS MACHINE COMPUTING

Embodied AI signifies a transformative convergence of AI and robotics, embedding intelligence directly into physical forms like robots and intelligent machines, which enables these systems to engage in meaningful interactions with their environments. Unlike conventional AI, which operates primarily in abstract data spaces, Embodied AI grounds itself in real-world interactions, seamlessly integrating perception, cognition, and action. These concepts were already advanced in the 90s but only in the past 10 tears practical embodiments of AI have emerged. This section traces the foundational milestones that have driven Embodied AI's progress and offers insights into its future trajectory.

A critical component in the development of advanced embodied AI systems is the creation of foundational models that can generalize across a wide range of tasks and environments. These foundational models, which are designed to learn adaptive strategies rather than isolated skills, form the core of effective Embodied AI systems. Drawing inspiration from human cognitive abilities, these models empower machines to perform context-sensitive tasks without the need for exhaustive, task-specific programming.

We propose that an ideal foundational model for embodied AI should meet several key criteria. First, it should be capable of learning from complex instructions, demonstrations, and feedback without the need for manually crafted optimization techniques. Second, it must demonstrate high sample efficiency during the learning and adaptation processes. Third, the model should be able to continuously learn from contextual information, thereby avoiding catastrophic forgetting. Given these criteria, we conclude that the meta-learning + General-Purpose In-Context Learning (GPICL) approach is more suited for Embodied AI systems. However, before committing to this approach, it is essential to examine the tradeoffs between these two strategies.

To ensure the computational efficiency required for Embodied AI applications, a new architecture is necessary. This architecture must integrate multimodal sensors, offer efficient support for core robotic functions, and enable real-time processing for model-based robotic applications. By addressing these needs, the proposed architecture will provide the foundation for next-generation Embodied AI systems capable of operating seamlessly in complex environments.

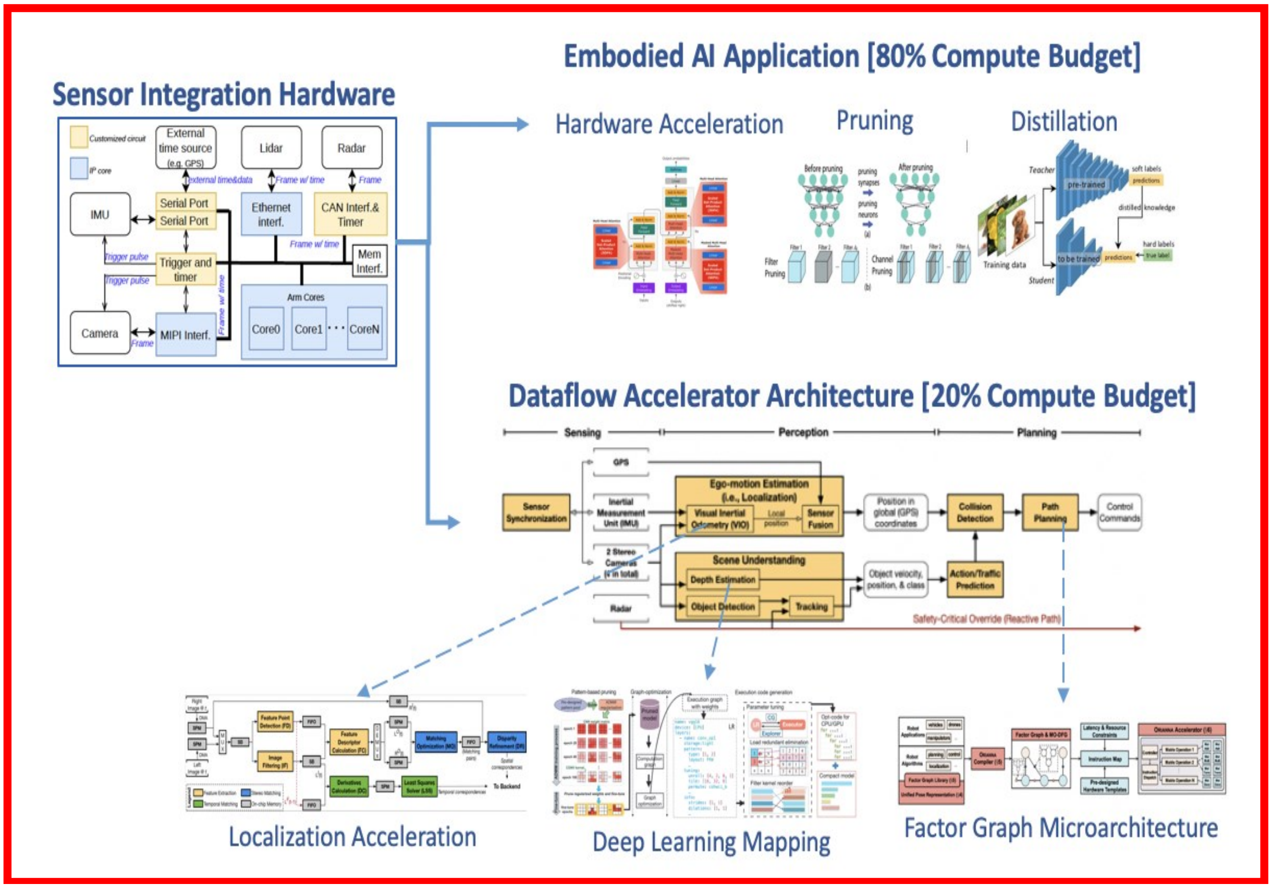


Figure ES-58. Computer Architecture for Embodied AI

Sensor Integration and Synchronization:

The effective integration and synchronization of sensors is paramount, particularly in robots equipped with multiple multimodal sensors. The quality of sensor integration directly influences the accuracy and reliability of the sensor data. To ensure precise data acquisition and integration, the hardware module must support the integration of over ten sensors and provide a unified timing source. This synchronization ensures the alignment of data streams, enabling seamless processing and accurate perception of the robot's environment.

Robotic systems typically rely on the dataflow paradigm for computational tasks. To optimize computational efficiency in embodied AI, it is essential to implement a dataflow accelerator architecture that accelerates core robotic functions.

Each embodied AI application operates as an independent AI agent responsible for understanding and executing complex tasks, often through visual language models (VLMs) or visual language action models (VLAMs).

Data serves as a crucial tool for monetization in both the internet and robotics sectors. Here, we explore the strategic value of data in Embodied AI , drawing on the internet sector as a historical benchmark.

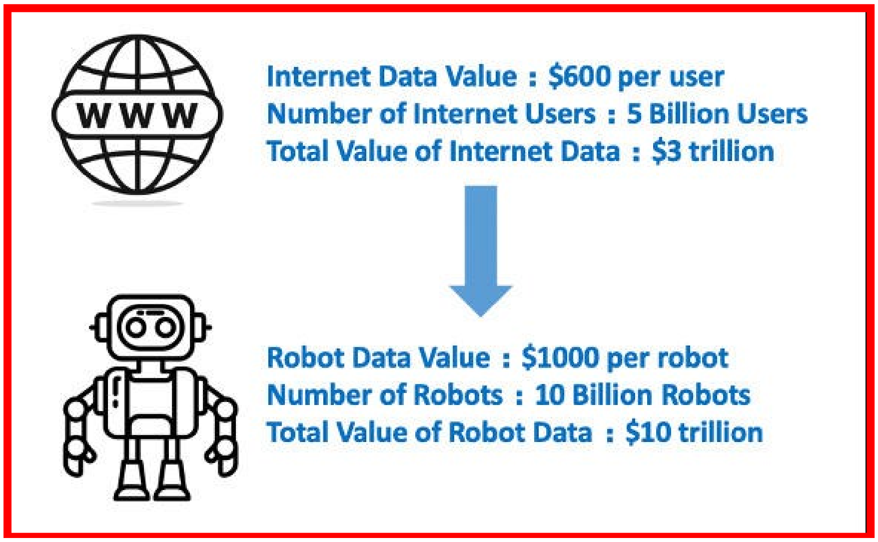


Figure ES-59. Data Monetization—Internet vs. Embodied AI

In the internet industry, companies primarily capitalize on user data for targeted advertising and personalized content delivery. This tailored approach not only drives sales but also increases user engagement, potentially leading to higher subscription fees and more frequent usage. In contrast, in the embodied AI sector, data plays a critical role in training deep learning models that enhance and optimize the capabilities of robotic systems.

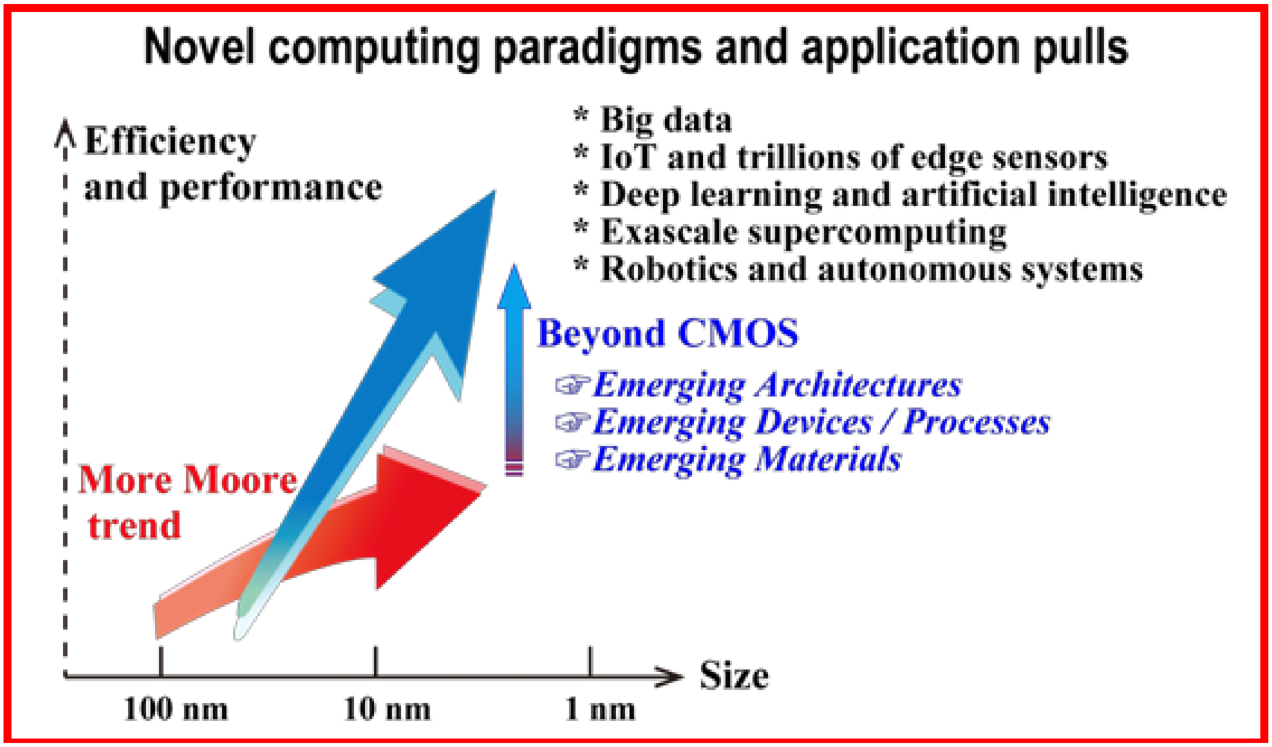
Assuming market saturation with over 10 billion robots and considering the estimated cost of $35,000 per robot post-commercialization, we conservatively estimate that robotic companies would allocate around 3% of the robot's cost towards data collection and generation. This investment would be aimed at advancing embodied AI capabilities, leading to an estimated market value of embodied AI data surpassing $10 trillion—three times the value of internet data.

Outlook

The potential of embodied AI to reshape industries, enhance human-robot collaboration, and provide societal benefits is vast. By advancing foundational models, robust computing systems, and data-driven frameworks, embodied AI is poised to unlock new levels of machine autonomy and adaptability. However, with these advancements come challenges, including data privacy concerns, energy consumption of embodied AI systems, regulatory requirements, and ethical implications.

9.8. HIGHLIGHTS OF BEYOND CMOS

An overarching goal of this chapter is to survey, assess, and catalog viable emerging devices and novel architectures for their long-range potential and technological maturity and to identify the scientific/technological challenges gating their acceptance by the semiconductor industry as having acceptable risk for further development. This chapter also surveys beyond-CMOS devices for more than Moore (MtM) applications.



(Courtesy of Japan beyond-CMOS Group)

Figure ES-60. Relationship of More Moore, Beyond CMOS, and Novel Computing Paradigms and Applications

The relationship between these domains is schematically illustrated in Figure ES-60. Novel computing paradigms and application pulls (e.g., big data, Internet of Things (IoT), artificial intelligence, autonomous systems, exascale computing) introduce higher performance and efficiency requirements, which is increasingly difficult for the saturating More Moore technologies to fulfill. Beyond-CMOS technologies may provide the devices, processes, and architectures needed for the new era of computing.

DEVICE TECHNOLOGIES

One challenge is the need of a new memory technology that combines the best features of current memories in a fabrication technology compatible with CMOS process flow and that can be scaled beyond the present limits of SRAM and FLASH. This would provide a memory device fabrication technology required for both stand-alone and embedded memory applications. The ability of a chip to execute programs is limited by interaction between the processor and the memory, and scaling does not automatically solve this problem. The current evolutionary solution is to increase cache memory, thereby increasing the floor space that SRAM occupies on a chip. However, this trend eventually leads to a decrease of the net information throughput. Volatility of semiconductor memory requires external long-term storage media that tend to be slow to access (e.g., magnetic hard drives, optical CD, etc.). Therefore, development of electrically accessible non-volatile memory with high speed and high density would initiate a revolution in computer architecture. This development would provide a significant increase in information throughput beyond the traditional benefits of scaling.

Another challenge is to sustain scaling of CMOS logic technology. One approach to continuing performance gains as CMOS scaling matures is to replace the strained silicon MOSFET channel (and the source/drain) with an alternate material offering a higher potential quasi-ballistic-carrier velocity and higher mobility than strained silicon. Introduction of non-silicon materials into the channel and source/drain regions of an otherwise silicon MOSFET is fraught with difficult challenges. These challenges include fabrication of high-quality (i.e., defect free) channel and source/drain materials on non-lattice matched silicon, minimization of band-to-band tunneling in narrow bandgap channel materials, fabrication of high-κ gate dielectrics on the new channel materials, and elimination of Fermi level pinning in the channel/gate dielectric interface. Additional challenges are to sustain the required reduction in leakage currents and power dissipation in these ultimately scaled CMOS and to introduce new materials into MOSFET while simultaneously minimizing variations in critical dimensions and statistical fluctuations in the doping concentrations.

Table ES-17. Beyond CMOS Difficult Challenges

Difficult Challenges	*Summary of Issues and Opportunities*
Scale high-speed, dense, embeddable, volatile/non-volatile memory technologies to replace SRAM and FLASH in appropriate applications.	The scaling limits of SRAM and FLASH in two dimensional (2D) are driving the need for new memory technologies to replace SRAM and FLASH memories. Identify the most promising technical approach(es) to obtain electrically accessible, high-speed, high-density, low-power, (preferably) embeddable volatile and non-volatile memories. The desired material/device properties must be maintained through and after high temperature and corrosive chemical processing. Reliability issues should be identified and addressed early in the technology development.
Extend CMOS scaling	Develop new materials to replace silicon (or III-V, Ge) as alternate channel and source/drain to increase the saturation velocity and to further reduce V_{dd} and power dissipation in MOSFETs while minimizing leakage currents Develop means to control the variability of critical dimensions and statistical distributions (e.g., gate length, channel thickness, S/D doping concentrations, etc.) Accommodate the heterogeneous integration of dissimilar materials. The desired material/device properties must be maintained through and after high temperature and corrosive chemical processing. Reliability issues should be identified and addressed early in this development.
Continue functional scaling of information processing technology substantially beyond that attainable by ultimately scaled CMOS.	Invent and reduce to practice new information processing technologies with the potential to replace CMOS as the performance driver. Ensure that new information processing technologies have compatible memory technologies and interconnect solutions. New information processing technologies must be compatible with system architectures that can fully utilize new devices. Non-binary data representations or non-Boolean logic may be required to employ new devices for information processing, which will drive the need for new system architectures. Bridge the gap that exists between materials behaviors and device functions. Accommodate the heterogeneous integration of dissimilar materials. Reliability issues should be identified and addressed early in the technology development.
Extend ultimately scaled CMOS as a platform technology into new domains of functionalities and applications ("more than Moore, MtM").	Discover and reduce to practice new device technologies and primitive-level architectures to provide optimized special-purpose functional accelerator functions heterogeneously integrable with CMOS. Provide added value by incorporating functionalities that do not necessarily scale according to "Moore's Law". Heterogeneous integration of digital *and* non-digital functionalities into compact systems that will be the key driver for a wide variety of applications, such as communication, automotive, environmental control, healthcare, security, and entertainment.
Bridge the gap between emerging devices and novel architectures and computing paradigms.	Identify suitable opportunities in unconventional architectures and computing paradigms that can utilize unique characteristics of emerging devices. Identify emerging devices that can implement computing functions and architectures more efficiently than CMOS and Boolean logic.

DEVICES FOR CMOS EXTENSION

Carbon Nanotube FETS

Semiconductor carbon nanotubes (CNTs) have a naturally ultra-thin body of around 1 nm and a bandgap large enough for room temperature operation of transistors, and both electrons and holes have extremely high mobility. Owing to these structural and electrical advantages, CNTs offer a solution when channel length is scaled down to the sub-10 nm dimensional scale. Furthermore, they can be deposited using low-temperature processes, and it is also possible to integrate CNT integrated circuits three-dimensionally on top of Si VLSIs.

In the past several years, significant advances have been made in improving CNT thin film formation, in enhancing device performance in CNT FETs, and in integration. These include realizing (1) a purity of semiconductor CNTs of

>99.9999%, (2) high-density aligned semiconductor CNT films of >100 CNTs/μm on the entire surface of a 4-inch Si wafer, (3) precise control of positioning at a 10-nm pitch using DNA, (4) symmetrical characteristics in NMOS and PMOS via electrostatic doping with YO_x/AlN in semiconductor CNTs with a relatively large band gap, (5) a low-threshold voltage CMOS with a foundry-compatible manufacturing process, (6) single-CNT FETs of a 5-nm gate length with SS of 73 mV/dec. and estimated EDP of 6.95×10^{-31} Js, (7) CNT FETs with a high-density aligned CNT film with a current density of 2.24 A/mm, (8) 5-stage ring oscillator with an oscillation frequency of 5.5 GHz, (9) 50-nm CNT FET with f_T = 540 GHz and f_{max} = 306 GHz, (10) a 16-bit RISC-V microprocessors using CNT FETs, (11) three-dimensional integration of CNT ICs on Si logic ICs, and (12) manufacturing highly-uniform CNT FETs on a 200-mm wafer with commercial silicon manufacturing facilities.

There remains a need for further research toward improving other material- and device-level aspects, including methodology of purity evaluation for semiconductor CNTs, wafer-scale aligned CNT film with a controlled pitch, further reduction of contact resistance at small contact lengths, reduction in variability, control of gate dielectric interfaces and properties, and the experimental study of devices and circuits fabricated using the most scaled and relevant device structures and materials. In short, much work remains for CNT FETs, but they have some of the most substantial (and already demonstrated) potential in high-performance, low-voltage, sub-10 nm scaled transistor applications.

2D Material Channel FETs

2D Ultra-thin two-dimensional (2D) layered semiconductors, such as certain transition metal dichalcogenides (TMDs), are promising candidates as future channel materials in field-effect transistors (FETs) for very large-scale integration (VLSI)[2]. TMDs, in general, are characterized by the chemical formula MX_2, where M is a transition metal element and X is a chalcogen. They can be metallic, half-metallic, semiconducting, or superconducting depending on their compositions. MoS_2, WS_2, $MoSe_2$, WSe_2 and $MoTe_2$ with thicknesses from 1 to 4 layers are the most popular 2D TMDs explored for FETs; thanks to their bandgaps ranging from 1.6 to 2 eV, these materials can (unlike graphene) achieve low currents in the off state[3]. The synthesis of semiconducting graphene nanoribbons and carbon nanotubes has been also demonstrated, but their processing is much more complex, variability is much higher, and stability is significantly worse[4]. The ultra-thin body thicknesses (t_{body}) provided by the TMD channel (down to ~0.7 nm for monolayers)[5] improves electrostatic control over the on/off switching process of the FET, leading to less vertical parasitic current towards the substrate, less horizontal phonon scattering, and reduced short channel effects[6].

MoS_2 and WS_2 exhibit n-type characteristics under typical gate field conditions due to a much preferred line-up of the source/drain metal Fermi levels with the conduction band edge, while WSe_2 allows for hole injection at sufficiently negative gate voltages due to a near mid-gap line-up of the source/drain metal Fermi levels.[7] This effect is not related to doping, but to the specifics of electron and/or hole injection from the contacts. The actual doping level determines the threshold voltage of the devices, as in any conventional transistor, but the absence of the so-called hole-branch in MoS_2 and WS_2 Schottky barrier FETs is a result of the suppressed hole injection from the source/drain contacts. This emphasizes the need to understand and control source/drain contacts to the TMD channel, which is key to lowering contact resistances in TMD FETs to unravel their intrinsic performance specs. Due to limited access to reliable doping approaches for TMDs, particular attention has been paid to engineering metal source/drain contacts to improve the current injection. Bismuth has recently been reported as enabling ~mA/□m on-currents at drain voltages of around 1V.[8] More work is needed to understand how to unlock the intrinsic channel properties, employing improved contact engineering schemes.

Spin FET and Spin MOSFET Transistors

Spin-transistors are classified as "non-conventional charge-based extended CMOS devices and can be further divided into two categories: the spin-FETs and spin-MOSFETs. The structures of both types of spin transistors consist of a ferromagnetic source and a ferromagnetic drain, which act as a spin injector and a detector, respectively. Although the devices have similar structures, they have quite different operating principles. In spin-MOSFETs, the gate has the same current switching function as in ordinary transistors, whereas in the spin-FETs, the gate acts to control the spin direction via the Rashba spin-orbit interaction. Both types of devices behave as a transistor and function as a magnetoresistive device. The important features of spin transistors are that they allow a variable current to be controlled by the magnetization configuration of the ferromagnetic electrodes (spin-MOSFETs) or the spin direction of the carriers (spin-FETs), and they offer the capability for non-volatile information storage using the magnetization configurations. These features are very useful for energy-efficient, low-power circuit architectures that cannot be achieved by ordinary CMOS circuits. Non-volatile logic and reconfigurable logic circuits have been proposed using the spin-MOSFET and the pseudo-spin-MOSFETs, which are suitable for power-gating systems with low static energy.

EMERGING DEVICE-ARCHITECTURE

Many new emerging Beyond-CMOS devices will require co-design between devices and higher levels of computer design (e.g., circuit, architecture and application). These emerging devices are not intended simply as "drop-in" replacements for standard CMOS devices, but will require new types of circuit designs, new functional module architectures, and even new software to best utilize the new devices' capabilities. Novel design issues spanning the device and architecture levels especially need to be considered when adopting new low-level computing paradigms. In such situations, devices may be organized in radically new ways to carry out computation in a very different manner from what we may consider the most "conventional" computing paradigm, which has relied on von Neumann architectures using standard combinational and sequential irreversible Boolean logic.

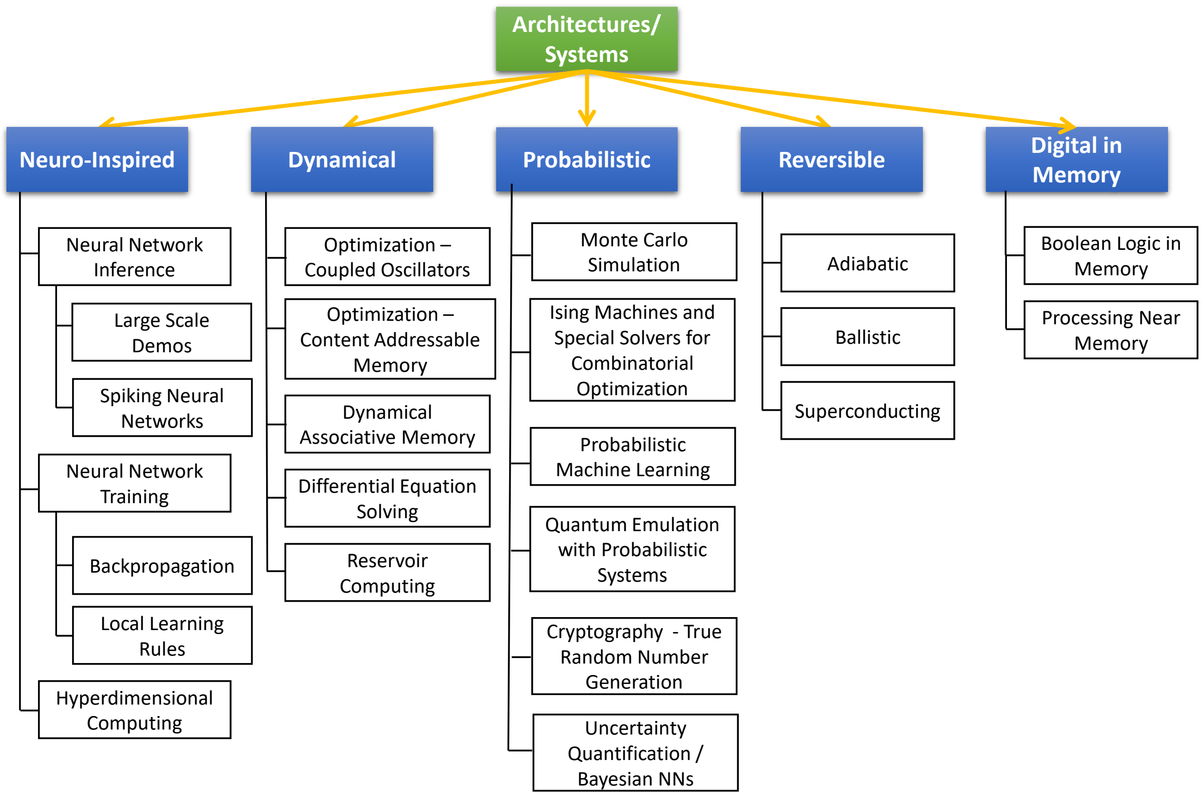


Figure ES-61. Architectures/Systems for novel computing paradigms discussed in the Beyond CMOS Chapter requiring codesign with emerging devices

Examples of unconventional or alternative computing architectures and systems that comprise the following: •

- Neuro-Inspired Systems These computing architectures take direct inspiration from the brain to develop more efficient systems. Local learning rules are a neuro-inspired method of training neural networks that keeps all communication local.
- Hyperdimensional computing is a cognitive computing model based on the high dimensional properties of neural circuits in the brain.
- Spiking neural networks minimize communication energy by using sparse spike-based communications.
- Reservoir computing nonlinearly transforms time dependent signals into a new, random high-dimensional basis and then a conventional machine learning algorithm is used to make predictions from the new basis set.
- Dynamical Systems: Analog dynamical systems can be used to solve a variety of problems. Optimization problems can be solved using simulated annealing or coupled-oscillator-based approaches Dynamical systems can be used to encode associative memories or to solve differential equations. Chaotic logic can theoretically enable sub-kT computing
- Probabilistic Systems: Devices and circuits that produce truly nondeterministic or random outputs at the hardware level may be useful for accelerating probabilistic algorithms such as Monte Carlo or simulated annealing, for generating secure cryptographic keys, and for other applications.
- Reversible Systems: Computing systems that approach logical and physical reversibility offer the potential to greatly exceed the energy efficiency of all other approaches for general digital computation. While devices for reversible computing may perform conventional functions (such as switching or oscillating), they should be optimized to use quasi-reversible physical processes such as near-adiabatic state transitions, near-ballistic signal propagation, highly elastic interactions, and highly underdamped oscillations. For maximal efficiency, circuits and architectures must approach reversibility at the logical as well as physical level. Careful fine-tuning and optimization of analog circuit characteristics (e.g., resonator quality factors, elasticity of ballistic interactions) remains a difficult and crucially important engineering challenge that must be met to fully realize the promise of the reversible computing paradigm. In the meantime, potentially commercially viable near-term applications of reversible computing are beginning to emerge for specialized cryogenic applications. •
- Digital in-Memory Systems Boolean logic can be performed in memory arrays and digital logic is being embedded closer and closer to memory to reduce data movement costs.
- Quantum computing (CEQIP chapter)—Quantum computing offers the potential to carry out exponentially more efficient algorithms for a variety of specialized problem classes. Devices for quantum computing are very different from conventional devices, and fine-tuning device characteristics to avoid decoherence while organizing them effectively into scalable architectures has so far proved to be a formidable engineering challenge. The 2025 IRDS Cryogenic Electronics and Quantum Information Processing (CEQIP) roadmap chapter will address quantum computing.

The following figure gives a simplified overview of the key parameters for different technology and architectural approaches (for more details see the 2024 Beyond CMOS chapter.):

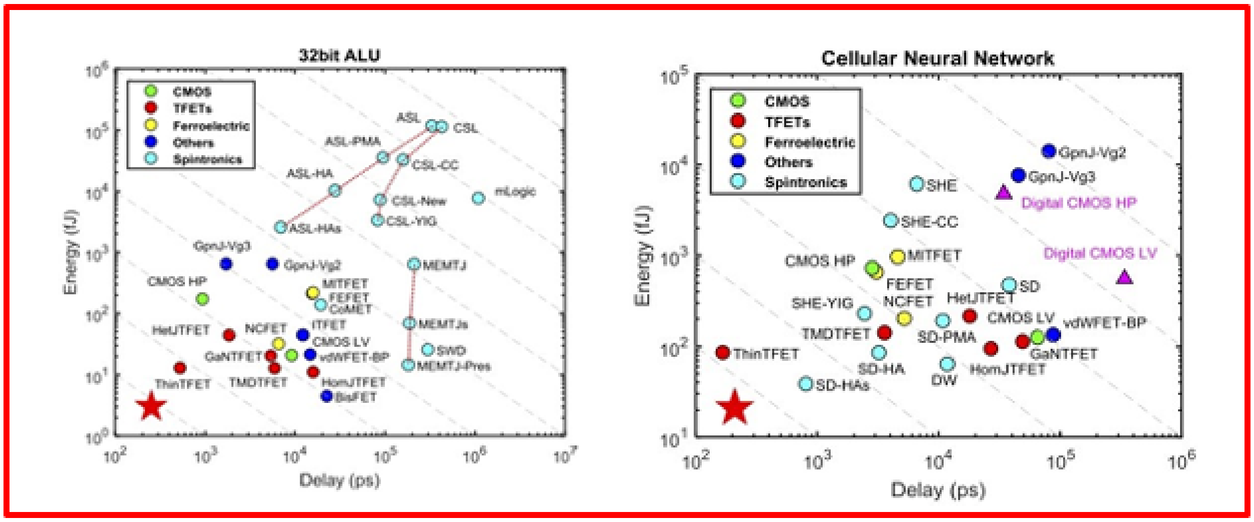


Figure ES-62. Energy delay relation for several architectures

BEYOND CMOS CONCLUSION

The "Beyond CMOS" chapter systematically surveys emerging memory and logic devices novel technologies and alternative architectures and computing paradigms to explore potential solutions beyond the conventional scaling of CMOS technologies. Although high performance at low power consumption has been a primary objective of beyond-CMOS devices, novel functionalities and applications have become increasingly important. The recent emergence of energy-efficient data-intensive cognitive applications is also shifting the emphasis from high-precision computing solutions to novel computing paradigms with massive parallelism and bio-inspired mechanisms. Research opportunities exist in the co-optimization of beyond-CMOS devices and architectures to explore unique device characteristics and architectural designs.

Although a beyond-CMOS device competitive against CMOS FET has not been identified, beyond-CMOS devices with dramatically enhanced scalability and performance while simultaneously reducing the energy dissipation per functional operation would still be fundamentally important and a worthwhile research objective. In considering the many disparate new approaches proposed to provide order of magnitude scaling of information processing beyond that attainable with ultimately scaled CMOS, the following set of guiding principles are proposed to provide a useful structure for directing research on "Beyond CMOS" information processing technology.

9.9. METROLOGY HIGHLIGHTS

INTRODUCTION

The Metrology Chapter identifies emerging measurement challenges and knowledge gaps in devices, systems, and integration in the semiconductor industry and describes research and development pathways for meeting them. This includes, but is not limited to, measurement needs for CMOS transistors, novel devices beyond CMOS, technologies for interconnects and chiplets architectures, and materials characterization for process development in integrated devices and systems. This also includes advances in metrology required for research and development, process control and device / system performance assurance in manufacturing environments.

Metrology is needed to support the rapid far-reaching changes and evolution in the semiconductor industry. Besides architectural changes in the computational module (i.e., front-end-of-the-line (FEOL)), new and major changes are being introduced in the communication / interconnect module (i.e., BEOL).

It seems inevitable to use more than one type of measurement, tool, and method, and take advantage of the rich, multi-faceted set of data by combining the most reliable information gained from each source. This combined (complementary, hybrid) metrology would be necessary for sub-10 nm dimensional metrology. Hybrid metrology has shown promising advantages in significantly reducing the overall uncertainties, i.e., augmenting the confidence in the results, and in the optimization of various complementary measurement methods for cost.

An application of combined metrology is the Reference Measurement System (RMS), where one of the instruments provides traceability to the others. An RMS could be a single instrument or a set of several instruments. An RMS is well characterized using the best the science and technology dimensional metrology can offer, applied physics, sound statistics, related standards, and proper handling of all measurement error contributions based on the best protocols available.

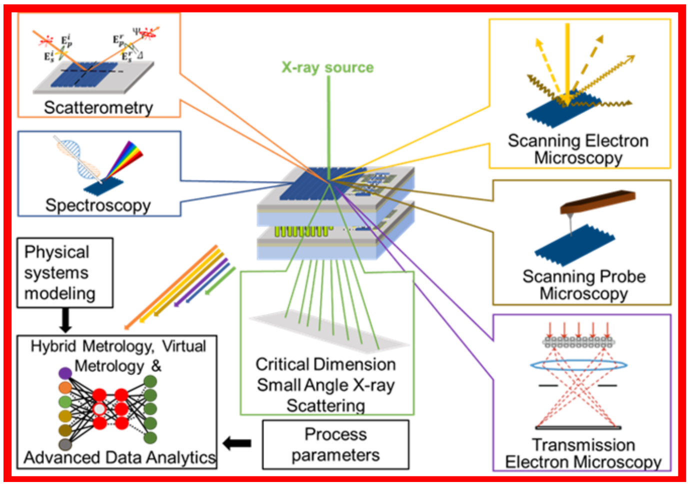


Figure ES-63. The RMS menu of combined technologies to achieve the metrology goal

The RMS methodology has been well characterized, it is more accurate, perhaps by an order of magnitude, and more precise than any instrument in a production fab. An RMS must be sufficiently stable so that other measurement systems can be related to it. An RMS can be used to track measurement discrepancies among the metrology instruments of a fab, control the performance and matching of production metrology instruments over time, and introduce absolute traceability to the measurement process if desired.

METROLOGY CONCLUSIONS

The large number of parameters needed to fully characterize complex logic and memory structures means that no single instrument has the capability, resolution, or low levels of uncertainty required to provide the needed information. As such, the use of hybrid or complementary metrologies, where multiple instruments are used to evaluate a set of parameters based on their capabilities, would increase. The question of which instrument combinations, analysis, and modeling techniques to use for specific measurements is still a subject of intense research. However, preliminary results are encouraging. Investments in developing platforms with multiple technologies would be helpful. At least, having tools available that enable the combination and comparison of data from multiple technologies would be a good starting point.

9.10. ENVIRONMENT, SAFETY, HEALTH AND SUSTAINABILITY (ESHS): ENVIRONMENTAL SUSTAINABILITY OF THE SEMICONDUCTOR FACILITIES (ESSF)

INTRODUCTION

The ESH/S IFT (Environmental, Safety, Health and Sustainability International Focus Team) comprises of a number of focus teams, driven by current ESH/S priority. This includes Environmental Sustainability of the Semiconductor Facilities (ESSF), Supply Chain, Materials Fab Process, and Product Content (circularity).

ESH/S IFT has recognized the urgency and the importance of dealing with this challenge, forming a focus team to analyze technology gaps and opportunities in the space of Environmental Sustainability of the Semiconductor Facilities. This document is a sub-chapter of ESH/S chapter and produced as an individual document to support the growing focus on this topic and help the audience with specific technology roadmap set of materials in the area. The forum operated throughout all time zones, regions, and geographies, including major semiconductor manufacturers, tool suppliers, and facility technology providers to ensure completeness of the assessment and the input into the roadmap.

The ESSF Roadmap seeks to provide the supply chain with the description of industry needs and the justification for developing new solutions for sustainable operation of facilities especially related to utilization of energy and water that will enable sustainable industry growth. Through a collaborative industry process, tangible roadmap targets for solutions in high-risk areas have been developed including rationales for these solutions anchored by key performance indicators (KPIs). The document scope includes all systems within semiconductor manufacturing facilities with meaningful water and energy footprints. Energy includes electricity, fuel, and other utilities that require significant amounts of energy to produce, including pure gases, compressed Clean Dry Air (CDA), and chemicals.

In addition to the technology roadmap definition supporting supply chain justification for the new technology development, the ESSF Roadmap is driving education and alignment within the industry on the opportunities related to the available technological solutions, allowing for improved environmental sustainability performance. Some of this information can be found in the section of Potential Solutions.

- Exhaust (e.g., make-up air energy demand),
- Ultrapure water (UPW) and Hot UPW (HUPW) (e.g., raw water treatment, heat recovery options),
- Air Conditioning (e.g., cleanroom heating, ventilation, and air conditioning (HVAC), energy and variable heat transfer methods).

As an example, water consumption in semiconductor facilities will typically fall into these four main categories, which are presented in the ESSF chapter:

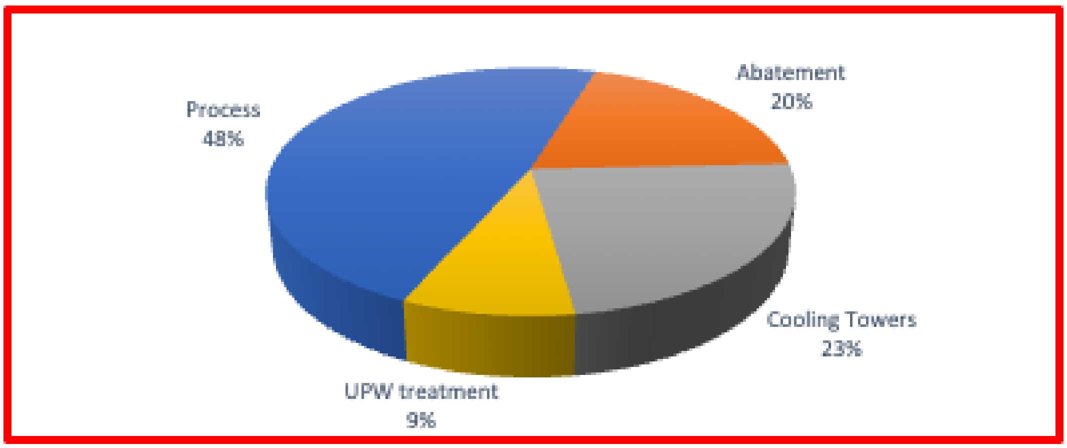


Figure ES-64. Semiconductor fab water consumption

Furthermore, the energy roadmap focus is on, but not limited to, every type of "utility" considered by SEMI S23 ECF (energy conversion factors) definitions.[9] The extension of energy management programs beyond small teams from one department within the organization to organization wide with senior management support is key. Many semiconductor companies have implemented ISO 50001:2018, Energy Management System Standard, which provides a framework for this expansion. A traditional absence of energy usage data on significant energy using equipment to provide indicators to drive action is a key industry gap and a process needs to be developed to address this.

A key opportunity for the industry is to highlight high energy consumption equipment and tools for easy targeting and to promote for less energy users with innovation in similar functions.

ESSF CONCLUSIONS

Growing semiconductor facilities drive increasing environmental footprint, including high energy demand, high freshwater withdrawal, carbon emission, and other greenhouse gases (GHG) and waste related emissions. Whereas in the past the sustainability performance was associated with general social responsibility and cost related benefits, it is now becoming mission critical for the industry. The nexus between the environmental performance indicators is getting more and more obvious. This chapter evaluated the technology drivers, boundaries, and other considerations from the point of view of sustainable industry growth and future technology enabling. The drivers include semiconductor technology, environmental social governance (ESG), climate change, and the growing chip demand. These drives create both urgency and difficulty to deal with the sustainability management.

9.11. YIELD ENHANCEMENT

INTRODUCTION

The Yield Enhancement (YE) chapter is dedicated to ensuring that semiconductor wafer processing environment is best equipped for identifying, reducing, and avoiding yield-relevant defects and contamination.

Yield in most industries has been defined as the number of working die produced divided by the number of potential dice on a wafer. In the semiconductor industry, yield is represented by the functionality and reliability of integrated circuits produced on the wafer surfaces. During the manufacturing of integrated circuits yield loss is caused, for example, by defects, faults, process variations, and design. The relationship of defects and yield, and an appropriate yield-to-defect correlation, is critical for yield enhancement.

Yield Enhancement displays the current advanced and next generation future requirements for high yielding manufacturing of More Moore as well as More than Moore products separated in "critical process groups" including microelectromechanical (MEMS), back-end processes, e. g., packaging. Consequently, the inclusion of material specifications for Si, SiC, GaN, etc., are considered.

Yield Enhancement focuses on random defects, related to the areas of technology responsible for contamination control, as shown in Figure ES-65.

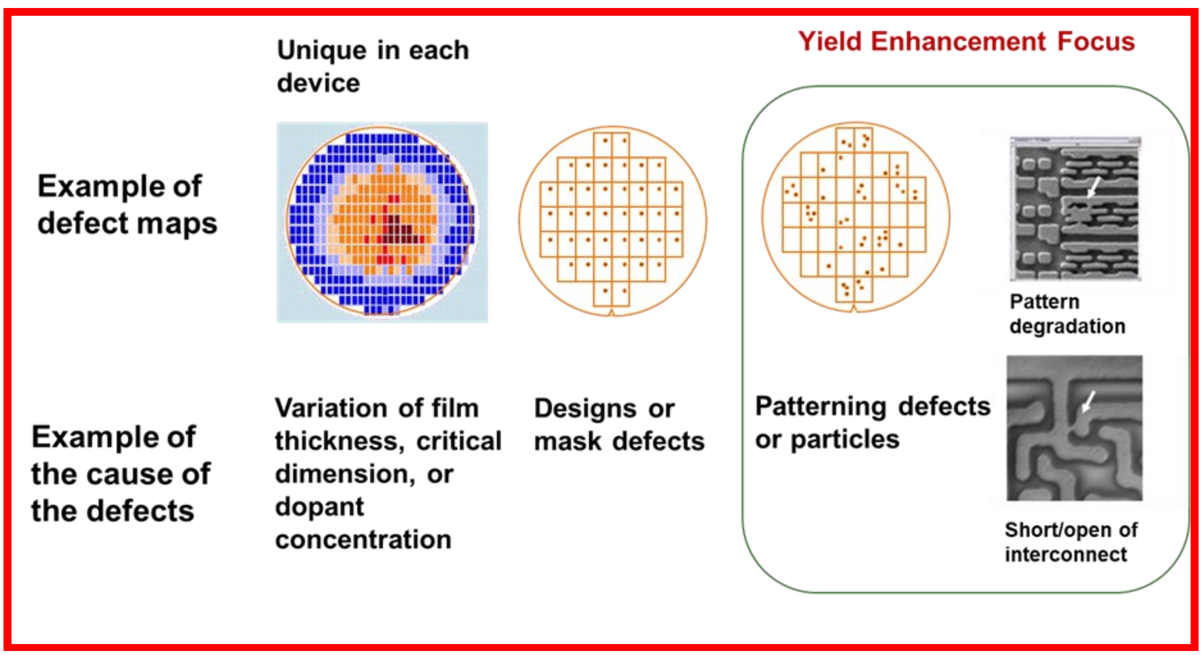


Figure ES-65. Random yield area of focus

The YE chapter has two focus topics: "Substrate Environment Contamination Control" (SECC) and "Characterization, Inspection and Analysis (CIA)". These two topics include the involvement of device design, front-end process technology, interconnect processes, lithography, metrology, process integration, test, and facility infrastructures.

For 2024, the IRDS Yield Enhancement Chapter Team adjusted its approach for risk and gap analysis. In the past, risk for YE was based on the device; geometrical device scaling gradually becoming smaller year over year. This was relatively simple based on the limited geometrical changes occurring. Now as we enter a more complex introduction of gate all around, 3D, and heterogeneous integration along with new materials being introduced, we are entering a new methodology that involves a much closer collaborative interaction process with the More Moore and device process manufacturing experts. This change in methodology will add in-depth discussion to fully understand the mechanisms within the manufacturing modules and how they connect to process tools, materials, systems, and environment that impact contamination and yield. Yield technology drivers are changing. While in past memory was the primary driver of the Yield Enhancement roadmap due to smaller critical dimension, now Logic has become and will likely continue to be the leading driver in the future. Logic was chosen to be a driver for Yield roadmap for the following reasons (refer to the IRDS More Moore chapter for additional information):

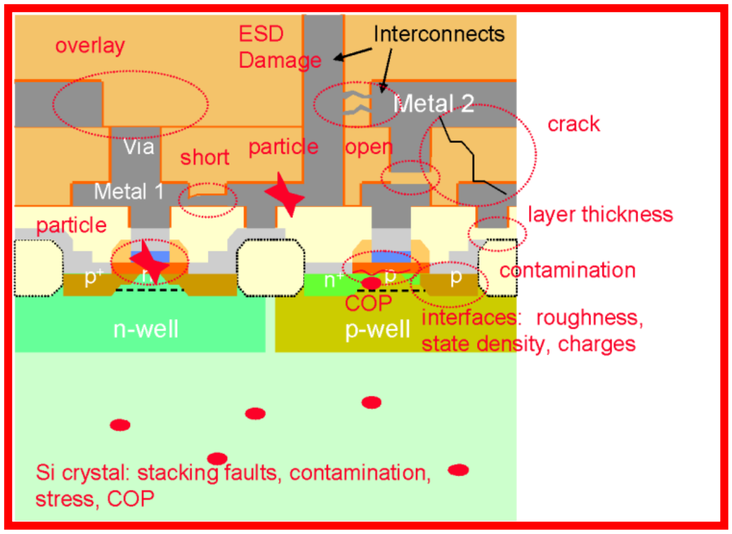


Figure ES-66. Causes of yield losses

In the manufacture of integrated circuits yield loss is related to a variety of sources. During processes such as implantation, etching, deposition, planarization, cleaning, lithography, etc. failures responsible for yield loss occur. Several examples of contaminations and mechanisms responsible for yield loss are listed in the following: a) airborne molecular contamination (AMC) or particles of organic or inorganic matter caused by the environment or by the tools; b) process induced defects as scratches, cracks, and particles, overlay faults, and stress; c) process variations resulting, e.g., in differing doping profiles or layer thicknesses; d) the deviation from design, due to pattern transfer from the mask to the wafer, results in deviations and variations of layout and critical dimensions; and e) diffusion of atoms through layers and in the semiconductor bulk material. Figure illustrates the YE scope.

Yield Enhancement forum interacts with the teams that either provide definitions about defects and factors affecting manufacturing yield or can benefit from the information generated by the forum for their respective roadmap development. The SECC forum of Yield Enhancement provides input into SEMI Standards development by triggering new standards development or existing standard updates in the areas associated with the respective technology challenges. The following diagram illustrates the cross-team linkage.

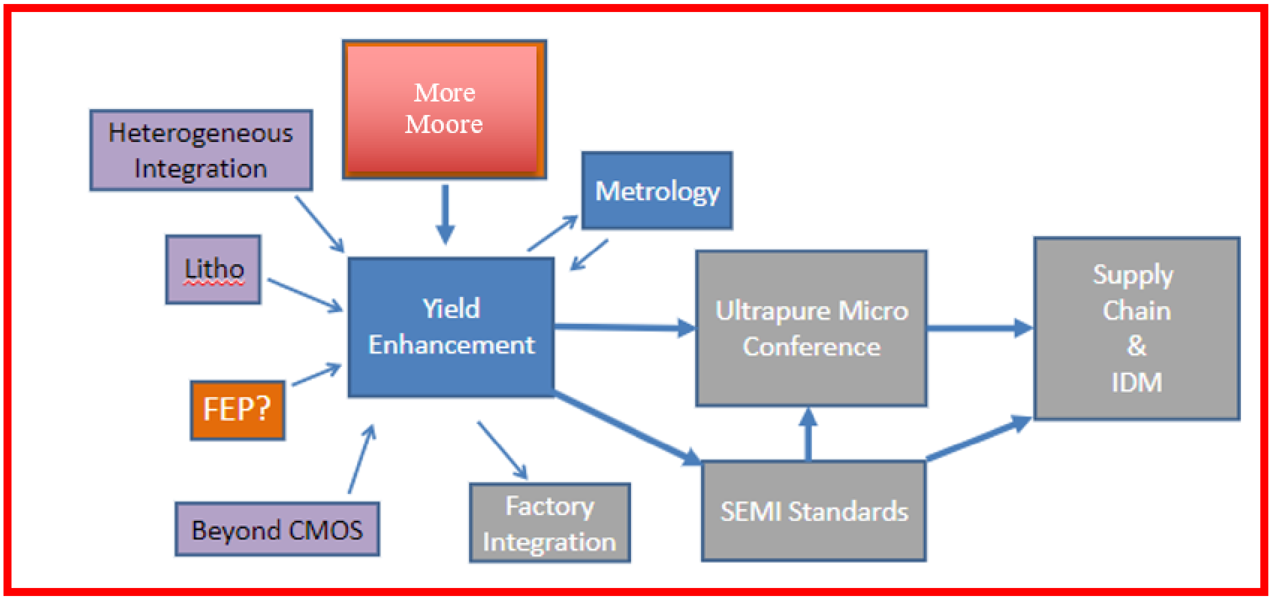


Figure ES-67. Yield Enhancement cross team linkage diagram

YIELD ENHANCEMENT CONCLUSIONS

Most advanced semiconductor technologies and particularly logic (the new yield technology driver) have reached the point when yield may become a constraining factor of the future shrinkage of the critical dimensions. This is because the defect metrology of both critical substrates (wafer, mask, lithography optics, etc.) and materials has reached their limits. This affects the ability to prevent, predict, and control defects in the manufacturing facility. This requires new systematic approaches to continue enabling future technology in accordance with the IRDS roadmap definitions. Such new approaches should include combination of the following measures:

- Proactive measurement of contamination control of the critical materials used throughout the supply chain.
- Leveraging data analytics to correlate process variation in production with any deviations in critical parameters.
- Employ prediction modeling and experimentation to help with decisions on the choice of the technologies and method of their applications at all levels of the manufacturing facility.
- Standardize quality control using SEMI and other standards.
- Take advantage of collaborative development in the industry via IRDS, SEMI, and other research institutions to drive most productive technology solutions related to the SECC. Assist emerging metrology to commercialize via benchmarking studies and independent third-party qualifications.

9.12. FACTORY INTEGRATION

No new update is introduced in 2024. The Factory Integration (FI) chapter of the 2023IRDS is dedicated to ensuring that the microelectronics manufacturing infrastructure contains the necessary components to produce items at affordable cost and high volume. Realizing the potential of Moore's Law requires taking full advantage of device feature size reductions, new materials, yield improvement to near 100%, wafer size increases, and other manufacturing productivity improvements. This in turn requires a factory system that can fully integrate additional factory components and utilize these components collectively to deliver items that meet specifications determined by other IRDS international focus teams as well as cost, volume, and yield targets. Preserving the decades-long trend of 30% per year reduction in cost per function also requires capturing all possible cost reduction

Societal driving forces and trends such as mobile devices and the internet of things are impacting all areas of the IRDS, however, these factors impact the evolution of FI from two perspectives, namely:

Requirements they place on product technologies that are delineated in roadmaps associated with other focus areas; these technology requirements indirectly influence FI in terms of tighter process requirements with acceptable yields, throughputs, and costs. Requirements they place on FI technologies that directly impact FI in terms of aligning with these trends and effectively leveraging these capabilities.

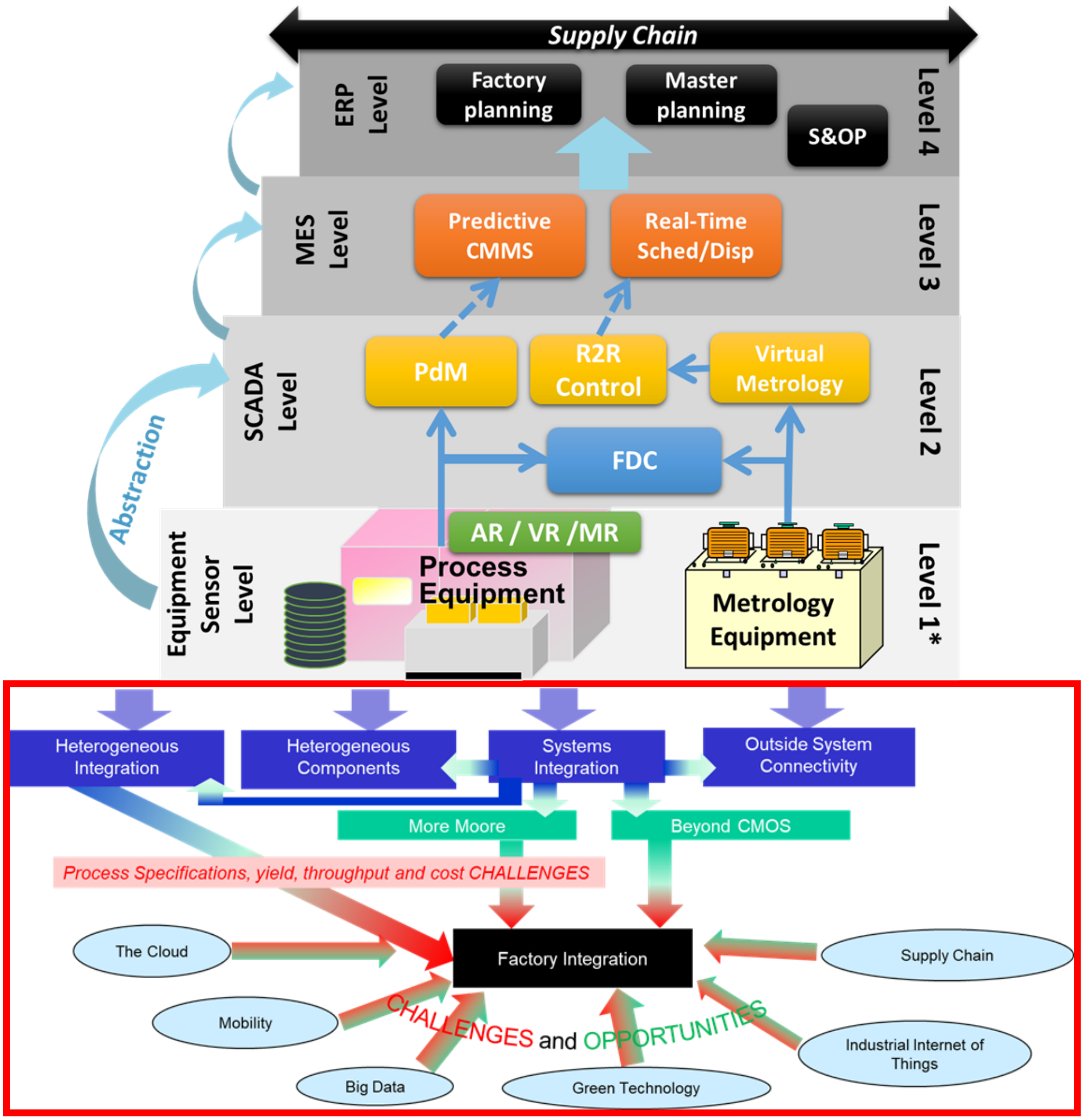


Figure ES-68. Societal forces impacting challenges and opportunities in FI

DIGITAL TWIN STATE-OF-THE-ART AND VISION

Many fabs today are already at the forefront of the digital twin (DT) revolution, having pioneered many DT technologies and providing testament to DT success. As shown in Figure ES-69, existing pervasive technologies, such as run-to-run (R2R) control and real-time Scheduling and Dispatch (S/D), and emerging technologies such as Predictive Maintenance (PdM) and Virtual Metrology (VM) are key members of the DT family.

The vision for DT is a framework of DT classes existing at all levels in the ISA-95 infrastructure (see Figure ES-69) with benefits resulting from the utilization of DTs instances in isolation (e.g., R2R control of a process), but also in collaboration with other instances in the same class (e.g., coordinated R2R controllers across chambers for improved chamber matching), with other classes (e.g., coordination of R2R control with scheduling/dispatch—S/D in order to incorporate yield objectives into S/D decisions), and within and with other SM components (e.g., coordination of PdM predictions with supply chain management). DTs can exist at any level, with much of the benefit from digital twins arising from the abstract of DTs to higher levels and integration with other DTs. The various classes within the framework will be developed at different rates depending on need, level of technical challenge, influence from other industries, etc., and flexible DT framework that supports interchange and interoperability of DT instances and classes will be a longer-term technical challenge.

1. Requirements they place on product technologies that are delineated in roadmaps associated with other focus areas; these technology requirements indirectly influence FI in terms of tighter process requirements with acceptable yields, throughputs, and costs.

2. Requirements they place on FI technologies that directly impact FI in terms of aligning with these trends and effectively leveraging these capabilities.

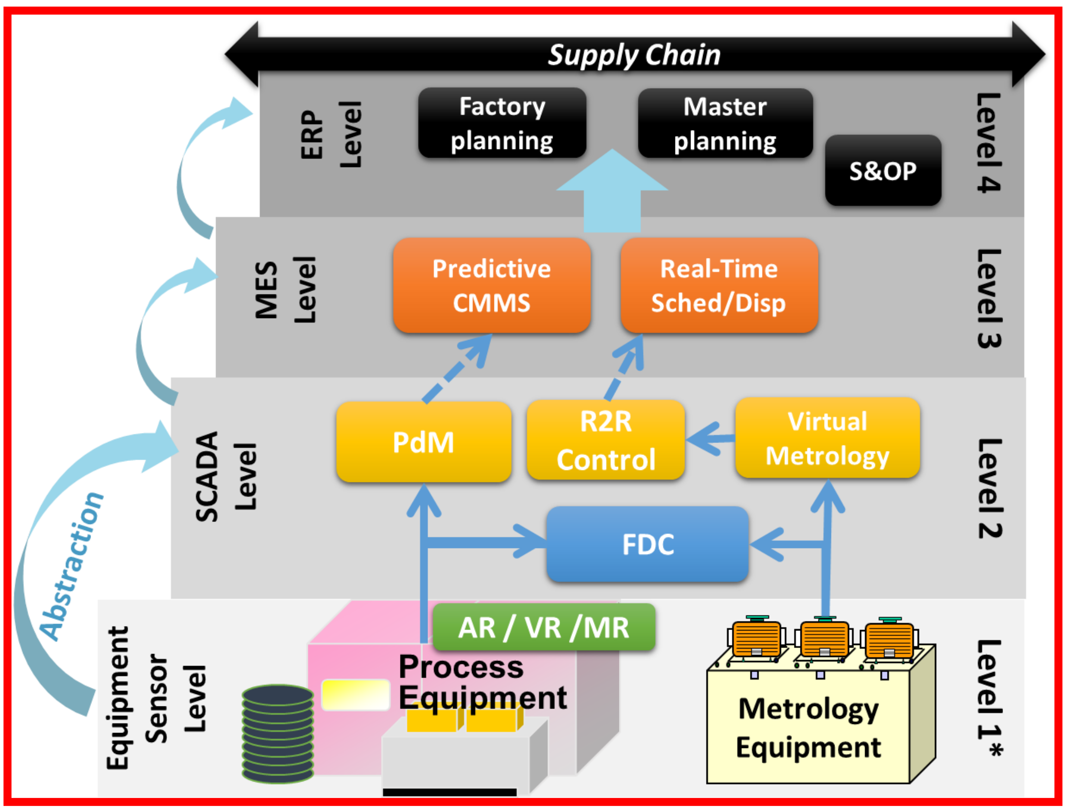


Figure ES-69. Digital Twin representation from the perspective of the International Society of Automation (ISA-95) Levels[10]

10. ACRONYMS AND ABBREVIATIONS

Term	Definition
1G-4G	First Generation to Fourth Generation
3D NAND	Three Dimensional NAND Memory (Flash Memory)
3DVLSI	3D Very Large Scale Integration
3GPP	Third Generation Partnership Project
5G	Fifth Generation
A&AI	Active & Available Inventory
A2A	All to All
AAA	Authentication, Authorization, and Accounting
ACK/NAK	Acknowledgment/Negative Acknowledgment
ADAS	Advanced Driver Assistance Systems
ADC	Analog to Digital Converter
ADP	Application Domain Profile of IEEE 2413
ADS	Augmented Data System
AFM	Atomic Force Microscope, Atomic Force Microscopy
A-GPS	Assisted GPS
AI	Artificial Intelligence
ALD	Atomic Layer Deposition Techniques
AMC	Airborne Molecular Contamination
AMD	Advanced Micro Devices
AMS	Analog/Mixed Signal
AOC	Active Optical Cables
APC	Airborne Particular Contamination or Advanced Process Control
API	Application Programming Interface
APIMS	Atmospheric Pressure Ionized Mass Spectroscopy
APT	Atom Probe Tomography
AR	Augmented Reality or Aspect Ratio
ARC	Antireflective Coating
ARM	Advanced RISC Machines and originally Acorn RISC Machine
ARPES	Angle-Resolved Photoemission Spectroscopy
As	Arsenic
ASIC	Application-Specific Integrated Circuit
ASP	Average Selling Price
ATE	Automatic Test Equipment
ATPG	Automatic Test Pattern Generation
AWS	Amazon Web Services
B	Boron
B2B	Business to Business
B2C	Business to Consumer
BBU	Base Band Unit
BC	Beyond CMOS IFT
BDS	Broadband Dielectric Spectroscopy
BE	Back End
BEOL	Back End of Line
BER	Bit Error Rate
BKC	Best Known Configurations
BS	Base Station
BSS	Business Support System
BV_{CBO}	Breakdown Voltage Collector-Base
BV_{CEO}	Breakdown Voltage Collector-Emitter
BWP	Bonded Wafer Pairs

C/U/D/M	Control Plane/User Plane/Data Plane/Management Plane
CAGR	Compound Annual Growth Rate
CAPEX	Capital Expenditure
CASS	Circuits and Systems Society
CBED	Convergent Beam Electron Diffraction
CBRS	Citizen Band Radio Services
CC	Cloud Computing
CCCS	Critical Contamination Control Specifications
CCD	Charge Coupled Device
CCIX	Cache Coherent Interconnect for Accelerators
CD	Critical Dimension
CDA	Clean Dry Air
CD-AFM	Critical Dimension-Atomic Force Microscopy
CDMA	Code Division Multiple Access
CD-SAXS	Critical Dimension-Small Angle X-Ray Scattering
CD-SEM	Critical Dimension-Scanning Electron Microscopy
CEDA	Council on Electronic Design Automation
CEP	Cloud Endpoint
CEQIP	Cryogenic Electronics and Quantum Information Processing IFT
CFP	C-Form Factor Pluggable
CGS	Coherent Gradient Sensing
CHIPS	Creating Helpful Incentives to Produce Semiconductors
CIA	Characterization, Inspection and Analysis
CIM	Computer Integrated Manufacturing
CML	Current Mode Logic
CMOS	Complementary Metal-Oxide Semiconductor
CMP	Chemical Mechanical Polishing or Chemical Mechanical Planarization
CN	Core Network
CNCF	Cloud Native Computing Foundation
CNFs	Cloud-Native Network Functions
CNT	Carbon Nanotube
CO	Central Office
COBO	Consortium for On Board Optics
ComSoc	Communications Society
CoO	Cost Of Ownership
COTS	Commercial Off-the-Shelf
CP	Control Plane
CPC	Condensation Particle Counter
CPRI	Common Public Radio Interface
CPS	Cyber-physical System
CRD	Custom Resource Descriptors
CRDS	Cavity Ring-Down Spectroscopy
CRM	Certified Reference Materials
CS	Computer Society
CSC	Council on Superconductivity
CU/DU	Centralized Unit/Distributed Unit
CVD	Chemical Vapor Deposition
CVS	Cyclic Voltammetric Stripping
CXL	Compute Express Link
D2D	Device to Device or Die to Die
D2W	Die to Wafer
DAC	Digital to Analog Converter
DBM	Design-Based Metrology

DC	Data Center
DCSA	Data Collection Service and Analytics
DeMUX	Demultiplexer
DevOps	Development and Information Technology Operations
DFFT	Discrete Fourier Transform
DFM	Design for Manufacturing, Design for Manufacture, Design for Manufacturability
DFT-s-OFDM	Discrete Fourier Transform Spread Orthogonal Frequency Division Multiplexing
DL	Downlink
DLTS	Deep Level Transient Spectroscopy
DMD	Digital Micro Mirror Devices
DP	Data Plane
DPU	Data Processing Unit
DRAM	Dynamic Random Access Memory
DRAM	Dynamic RAM
DSA	Domain Specific Application or Directed Self Assembly
DT	Digital Twin
DTC	Deep Trench Cap
DTCO	Design-Technology Co-Optimization
DUV	Deep Ultraviolet
DWBA	Distorted Wave Born Approximation
EAP	Edge Automation Platform or Electrically Active Particle
EBSD	Electron Backscatter Diffraction
EBSEM	Electron Beam Scanning Electron Microscopy
ECB	European Central Bank
ECB	European Central Bank
eCPRI	Ethernet Common Public Radio Interface
ECT	Electro-Capacitive Tomography
ECU	Engine Control Unit
ECU	Engine Control Unit
EDS	Energy Dispersive X-Ray Spectroscopy, or Energy-Dispersive Spectroscopy, or Electron-Dispersive Spectroscopy, or Energy Dispersive Spectrometers or Electron Devices Society
EELS	Electron Energy Loss Spectroscopy
EEP	Edge Endpoint
ELS	Energy Loss Spectroscopy
eMBB	Enhanced Mobile Broadband
EM	Electromigration
EMI	Electromagnetic Interference
EMT	Emergency Medical Technicians
eNB	Evolved Node B
EO	Electronic to Optical
EOM	Electro-Optic Modulator
EPC	Evolved Packet Core
EPD	Etch Pit Density
EPE	Edge Placement Error
EPS	Electronics Packaging Society
ESA	Electrostatic Attraction
ESD	Electrostatic Discharge
E-SEM	Environmental-Scanning Electron Microscopy
ESG	Environmental, Social, and Governance
ESH/S	Environment, Safety, Health, and Sustainability IFT
ESI	European SINANO Institute
ESSF	Environmental Sustainability of Semiconductor Facilities (part of ESH/S)
ETSI	European Telecommunications Standards Institute
EUV	Extreme Ultraviolet

EUV AIMS	EUV Aerial Image Measurements Systems, Aerial Image Microscope
EUVL	Extreme Ultra Violet Lithography
EV	Electric Vehicle
EXAFS	Extended X-Ray Absorption Fine Structure
FDD	Frequency-Division Duplex
FDMA	Frequency Division Multiple Access
FDSOI	Fully Depleted Silicon On Insulator
FE	Front End
FE-AES	Field Emission Auger Electron Spectroscopy
FEOL	Front End of Line
FEP	Front End Processes
FET	Field Effect Transistor
FFT-SSRM	Fast Fourier Transform-Scanning Spreading Resistance Microscopy
FI	Factory Integration IFT
FIB	Focused Ion Beam
FID	Flame Ionization Detector
finFET	Fin Structure Field Effect Transistor
fmax	Frequency Where Unilateral Gain (U) Becomes Unity, or Zero dB
FMC	Fixed Mobile Convergence
FMEA	Failure Mode and Effects Analysis
FMP	Fleet Matching Precision
FOM	Figure Of Merit
FOUPs	Front Opening Unified Pods
FPGA	Field-Programmable Gate Array
FSD	Full Self-Driving
f_T	Transition Frequency (Where the Current Gain Goes to Unity, or Zero dB)
FTIR	Fourier Transform Infrared Spectroscopy
FTTX	Fiber to X
GAA	Gate All Around
Gbps	Gigabits Per Second
GC	Gas Chromatography
GCMS	Gas Chromatography–Mass Spectrometry
GDP	Gross Domestic Product
GDS	Graphic Database System
GEO	Geostationary Satellite
GeSi	Ge Rich Silicon Germanium
GFAAS	Graphite Furnace Atomic Absorption Spectroscopy
GHG	Greenhouse Gases
GHz	Gigahertz
GI-SAXS	Grazing Incidence Small Angle X-Ray Scattering
g_m	Transconductance
GPICL	General-Purpose In-Context Learning
GPU	Graphics Processing Unit
GSM	Global System for Mobile Communications
GSMA	GSM (Groupe Speciale Mobile) Association
HA	High Availability
HAADF	High Angle Annular Dark Field
HAR	High Aspect Ratio
HBM	High Bandwidth Memory
HBT	Heterojunction Bipolar Transistor
HCI	Hyper-Converged Infrastructure
HDD	Hard Disk Drive
HDMIM Cap	High Density Metal-Insulator-Metal Cap

HeIM	Helium Ion Microscopy
HEMT	High Electron Mobility Transistor
HEPA	High-Efficiency Particulate Air
HIM	Helium Ion Microscopy
HIR	Heterogeneous Integration Roadmap
HMW	High Molecular Weight
HOM	High Order Modulation
HPC	High Performance Computing
HSS	Home Subscriber Services
HUPW	Hot Ultra Pure Water
HVM	High Volume Manufacturing
HV-SEM	High Voltage Scanning Electron Microscopy
I/O	Input-Output
IBM	International Business Machines
IC	Integrated Circuit
ICN	Integrated Cloud Native
ICP-MS	Inductively Coupled Plasma Mass Spectrometry or Ion Coupled Plasma Mass Spectrometry
IDM	Integrated Device Manufacturer
IEA	International Energy Agency
IEDM	International Electron Devices Meeting
IEEE	Institute Of Electrical and Electronics Engineers
IETF	Internet Engineering Task Force
IFFT	Inverse Fast Fourier Transform
IFR	International Federation of Robotics
IFT	International Focus Team (of the IRDS)
IMF	International Monetary Fund
IMS	Ip Multimedia Subsystem or Ion Mobility Spectrometry
INEMI	International Electronic Manufacturing Initiative
IoE	Internet of Everything
IoT	Internet of Things
IP	Internet Protocol, or Intellectual Property
IR	Infrared
IRC	International Roadmap Committee (of the IRDS)
IRDS	International Roadmap for Devices and Systems
ISA	International Society of Automation
ISC	Integrated Stacked Cap
ISG	Industrial Specification Group
ISO	International Standards Organization
ISP	Internet Service Provider
ISSCC	International Solid-State Circuits Conference
ITRS	International Technology Roadmap for Semiconductors
ITS	Intelligent Transport System
ITU	International Telecommunication Union
ITU-T	Itu Telecommunication Standardization Sector
Jc	Collector Current Density
Kbps	Kilobits Per Second
KPI	Key Performance Indicator
L	Lithography IFT
LAA	Licensed Assisted Access
LAN	Local Area Network
LC	Local Cloud Associated With CO or Edge (Near as Well Far)
LCOCD	Liquid Chromatography: Organic Carbon Detection
LDPC	Low Density Parity Code, Low-Density Parity-Check

LDPE	Licensed Data Payload Encryption
LEL	Living Edge Lab
LEO	Low Earth Orbit
LER	Line Edge Roughness
LFE	Linux Foundation Edge (Akraino Project Belongs to LFE)
LGAA	Lateral Gate All Around
LiDAR	Light Detection and Ranging
LiFi	Light Fidelity
LLBSE-SEM	Low-Loss Backscattered Electrons SEM
LMW	Low Molecular Weight
LNA	Low Noise Amplifier
LNP	Low-Noise Power
LoRaWAN	Long-Range WAN
LPC	Laser Particle Counter
LPWAN	Low-Power WAN
LTE	Long-Term Evolution
LWR	Line Width Roughness, Line Width Reduction
M	Metrology IFT
M2M	Machine to Machine
MAC	Medium Access Control
MAG	Maximum Available Gain or Magnetics Society
MANO	Management and Network Orchestration
MBA	Multi-Beam Antenna
MBIR	Model-Based Infrared
Mbps	Megabits Per Second
MDC	Mobile Data Centers, Micro Data Centers
MDM	Mode Division Multiplexing
MEC	Multi Access Edge Computing or Multi-Access Edge Cloud
MEF	Mask Error Factor, Mask Error Factors, Mask Error Fault
MEMS	Micro-Electro-Mechanical Systems
MFC	Mass Flow Controller
MFM	Multiple Feature Measurement, Mass Flow Meter
MHEMT	Metamorphic High Electron Mobility Transistor
MIMO	Multiple Input, Multiple Output
ML	Machine Learning
MM	More Moore IFT
MMDC	Mobile MDC
mMIMO	Massive MIMO
mMTC	Massive Machine-Type Communication
mmWave	Millimeter Wave
MNO	Mobile Network Operator
MOL	Middle of Line
MOS	Metal-Oxide Semiconductor; Metal-Oxide-Silicon
MPS	Mean Particle Size
MR	Merged Reality
MS	Mass Spectrometry
MtM	More than Moore IFT
MUX	Multiplexer
MVNO	Mobile Virtual Network Operators
NA	Numerical Apeture
NaaS	Network as a Service
NB & SB	North Bound & South Bound
NBD	Nanobeam Diffraction

NBED	Nanobeam Electron Diffraction
NB-IOT	Narrow Band – Internet of Things
NEP	Network Equipment Operators
NEXAF	Near Edge X-Ray Absorption Fine Structure
NF	Noise Figure (Db) (That Indicate Degradation Of the Signal-to-Noise Ratio)
NFV	Network Function Virtualization
nFET	NMOS field effect transistors
NFVi	Network Function Virtualization Infrastructure
NFVO	NFV Orchestrator
NGC	Next Generation Core
NGCO	Next-Generation Central Office
NGDC	Next Gen Data Centers
NGMN	Next Generation Mobile Networks
NIC	Network Interface Card
NIST	National Institute Of Standards and Technology
NMI	National Metrology Institutions
NMOS	Negative Channel Metal-Oxide Semiconductor
NOMA	Non-Orthogonal Multiple Accesses
N-PED or NPED	NBED in Precession Mode
NPN	N-Type-P-Type-N-Type Bipolar Transistor
NPU	Network Processing Unit or Neural Processing Unit
NR	New Radio
NRZ	Non-Return to Zero
NS	Network Slicing
NSA	Non-Standalone
NTC	Nanotechnology Council
NTN	Non-Terrestrial Network
NTRM	NIST Traceable Reference Materials
NTRS	National Technology Roadmap for Semiconductors
NUMA	Non-Uniform Memory Access
NVME	Non-Volatile Memory Express
O=C=S	Type Of Chemical Compound Containing Oxygen, Carbon, and Sulfur
OAM	Orbital Angular Momentum
OCD	Optical Critical Dimension
OE	Optical to Electronic
OEC	Open Edge Computing
OEMs	Original Equipment Manufacturer
OFDM	Orthogonal Frequency-Division Multiplexing
OMEC	Open Mobile Edge Cloud
ONAP	Open Networking Automation Platform
ONF	Open Networking Foundation
OOM	ONAP Operation Manager
OPC	Optical Proximity Correction
OpenNESS	Open Network Edge Services Software
OPEX	Operational Expenditure
OPNFV	Open Platform Network Virtualization
OSC	Outside System Connectivity IFT
OSM	Order and Service Management
OSS	Operational Support System
OTA	Over the Air
OTT	Over the Top
P	Phosphorus
P/T	Precision to Tolerance

PA	Power Amplifier
PAM	Pulse Amplitude Modulation
PCB	Printed Circuit Board
pCMP	Post-CMP or Post Chemical Mechanical Planarization
PdM	Predictive Maintenance
PDM	Polarization Division Multiplexing
PELS	Power Electronics Society
pFET	PMOS field effect transistors
PGW	Packet Gateway
PHEMT	Pseudomorphic High Electron Mobility Transistor
PHY	Physical Layer
PI	Packaging Integration IFT
PID	Photo Ionization Detector
PMOR	Photo Modulated Optical Reflectance
PMOS	Positive Channel Metal-Oxide Semiconductor
PoC	Proof of Concept
POD	Point of Delivery or Point of Distribution
POE	Point of Entry
POP	Point of Process
POS	Point of Supply
POU	Point-of-Use
POUP	POU Purification
PPAC	Power, performance, area, and cost
PPS	Packets Per Second
PSM	Phase Shift Mask
PTP	Point-to-Point
Qbits	Quantum Bits
QC	Quantum Computing
QEMU	Quick Emulator
QKD	Quantum Key Distribution
QoE	Quality of Experience
QoQ	Quarter on Quarter
QoS	Quality of Service
QSFP	Quad Small Formfactor Pluggable
R2R	Run-to-Run
RA	Reference Architecture
RAM	Random Access Memory
RAN	Radio Access Network
RCI	Rebooting Computing Initiative
RCWA	Rigorous Coupled Wave Analysis
RE	Range Extension
RF	Reference Framework, Radio Frequency
RI	Reference Implementation
RI-EAP	Reference Implementation of EAP
RII-EAP	Reference Intelligent Infrastructure for EAP (IEEE 1934.EAP)
RM	Reference Models
RMS	Reference Measurement System, Root Mean Square;
RNIS	Radio Network Information Service
RO	Ring Oscillator
ROADM	Reconfigurable Optical Add Drop Multiplexer
ROI	Return On Investment
RRH	Remote Radio Head
RS	Reliability Society

RS-EAP	Reference Stack for EAP (IEEE 1934.EAP)
RSRP	Reference Signal Received Power
RTOS	Real-time Operating Systems
RTT	Real-Time Technology
RU	Radio Unit
RWM	Reference Workload Models (IEEE 1934.EAP)
S/D	Scheduling and Dispatch
SA	Systems and Architectures IFT
SAM	Scanning Acoustic Microscopy, Scanning Auger Microscopy, Scanning Acoustical Microscopy, Self-Assembled Monolayer
SANS	Small Angle Neutron Scattering
SAXS	Small Angle X-Ray Scattering
SDD	Silicon Drift Detectors
SDM	Space Division Multiplexing
SDN	Software Defined Network
SDO	Standards Developing Organization Or Standards Development Organization
SDRJ	Systems and Devices Roadmap of Japan
SECC	Surface Environment Contamination Control
SEM	Scanning Electron Microscopy Or Scanning Electron Microscope
SEM-EDS	Scanning Electron Microscopy - Energy-Dispersive X-Ray Spectroscopy
SEMI	Semiconductor Equipment and Materials International
SERDES	Serializer Deserializer
SFQ	Single Flux Quantum
SIA	Semiconductor Industry Association
SiGe	Silicon Germanium Compounds
SIGFOX	A Wireless Interface, Cellular Style, Long Range, Low Power, Low Data Rate Form of Wireless Communications Developed to Provide Wireless Connectivity for Devices Like Remote Sensors, Actuators and Other M2M and IoT Devices
SIM	Subscriber Identification Module
SIMS	Secondary Ion Mass Spectroscopy
SLA	Service Level Agreements
SMC	Surface Molecular Contaminant
SMLY	Systematic Mechanisms Limited Yield
SOA	Semiconductor Optical Amplifier
SOI	Silicon On Insulator
SON	Self-Optimizing Network
SP	Service Provider
SPC	Statistical Process Control
SPM	Scanning Probe Microscopy, Scanning Probe Microscope
SPV	Surface Photovoltage
SRC	Semiconductor Research Corporation
SRM	Standard Reference Materials
SSCS	Solid State Circuit Society
SSD	Solid State Drive
SSi	Strained Silicon
SSOI	Strained Silicon On Insulator
SSRM	Scanning Spreading Resistance Microscopy
STEM	Scanning Transmission Electron Microscope/Microscopy or Science, Technology, Engineering, and Mathematics
STI	Shallow Trench Isolation
SUT	System Under Test
Tbps	Terabits Per Second
TCAD	Technology Computer Aided Design
TD	Thermal Desorption

TDD	Time-Division Duplex
TDDB	Time-Dependent Dielectric Breakdown
TD-LTE	Time Division-Long Term Evolution
TDMA	Time Division Multiple Access
TD-SCDMA	Time Division-Synchronous Code Division Multiple Access
TEM	Transmission Electron Microscopy Or Transmission Electron Microscope
TEM/EDS	Transmission Electron Microscopy Energy-Dispersive X Ray Spectroscopy
TEOS	Tetraethyl-Orthosilicate
TERS	Tip Enhanced Raman Scattering
TFET	Tunnel Field-Effect Transistor
THz	Terahertz
TIP	Telecommunication Infrastructure Platform
TMD	Transition Metal Dichalcogenides
TMU	Total Measurement Uncertainty
TOC	Total Organic Carbon Content
TOF	Time-Of-Flight
ToF-SIMS	Time-Of-Flight - Secondary Ion Mass Spectroscopy
TP	Technical Profile Of IEEE 2413
TPU	Tensor Processing Unit
T-SAXS	Transmission Small Angle X-Ray Scattering
TSDSI	Telecommunications Standards Development Society India
TSMC	Taiwan Semiconductor Manufacturing Company
TSOM	Through-Focus Scanning Optical Microscopy
TSVs	Through Silicon Vias
TTI	Transmission Time Interval
TTV	Total Thickness Variation
TUG	Telecom Users Group
TuT	Tools Under Test
Tx/Rx	Transmit and Receive
TXRF	Total Reflectance X-Ray Fluorescence, Total Reflection X-Ray Fluorescence
UAV	Unmanned Aerial Vehicles
uCPE	Universal Customer Premise Equipment
UE	User Equipment
UF	Ultra-Filtration
UL	Uplink
UP	User Plane
UPW	Ultrapure Water
URLLC	Ultra-Reliability Low-Latency Connection
UV	Ultraviolet
V2I	Vehicle to Infrastructure
V2V	Vehicle to Vehicle
vBBU	Virtual Base Band Unit
vBNG	Virtual Broadband Network Gateway
VC-D	Vibrational Circular Dichroism
VCO	Voltage-Controlled Oscillator
VCSEL	Vertical Cavity Surface Emitting Laser
vEPC	Virtual Evolved Packet Core
VGA	Video Graphics Array
VGAA	Vertical Gate All Around
VLAM	Visual Language Action Models
VLM	Visual Language Models
VLSI	Very Large Scale Integration
VM	Virtual Metrology

VMB	Valve Manifold Box
VMP	Valve Manifold Post
VNF	Virtual Network Function
VPD-ICPMS	Vapor Phase Decomposition-Inductively-Coupled Plasma Mass Spectrometry
VR	Virtual Reality
vRAN	Virtualized Radio Access Network
W2W	Wafer to Wafer
WAN	Wireless Area Network
WCDMA-HSPA	Wideband Code Division Multiple Access (WCDMA) and High-Speed Packet Access (HSPA)
WDM	Wavelength Division Multiplexing
WECC	Wafer Environment Contamination Control
WG	Working Group
WRC	World Radiocommunication Conferences
WSTS	World Semiconductor Trade Statistics
XCDA	Extreme Clean Dry Air
XPS	X-Ray Photoelectron Spectroscopy
XRD	X-Ray Diffraction
XRF	X-Ray Fluorescence
XRR	X-Ray Reflectivity
XSR	Insulation Displacement Connectors Are Space-Saving, 0.6mm Pitch Connectors Suitable for Thin Wires.
YE	Yield Enhancement
YOY	Year over Year

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